

Sharp Generalization for Shallow Neural Networks with Channel Attention

Yingzhen Yang

Arizona State University

YINGZHEN.YANG@ASU.EDU

Abstract

We study nonparametric regression using an over-parameterized two-layer neural network with channel attention, where training features are drawn from an arbitrary continuous distribution on the unit sphere in $\mathbb{R}^d$, and the target function lies in a standard interpolation space. We show that early-stopped gradient descent achieves a sharp regression risk of $\mathcal{O}(\varepsilon_n^2)$, where ε_n is the critical population rate of the induced attention kernel, improving upon the state-of-the-art [57] for distribution-free spherical covariates. When the covariate distribution satisfies an eigenvalue decay with parameter 2α and $\alpha > 1/2$, the rate becomes $\mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}})$ under spectral bias assumptions, improving over the nearly-optimal rate $\mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}}) \log^2(1/\delta)$ [30], where n is the sample size and $\delta \in (0, 1)$. This is, to our knowledge, the first work establishing a theoretical advantage of channel attention for nonparametric regression. Our analysis shows that channel attention aligns with spectrally biased targets and induces a novel attention kernel. We decompose the network output at each gradient descent step into an RKHS component of this kernel and a small L^∞ residual, and combine this with local Rademacher complexity to obtain sharp bounds. Our results further show that channel attention changes the training dynamics of the vanilla network without attention and enables escape from the linear NTK regime of the vanilla network, yielding better generalization than vanilla networks with lower kernel complexity, supported by simulations on synthetic and real data.

1. Introduction

The success of deep learning [28] has spurred theoretical studies on optimization and generalization of DNNs. Prior works show GD/SGD can drive training loss to zero under various settings [2, 3, 14, 15, 45, 68], while generalization theory provides guarantees on gradient-based methods. The Neural Tangent Kernel (NTK) [22] models over-parameterized networks as kernel methods, enabling tractable first-order analysis near initialization [3, 7, 17], with extensions capturing feature learning [56]. Beyond NTK, higher-order and alternative approaches, e.g., QuadNTK [4], hybrid methods [33], and mean-field analyses [13, 47], study feature learning effects. The literature also examines nonparametric regression with noisy data using DNNs. They achieve minimax rates for smooth [6, 23, 41, 63, 66] and non-smooth [21] targets, though often without algorithmic guarantees or relying on architectures not realized by GD. Recent works [20, 30, 46, 57, 58] analyze generalization of GD/SGD-trained DNNs. This paper studies the theoretical advantage of channel attention for nonparametric regression, related work in attention mechanisms is reviewed in Section D.4.

Summary of Contributions. We study nonparametric regression using an over-parameterized two-layer NN with XCA-style channel attention [1] trained by GD. For target functions in an interpolation space with spectral bias (Section 2.2), early stopping yields a sharp risk bound $\mathcal{O}(\varepsilon_n^2)$ under

arbitrary continuous spherical covariate distributions, where ε_n is the critical population rate of the induced kernel. The sharpness of $\mathcal{O}(\varepsilon_n^2)$ is reflected from the following two aspects: (i) without distributional assumptions, it improves upon state-of-the-art results [57]; (ii) under polynomial eigenvalue decay rate (EDR), it is minimax optimal and sharper than [30] for the same setting. See Section 3 and Table 1. Our work is among the first to show theoretical benefits of channel attention for nonparametric regression in interpolation spaces, enabling learning of spectrally biased targets [3, 8, 11, 38, 53, 54]. Channel attention induces a new “attention kernel,” distinct from NTK, with reduced complexity and analysis beyond the linear NTK regime.

Novelty and Significant Differences form [57]. Our novelty lies in analyzing GD-trained networks with channel attention via the induced attention kernel and establishing sharp regression rates using local complexity in its RKHS. While [57] applies local complexity to vanilla networks, we extend it to the attention kernel setting. Key challenges include proving uniform convergence to the attention kernel during training and conducting local complexity analysis in its RKHS. Details are in Section B.2.

The paper is organized as follows. Section 2 introduces the problem setup, and Section 3 presents main results. In the appendix, Section A describes GD training with channel attention, and Section B outlines our novel proof strategy and roadmap, with detailed proofs in the remaining parts of the appendix. Simulation results are presented in Section G of the appendix.

Notations. Bold letters denote matrices/vectors and lowercase letters denote scalars. A superscript (i) indicates the i -th column (e.g., $\mathbf{A}^{(i)}$), while subscripts denote rows/elements; $\vec{\mathbf{x}}_i$ denotes the i -th training feature. $\|\cdot\|_F$ and $\|\cdot\|_p$ denote the Frobenius and ℓ^p /matrix p -norms. $[m : n]$ denotes integers from m to n (with $[n] \equiv [1 : n]$), $\text{Var}[\cdot]$ the variance, $\mathbf{I}_n$ the identity, and $\mathbb{I}_{\{E\}}$ the indicator; A^c is the complement and $|A|$ the cardinality. $\text{vec}(\cdot)$ and $\text{tr}(\cdot)$ denote vectorization and trace. The unit sphere is $\mathbb{S}^{d-1} := \mathbf{x} \in \mathbb{R}^d : \|\mathbf{x}\|_2 = 1$; $L^2(\mathbb{S}^{d-1}, \mu)$ is the space of square-integrable functions with $\langle f, g \rangle_{L^2} := \int \mathbb{S}^{d-1} f(x)g(x)d\mu(x)$ and $\|f\|_{L^2}^2 := \int \mathbb{S}^{d-1} f^2(x)d\mu(x)$. $\mathbf{B}(\mathbf{x}; r)$ is the closed Euclidean ball. $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and $\|\cdot\|_{\mathcal{H}}$ denote the inner product and norm in $\mathcal{H}$. Asymptotics: $a = \mathcal{O}(b)$ or $a \lesssim b$, $\tilde{\mathcal{O}}, a = o(b)$, $a = w(b)$, and $a \asymp b$ or $a = \Theta(b)$ have standard meanings. We set $\mathcal{X} = \mathbb{S}^{d-1}$ with $\text{Unif}(\mathcal{X})$ the uniform distribution; for $g : \mathbb{S}^{d-1} \rightarrow \mathbb{R}$, $\|g\|_{\infty} := \sup_{\mathbf{x} \in \mathbb{S}^{d-1}} |g(\mathbf{x})|$, and L^{∞} is the class of functions with bounded $\|\cdot\|_{\infty}$. Constants may vary line to line. For RKHS $\mathcal{H}$, $\mathcal{H}(\mu_0)$ is the radius- μ_0 ball at the origin, and $\mathbb{E}_P[\cdot]$ denotes expectation under P .

2. Problem Setup

We introduce the problem setup for nonparametric regression using a neural network with channel attention in this section.

2.1. Two-Layer Neural Network

We are given training data $\left\{(\vec{\mathbf{x}}_i, y_i)\right\}_{i=1}^n$ with $\vec{\mathbf{x}}_i \in \mathcal{X}$, $y_i \in \mathbb{R}$, and assume $\vec{\mathbf{x}}_i \neq \vec{\mathbf{x}}_j$ for all $i \neq j$. Let $\mathbf{S} = \left\{\vec{\mathbf{x}}_i\right\}_{i=1}^n$, P_n be the empirical distribution over $\mathbf{S}$, and $\mathbf{y} = [y_1, \dots, y_n]^{\top} \in \mathbb{R}^n$. The responses satisfy $y_i = f^*(\vec{\mathbf{x}}_i) + w_i$ for $i \in [n]$, where $\{w_i\}_{i=1}^n$ are i.i.d. sub-Gaussian with mean 0 and variance proxy σ_0^2 , i.e., $\mathbb{E}[\exp(\lambda w_i)] \leq \exp(\lambda^2 \sigma_0^2 / 2)$ for any $\lambda \in \mathbb{R}$. Define $\mathbf{y} := [y_1, \dots, y_n]$, $\mathbf{w} := [w_1, \dots, w_n]^{\top}$, and $f^*(\mathbf{S}) := \left[f^*(\vec{\mathbf{x}}_1), \dots, f^*(\vec{\mathbf{x}}_n)\right]^{\top}$. The features are drawn i.i.d. from P with measure μ , where P is a continuous distribution on $\mathcal{X}$. We consider a two-layer NN with

channel attention:

$$f(\mathbf{W}, \mathbf{a}, \mathbf{x}) = \frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m a_r \sigma(\vec{\mathbf{w}}_{r'}^\top \mathbf{x}) \mathbf{A}_{r'r}, \quad (1)$$

where $\sigma(\cdot) = \max\{\cdot, 0\}$, $\mathbf{W} = \left\{ \vec{\mathbf{w}}_r \right\}_{r=1}^m$, $\vec{\mathbf{w}}_r \in \mathbb{R}^d$, and m is the width. The attention matrix is $\mathbf{A} = \sigma(\mathbf{W}(0), \mathbf{Q}) \sigma(\mathbf{W}(0), \mathbf{Q})^\top / (Nm)$, where $\sigma(\mathbf{W}(0), \mathbf{x}) \in \mathbb{R}^m$ has entries $[\sigma(\mathbf{W}(0), \mathbf{x})]_r = \sigma(\vec{\mathbf{w}}_r(0)^\top \mathbf{x})$. Here $\mathbf{a} \in \mathbb{R}^m$, $\mathbf{W}(0) = \left\{ \vec{\mathbf{w}}_r(0) \right\}_{r=1}^m$ with $\vec{\mathbf{w}}_r(0) \sim \mathcal{N}(\mathbf{0}, \kappa^2 \mathbf{I}_d)$, $\kappa = \Theta(1) \in (0, 1)$, and $\mathbf{Q} = \left\{ \vec{\mathbf{q}}_i \right\}_{i=1}^N$ are i.i.d. from P , independent of $\mathbf{W}(0)$, with $\sigma(\mathbf{W}(0), \mathbf{Q}) \in \mathbb{R}^{m \times N}$.

XCA-style Channel Attention [1] in the Two-Layer NN (1) and Generation of the Sample $\mathbf{Q}$. The channel attention in (1) follows XCA [1], applying self-attention across channels (viewed as tokens), so $\mathbf{A}$ encodes cross-channel attention weights. Depending on whether P is known, $\mathbf{Q}$ can be sampled exactly or approximately; details are deferred to Section D.1.

2.2. Kernel and Target Function for Nonparametric Regression

We define the kernel

$$K(\mathbf{u}, \mathbf{v}) := \kappa \cdot \frac{\mathbf{u}^\top \mathbf{v} (\pi - \arccos(\mathbf{u}^\top \mathbf{v})) + \sqrt{1 - (\mathbf{u}^\top \mathbf{v})^2}}{2\pi}, \quad (2)$$

for $\mathbf{u}, \mathbf{v} \in \mathcal{X}$, which is the NTK of the vanilla two-layer NN

$$f^{(\text{vanilla})}(\mathbf{W}, \mathbf{a}, \mathbf{x}) = \frac{1}{\sqrt{m}} \sum_{r=1}^m a_r \sigma(\vec{\mathbf{w}}_r^\top \mathbf{x}), \quad (3)$$

with $\mathbf{a}$ initialized to $\mathbf{0}$. K is PD. Let $\mathbf{K}_{ij} = K(\vec{\mathbf{x}}_i, \vec{\mathbf{x}}_j)$ and $\mathbf{K}_n = \mathbf{K}/n$ with eigendecomposition $\mathbf{K}_n = \mathbf{U} \Sigma \mathbf{U}^\top$, where $\{\hat{\lambda}_i\}_{i=1}^n$ are non-increasing and $\mathbf{K}_n$ is non-singular [15]. Let $\mathcal{H}_K$ be the RKHS of K . Since K is continuous on compact $\mathcal{X} \times \mathcal{X}$, the operator T_K is positive, self-adjoint, and compact, admitting eigenpairs $\{(e_j, \lambda_j)\}_{j \geq 0}$ with $T_K e_j = \lambda_j e_j$. Let $\{v_j = \sqrt{\lambda_j} e_j\}$ be an ONB of $\mathcal{H}_K$, and define $\mathcal{H}_K(\mu_0) = \{f \in \mathcal{H}_K : \|f\|_{\mathcal{H}} \leq \mu_0\}$, equivalently $\{f = \sum_{j \geq 1} \beta_j e_j : \sum_{j \geq 1} \beta_j^2 / \lambda_j \leq \mu_0^2\}$.

Target Function in an Interpolation Space with Spectral Bias. Neural networks exhibit spectral bias, favoring low-frequency components [3, 8, 11, 38, 53, 54]. Prior work shows over-parameterized NNs more easily learn low-degree polynomials, low-rank labels, or simple patterns [3, 8], motivating restriction of f^* to $\mathcal{H}_{K^{(\text{attn})}}(\mu_0)$. Define the attention kernel

$$K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') := \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{v}) K(\mathbf{v}, \mathbf{v}') K(\mathbf{v}', \mathbf{x}') d\mu(\mathbf{v}) \otimes \mu(\mathbf{v}'), \quad (4)$$

$$\hat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') := \frac{1}{N^2} \sum_{i,j=1}^N K(\mathbf{x}, \vec{\mathbf{q}}_i) K(\vec{\mathbf{q}}_i, \vec{\mathbf{q}}_j) K(\vec{\mathbf{q}}_j, \mathbf{x}'), \quad (5)$$

which arises from channel attention in (1). The operator $T_{K^{(\text{attn})}}$ shares eigenfunctions $\{e_j\}$ with T_K but has eigenvalues $\lambda_j^{(\text{attn})} = \lambda_j^3$ (Theorem 35). Thus $\mathcal{H}_{K^{(\text{attn})}}(\mu_0) = \{f = \sum_{j \geq 1} \beta_j e_j : \sum_{j \geq 1} \beta_j^2 / \lambda_j^3 \leq \mu_0^2\}$.

$\sum_{j \geq 1} \beta_j^2 / \lambda_j^3 \leq \mu_0^2\} \subseteq \mathcal{H}_K(\mu_0)$, implying stronger concentration on low-index eigenfunctions. This formalizes stronger spectral bias. Low-degree polynomials lie in the span $\{e_j\}_{j=1}^{m_{\ell+1}}$ and belong to $\mathcal{H}_{K(\text{attn})}(\mu_0)$ for suitable μ_0 . Moreover, $\mathcal{H}_{K(\text{attn})}(\mu_0) = [\mathcal{H}_K]^3(\mu_0)$, where $[\mathcal{H}_K]^{s'}(\mu_0) = \{\sum_{j \geq 1} a_j \lambda_j^{s'/2} e_j : \sum a_j^2 \leq \mu_0\}$. We focus on $s' = 3$ and $f^* \in [\mathcal{H}_K]^3(\mu_0)$. Hence, $f^* \in \mathcal{H}_{K(\text{attn})}(\mu_0) = [\mathcal{H}_K]^3(\mu_0) \subseteq \mathcal{H}_K(\mu_0)$ exhibits spectral bias.

The Task of Nonparametric Regression. Given data $\{(\vec{\mathbf{x}}_i, y_i)\}_{i=1}^n$, the goal is to learn $\hat{f}$ such that $\mathbb{E}_P[(\hat{f} - f^*)^2] \rightarrow 0$ at a fast rate, where $f^* \in [\mathcal{H}_K]^3(\mu_0)$, and $\hat{f}$ is the over-parameterized NN (1). Prior work establishes sharp convergence rates for kernel regression [39, 44, 60, 65]; in particular, [39] shows $\mathbb{E}_P[(\hat{f} - f^*)^2] \lesssim \varepsilon_{K,n}^2$, where $\varepsilon_{K,n}$ is the critical population rate (radius [50]) of K , achieved by GD with early stopping. This bound is minimax optimal in common settings (e.g., polynomial EDR of T_K with $f^* \in \mathcal{H}_K(\mu_0)$). We will show that the two-layer NN with channel attention (1) trained by GD attains sharper, minimax optimal risk bounds when $f^* \in [\mathcal{H}_K]^3(\mu_0)$.

3. Main Results

All results and discussions in this paper are presented under the fixed-dimension setting with $d \geq 5$, a widely adopted setting in prior works [20, 30, 46, 57, 58]. We present the first main result, Theorem 1, as follows, for arbitrary continuous distribution P supported on $\mathcal{X}$.

Theorem 1 Suppose P is an arbitrary continuous distribution on $\mathcal{X}$, $\delta \in (0, 1)$,

$$m \gtrsim \max\{(d^4 + \log^4 m)/\varepsilon_n^{16}, (d \log m)^3/\varepsilon_n^8, n\}, N \gtrsim \log(n/\delta)/\varepsilon_n^8, \quad (6)$$

and the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \mathbf{a}(t), \cdot)$ is trained by GD in Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$ with $T \leq \hat{T}$. Then for every $t \in [c_t T, T]$, with probability at least $1 - 2/n - \delta - \exp(-\Theta(n)) - 7 \exp(-\Theta(n \varepsilon_n^2))$ over $\mathbf{w}, \mathbf{S}, \mathbf{Q}, \mathbf{W}(0)$, the stopping time satisfies $\hat{T} \asymp \varepsilon_n^{-2}$, and

$$\mathbb{E}_P[(f_t - f^*)^2] \lesssim \varepsilon_n^2. \quad (7)$$

Theorem 1 establishes sharper risk bound than that in [57]. We emphasize that Theorem 1 renders sharper regression risk bound than the current state-of-the-art, [57, Theorem 5.1], and both works consider arbitrary continuous distribution P of the covariate. In particular, [57, Theorem 5.1] shows a sharp regression risk bound of $\mathcal{O}(\varepsilon_{K,n}^2)$ when training the vanilla network $f^{(\text{vanilla})}$ (3) without channel attention by GD, when $f \in \mathcal{H}_K(\mu_0)$. In the same distribution-free manner in the covariate, Theorem 1 shows the clear theoretical benefit of the XCA-style channel attention [1] on the vanilla network. That is, when the target function f^* lies in the interpolation space $[\mathcal{H}_K]^3(\mu_0) \subseteq \mathcal{H}_K(\mu_0)$ with spectral bias, a sharper regression risk bound of $\mathcal{O}(\varepsilon_n^2)$ is achieved when training the network (1) by GD. This is because $\varepsilon_n^2 \leq \varepsilon_{K,n}^2$ according to Proposition 13 in the appendix. The sharper risk bound by Theorem 1 is instantiated for certain distribution P in Theorem 2 below.

Applying Theorem 1 to the case where the distribution P admits a polynomial EDR for the kernel K (2) such that $\lambda_j \asymp j^{-2\alpha}$ for $\alpha > 1/2$ and all $j \geq 1$, we have the following theorem as a direct consequence of Theorem 1. Such polynomial EDR holds when P is the uniform distribution on $\mathcal{X}$, with $2\alpha = d/(d-1)$, which is shown by existing works such as [20, Lemma 3.1]. The proofs of Theorem 1 and Theorem 2 are deferred to Section E.3 of the appendix.

Theorem 2 Suppose the distribution P admits a polynomial EDR of $\lambda_j \asymp j^{-2\alpha}$ for $\alpha > 1/2$ and all $j \geq 1$, $\delta \in (0, 1)$,

$$m \gtrsim n^{48\alpha/(6\alpha+1)} d^4 \log^4 m, \quad N \gtrsim n^{\frac{24\alpha}{6\alpha+1}} \log(n/\delta), \quad (8)$$

and the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ is trained by GD in Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$ with $T \leq \hat{T}$. Then for every $t \in [c_t T : T]$, with probability at least $1 - 2/n - \delta - \exp(-\Theta(n)) - 7 \exp(-\Theta(n^{\frac{1}{6\alpha+1}}))$ over $\mathbf{w}, \mathbf{S}, \mathbf{Q}, \mathbf{W}(0)$, the stopping time satisfies $\hat{T} \asymp n^{\frac{6\alpha}{6\alpha+1}}$, and

$$\mathbb{E}_P [(f_t - f^*)^2] \lesssim n^{-\frac{6\alpha}{6\alpha+1}}. \quad (9)$$

Table 1: Comparisons with existing works on the regression risk bounds and assumptions for non-parametric regression using over-parameterized neural networks with algorithmic guarantees. The results listed are under a widely studied setup where $f^* \in \mathcal{H}_{\tilde{K}}$ and the responses $\{y_i\}_{i=1}^n$ are corrupted by i.i.d. Gaussian or sub-Gaussian noise. Here P denotes the distribution of the training features, and $\tilde{K}$ represents the kernel induced by the neural architecture and optimization method of each particular work. For all prior works, $\tilde{K}$ corresponds to the regular NTK, while in this work we instead have $\tilde{K} = K^{(\text{attn})}$. Both our work and [30] consider target functions satisfying $f^* \in [\mathcal{H}_K]^3$ under the polynomial EDR $\lambda_j \asymp j^{-2\alpha}$, and $2\alpha = d/(d-1)$ in [20, 30, 46]. Moreover, [30, Proposition 13] can be adapted to our setting with no bias/intercept learned in the first layer, leading to the polynomial EDR of $\lambda_j \asymp j^{-\frac{d}{d-1}}$ rather than $\lambda_j \asymp j^{-\frac{d+1}{d}}$.

Existing Works and Our Result	Distributional Assumptions	Eigenvalue Decay Rate (EDR)	Rate of Nonparametric Regression Risk
[26, Theorem 2]	No	–	$\sigma^2 + \mathcal{O}(n^{-\frac{2}{d-1}})$
[20, Theorem 5.2], [46, Theorem 3.11]	P is Unif ($\mathcal{X}$)	$\lambda_j \asymp j^{-\frac{d}{d-1}}$	$\mathcal{O}(n^{-\frac{2\alpha}{2\alpha+1}}) = \mathcal{O}(n^{-\frac{d}{2d-1}})$
[30, Proposition 13]	P is sub-Gaussian	$\lambda_j \asymp j^{-\frac{d}{d-1}}$	$\mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}}) \log^2(1/\delta)$
[57, Theorem 5.1]	Arbitrary continuous distribution on $\mathcal{X}$	No requirement for EDR	$\mathcal{O}(\varepsilon_{K,n}^2)$
[57, Corollary 5.2]	P admits the polynomial EDR $\lambda_j \asymp j^{-2\alpha}$	$\lambda_j \asymp j^{-\frac{d}{d-1}}$	$\mathcal{O}(n^{-\frac{2\alpha}{2\alpha+1}})$
Our Result (Theorem 1)	Arbitrary continuous distribution on $\mathcal{X}$	No requirement for EDR	$\mathcal{O}(\varepsilon_n^2)$
Our Result (Theorem 2)	P admits the polynomial EDR $\lambda_j \asymp j^{-2\alpha}$	$\lambda_j^{(\text{attn})} \asymp j^{-\frac{3d}{d-1}}$	$\mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}}) = \mathcal{O}(n^{-\frac{3d}{4d-1}})$

Minimax Optimality of the Risk Bound by Theorem 2. While the rate $\mathcal{O}(n^{-\frac{d}{2d-1}})$ in Hu et al. [20], Suh et al. [46], Yang and Li [58] is minimax optimal for kernel regression with NTK $\tilde{K}$ when $f^* \in \mathcal{H}_{\tilde{K}}(\mu_0)$, a faster rate is possible if f^* lies in the interpolation space $[\mathcal{H}_K]^3(\mu_0) \subseteq \mathcal{H}_K(\mu_0)$. In particular, if $f^* \in [\mathcal{H}_K]^{s'}(\mu_0)$, then regression with the attention kernel $K^{(\text{attn})}$ achieves the sharper rate $\mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}})$, which is minimax optimal over $\mathcal{H}_{K^{(\text{attn})}}(\mu_0)$ [44, 60, 65]. Our bound (9) improves upon [30, Proposition 13] by removing the $\log^2(1/\delta)$ factor. By Theorem 35, $K^{(\text{attn})}$ has EDR $\lambda_j^{(\text{attn})} = \lambda_j^3 \asymp j^{-6\alpha}$ for all $j \geq 1$, yielding a kernel complexity fixed point $\mathcal{O}(\varepsilon_n^2) = \mathcal{O}(n^{-\frac{6\alpha}{6\alpha+1}})$ and thus the sharper risk bound. Additional comparisons are given in Table 1, including works on interpolation spaces and general RKHS.

4. Conclusion

We study nonparametric regression via an over-parameterized two-layer neural network with channel attention, where the target lies in an interpolation space exhibiting spectral bias. We show that GD with early stopping achieves a sharp rate $\mathcal{O}(\varepsilon_n^2)$ for arbitrary continuous covariate distributions on the unit sphere in $\mathbb{R}^d$, which is minimax optimal when the distribution has polynomial eigenvalue decay. Our analysis introduces novel proof strategies, along with comparisons to state-of-the-art methods and supporting simulations.

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References

- [1] Alaaeldin Ali, Hugo Touvron, Mathilde Caron, Piotr Bojanowski, Matthijs Douze, Armand Joulin, Ivan Laptev, Natalia Neverova, Gabriel Synnaeve, Jakob Verbeek, and Hervé Jégou. Xcit: Cross-covariance image transformers. In Marc’Aurelio Ranzato, Alina Beygelzimer, Yann N. Dauphin, Percy Liang, and Jennifer Wortman Vaughan, editors, *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 20014–20027, 2021.
- [2] Zeyuan Allen-Zhu, Yuanzhi Li, and Zhao Song. A convergence theory for deep learning via over-parameterization. In *International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 242–252. PMLR, 2019.
- [3] Sanjeev Arora, Simon S. Du, Wei Hu, Zhiyuan Li, and Ruosong Wang. Fine-grained analysis of optimization and generalization for overparameterized two-layer neural networks. In *International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 322–332. PMLR, 2019.
- [4] Yu Bai and Jason D. Lee. Beyond linearization: On quadratic and higher-order approximation of wide neural networks. In *International Conference on Learning Representations*. OpenReview.net, 2020.
- [5] Peter L. Bartlett, Olivier Bousquet, and Shahar Mendelson. Local rademacher complexities. *Ann. Statist.*, 33(4):1497–1537, 08 2005.
- [6] Benedikt Bauer and Michael Kohler. On deep learning as a remedy for the curse of dimensionality in nonparametric regression. *Ann. Statist.*, 47(4):2261 – 2285, 2019.
- [7] Yuan Cao and Quanquan Gu. Generalization bounds of stochastic gradient descent for wide and deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett, editors, *Advances in Neural Information Processing Systems*, pages 10835–10845, 2019.

- [8] Yuan Cao, Zhiying Fang, Yue Wu, Ding-Xuan Zhou, and Quanquan Gu. Towards understanding the spectral bias of deep learning. In Zhi-Hua Zhou, editor, *International Joint Conference on Artificial Intelligence*, pages 2205–2211. ijcai.org, 2021.
- [9] A. Caponnetto and E. De Vito. Optimal rates for the regularized least-squares algorithm. *Foundations of Computational Mathematics*, 7(3):331–368, Jul 2007. ISSN 1615-3383. doi: 10.1007/s10208-006-0196-8.
- [10] Ziheng Chen, Yue Song, Xiaojun Wu, Gaowen Liu, and Nicu Sebe. Understanding matrix function normalizations in covariance pooling through the lens of riemannian geometry. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025.
- [11] Moulik Choraria, Leello Tadesse Dadi, Grigorios Chrysos, Julien Mairal, and Volkan Cevher. The spectral bias of polynomial neural networks. In *International Conference on Learning Representations*. OpenReview.net, 2022.
- [12] Krzysztof Marcin Choromanski, Valerii Likhoshesterov, David Dohan, Xingyou Song, Andreea Gane, Tamás Szepesvári, Peter Hawkins, Jared Quincy Davis, Afroz Mohiuddin, Lukasz Kaiser, David Benjamin Belanger, Lucy J. Colwell, and Adrian Weller. Rethinking attention with performers. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021.
- [13] Alexandru Damian, Jason D. Lee, and Mahdi Soltanolkotabi. Neural networks can learn representations with gradient descent. In Po-Ling Loh and Maxim Raginsky, editors, *Conference on Learning Theory, 2-5 July 2022, London, UK*, volume 178 of *Proceedings of Machine Learning Research*, pages 5413–5452. PMLR, 2022.
- [14] Simon S. Du, Jason D. Lee, Haochuan Li, Liwei Wang, and Xiyu Zhai. Gradient descent finds global minima of deep neural networks. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 1675–1685. PMLR, 2019.
- [15] Simon S. Du, Xiyu Zhai, Barnabas Póczos, and Aarti Singh. Gradient descent provably optimizes over-parameterized neural networks. In *International Conference on Learning Representations*, 2019.
- [16] Jun Fu, Jing Liu, Haijie Tian, Yong Li, Yongjun Bao, Zhiwei Fang, and Hanqing Lu. Dual attention network for scene segmentation. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pages 3146–3154. Computer Vision Foundation / IEEE, 2019.
- [17] Behrooz Ghorbani, Song Mei, Theodor Misiakiewicz, and Andrea Montanari. Linearized two-layers neural networks in high dimension. *Ann. Statist.*, 49(2):1029 – 1054, 2021.
- [18] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In Hugo Larochelle, Marc’Aurelio Ranzato, Raia Hadsell, Maria-Florina Balcan, and Hsuan-Tien Lin, editors, *Advances in Neural Information Processing Systems*, 2020.

- [19] Jiri Hron, Yasaman Bahri, Jascha Sohl-Dickstein, and Roman Novak. Infinite attention: NNGP and NTK for deep attention networks. In *Proceedings of the 37th International Conference on Machine Learning, ICML 2020, 13-18 July 2020, Virtual Event*, volume 119 of *Proceedings of Machine Learning Research*, pages 4376–4386. PMLR, 2020.
- [20] Tianyang Hu, Wenjia Wang, Cong Lin, and Guang Cheng. Regularization matters: A non-parametric perspective on overparametrized neural network. In Arindam Banerjee and Kenji Fukumizu, editors, *International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pages 829–837. PMLR, 2021.
- [21] Masaaki Imaizumi and Kenji Fukumizu. Deep neural networks learn non-smooth functions effectively. In Kamalika Chaudhuri and Masashi Sugiyama, editors, *International Conference on Artificial Intelligence and Statistics*, volume 89 of *Proceedings of Machine Learning Research*, pages 869–878. PMLR, 2019.
- [22] Arthur Jacot, Clément Hongler, and Franck Gabriel. Neural tangent kernel: Convergence and generalization in neural networks. In Samy Bengio, Hanna M. Wallach, Hugo Larochelle, Kristen Grauman, Nicolò Cesa-Bianchi, and Roman Garnett, editors, *Advances in Neural Information Processing Systems*, pages 8580–8589, 2018.
- [23] Yuling Jiao, Guohao Shen, Yuanyuan Lin, and Jian Huang. Deep nonparametric regression on approximate manifolds: Nonasymptotic error bounds with polynomial prefactors. *Ann. Statist.*, 51(2):691 – 716, 2023.
- [24] Juno Kim, Tai Nakamaki, and Taiji Suzuki. Transformers are minimax optimal nonparametric in-context learners. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [25] Vladimir Koltchinskii. Local rademacher complexities and oracle inequalities in risk minimization. *Ann. Statist.*, 34(6):2593–2656, 12 2006.
- [26] Ilja Kuzborskij and Csaba Szepesvári. Nonparametric regression with shallow overparametrized neural networks trained by GD with early stopping. In Mikhail Belkin and Samory Kpotufe, editors, *Conference on Learning Theory, COLT 2021, 15-19 August 2021, Boulder, Colorado, USA*, volume 134 of *Proceedings of Machine Learning Research*, pages 2853–2890. PMLR, 2021.
- [27] B. Laurent and P. Massart. Adaptive estimation of a quadratic functional by model selection. *The Annals of Statistics*, 28(5):1302 – 1338, 2000.
- [28] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521:436–444, 2015.
- [29] Michel Ledoux. *Probability in Banach Spaces [electronic resource] : Isoperimetry and Processes / by Michel Ledoux, Michel Talagrand*. Classics in Mathematics. Springer Berlin Heidelberg, Berlin, Heidelberg, 1st ed. 1991. edition, 1991.
- [30] Yicheng Li, Zixiong Yu, Guhan Chen, and Qian Lin. On the eigenvalue decay rates of a class of neural-network related kernel functions defined on general domains. *Journal of Machine Learning Research*, 25(82):1–47, 2024.

- [31] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. 2023.
- [32] Shahar Mendelson. Geometric parameters of kernel machines. In Jyrki Kivinen and Robert H. Sloan, editors, *Conference on Computational Learning Theory*, volume 2375 of *Lecture Notes in Computer Science*, pages 29–43. Springer, 2002.
- [33] Eshaan Nichani, Yu Bai, and Jason D. Lee. Identifying good directions to escape the NTK regime and efficiently learn low-degree plus sparse polynomials. In Sanmi Koyejo, S. Mohamed, A. Agarwal, Danielle Belgrave, K. Cho, and A. Oh, editors, *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*, 2022.
- [34] George Papamakarios, Eric T. Nalisnick, Danilo Jimenez Rezende, Shakir Mohamed, and Balaji Lakshminarayanan. Normalizing flows for probabilistic modeling and inference. *J. Mach. Learn. Res.*, 22:57:1–57:64, 2021.
- [35] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision, ICCV 2023, Paris, France, October 1-6, 2023*, pages 4172–4182. IEEE, 2023.
- [36] Hao Peng, Nikolaos Pappas, Dani Yogatama, Roy Schwartz, Noah A. Smith, and Lingpeng Kong. Random feature attention. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021.
- [37] Iosif Pinelis. *An Approach to Inequalities for the Distributions of Infinite-Dimensional Martingales*, pages 128–134. Birkhäuser Boston, Boston, MA, 1992. ISBN 978-1-4612-0367-4. doi: 10.1007/978-1-4612-0367-4_9.
- [38] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 5301–5310. PMLR, 09–15 Jun 2019.
- [39] Garvesh Raskutti, Martin J. Wainwright, and Bin Yu. Early stopping and non-parametric regression: an optimal data-dependent stopping rule. *J. Mach. Learn. Res.*, 15(1):335–366, 2014.
- [40] Danilo Jimenez Rezende and Shakir Mohamed. Variational inference with normalizing flows. In Francis R. Bach and David M. Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning, ICML 2015, Lille, France, 6-11 July 2015*, volume 37 of *JMLR Workshop and Conference Proceedings*, pages 1530–1538. JMLR.org, 2015.
- [41] Johannes Schmidt-Hieber. Nonparametric regression using deep neural networks with ReLU activation function. *Ann. Statist.*, 48(4):1875 – 1897, 2020.
- [42] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*.

- [43] Yue Song, Nicu Sebe, and Wei Wang. Why approximate matrix square root outperforms accurate SVD in global covariance pooling? In *2021 IEEE/CVF International Conference on Computer Vision, ICCV 2021, Montreal, QC, Canada, October 10-17, 2021*, pages 1095–1103. IEEE, 2021.
- [44] Charles J. Stone. Additive Regression and Other Nonparametric Models. *Ann. Statist.*, 13(2): 689 – 705, 1985.
- [45] Lili Su and Pengkun Yang. On learning over-parameterized neural networks: A functional approximation perspective. In *Advances in Neural Information Processing Systems*, pages 2637–2646, 2019.
- [46] Namjoon Suh, Hyunouk Ko, and Xiaoming Huo. A non-parametric regression viewpoint : Generalization of overparametrized deep RELU network under noisy observations. In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net, 2022.
- [47] Shokichi Takakura and Taiji Suzuki. Mean-field analysis on two-layer neural networks from a kernel perspective. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net, 2024.
- [48] Roman Vershynin. Introduction to the non-asymptotic analysis of random matrices. In Yonina C. Eldar and Gitta Kuttyniok, editors, *Compressed Sensing: Theory and Practice*, pages 210–268. Cambridge University Press, 2012.
- [49] Oriol Vinyals, Charles Blundell, Tim Lillicrap, Koray Kavukcuoglu, and Daan Wierstra. Matching networks for one shot learning. In Daniel D. Lee, Masashi Sugiyama, Ulrike von Luxburg, Isabelle Guyon, and Roman Garnett, editors, *Advances in Neural Information Processing Systems 29: Annual Conference on Neural Information Processing Systems 2016, December 5-10, 2016, Barcelona, Spain*, pages 3630–3638, 2016.
- [50] Martin J. Wainwright. *High-Dimensional Statistics: A Non-Asymptotic Viewpoint*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2019.
- [51] Qilong Wang, Banggu Wu, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. Ecanet: Efficient channel attention for deep convolutional neural networks. In *2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2020, Seattle, WA, USA, June 13-19, 2020*, pages 11531–11539. Computer Vision Foundation / IEEE, 2020.
- [52] Qilong Wang, Zhaolin Zhang, Mingze Gao, Jiangtao Xie, Pengfei Zhu, Peihua Li, Wangmeng Zuo, and Qinghua Hu. Towards a deeper understanding of global covariance pooling in deep learning: An optimization perspective. *IEEE Trans. Pattern Anal. Mach. Intell.*, 45(12): 15802–15819, 2023.
- [53] Yancheng Wang, Rajeev Goel, Utkarsh Nath, Alvin C. Silva, Teresa Wu, and Yingzhen Yang. Learning low-rank feature for thorax disease classification. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang, editors, *Advances in Neural Information Processing Systems*, 2024.

- [54] Yancheng Wang, Changyu Liu, and Yingzhen Yang. Diffusion on graph: Augmentation of graph structure for node classification. *Trans. Mach. Learn. Res.*, 2025, 2025.
- [55] F. T. Wright. A Bound on Tail Probabilities for Quadratic Forms in Independent Random Variables Whose Distributions are not Necessarily Symmetric. *Ann. Probab.*, 1(6):1068 – 1070, 1973.
- [56] Greg Yang and Edward J. Hu. Tensor programs IV: feature learning in infinite-width neural networks. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 11727–11737. PMLR, 2021.
- [57] Yingzhen Yang. Sharp generalization for nonparametric regression by over-parameterized neural networks: A distribution-free analysis in spherical covariate. In *International Conference on Machine Learning (ICML)*, 2025.
- [58] Yingzhen Yang and Ping Li. Gradient descent finds over-parameterized neural networks with sharp generalization for nonparametric regression. *arXiv preprint arXiv:2411.02904*, 2024. URL <https://arxiv.org/abs/2411.02904>.
- [59] Yingzhen Yang and Ping Li. Sharp generalization for nonparametric regression in interpolation space by over-parameterized neural networks trained with preconditioned gradient descent and early-stopping. 2025. URL <https://arxiv.org/abs/2407.11353>.
- [60] Yuhong Yang and Andrew Barron. Information-theoretic determination of minimax rates of convergence. *Ann. Statist.*, 27(5):1564 – 1599, 1999.
- [61] Yun Yang, Mert Pilanci, and Martin J. Wainwright. Randomized sketches for kernels: Fast and optimal nonparametric regression. *Ann. Statist.*, 45(3):991 – 1023, 2017.
- [62] Yuan Yao, Lorenzo Rosasco, and Andrea Caponnetto. On early stopping in gradient descent learning. *Constructive Approximation*, 26(2):289–315, Aug 2007. ISSN 1432-0940. doi: 10.1007/s00365-006-0663-2.
- [63] Dmitry Yarotsky. Error bounds for approximations with deep relu networks. *Neural Networks*, 94:103–114, 2017.
- [64] Zixiong Yu, Songtao Tian, and Guhan Chen. Divergence of neural tangent kernel in classification problems. In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net, 2025.
- [65] Ming Yuan and Ding-Xuan Zhou. Minimax optimal rates of estimation in high dimensional additive models. *Ann. Statist.*, 44(6):2564 – 2593, 2016.
- [66] Kaiqi Zhang and Yu-Xiang Wang. Deep learning meets nonparametric regression: Are weight-decayed dnns locally adaptive? In *International Conference on Learning Representations*. OpenReview.net, 2023.
- [67] Lin Zheng, Jianbo Yuan, Chong Wang, and Lingpeng Kong. Efficient attention via control variates. In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net, 2023.

- [68] Difan Zou and Quanquan Gu. An improved analysis of training over-parameterized deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett, editors, *Advances in Neural Information Processing Systems*, pages 2053–2062, 2019.

Appendix A. Training by Gradient Descent

Algorithm 1 Training the Two-Layer NN with Channel Attention (1) by GD

```

1:  $\mathbf{W}(T) \leftarrow \text{Training-by-GD}(T, \mathbf{W}(0))$ 
2: input:  $T, \mathbf{W}(0), \eta$ 
3: for  $t = 1, \dots, T$ 
4:   Perform the  $t$ -th step of GD by (11)-(12)
5: end for
6: Return  $\mathbf{W}(T)$ 

```

In the training process of our two-layer NN (1), only $\mathbf{W}$ is optimized with $\mathbf{a}$ randomly initialized to ± 1 with equal probabilities and then fixed. The following quadratic loss function is minimized during the training process:

$$L(\mathbf{W}) := \frac{1}{2n} \sum_{i=1}^n \left(f(\mathbf{W}, \mathbf{a}, \vec{\mathbf{x}}_i) - y_i \right)^2. \quad (10)$$

In the $(t+1)$ -th step of GD with $t \geq 0$, the weights of the neural network, $\mathbf{W}$ and $\mathbf{a}$, are updated by one-step of GD through

$$\text{vec}(\mathbf{W}_{\mathbf{S}}(t+1)) = \text{vec}(\mathbf{W}_{\mathbf{S}}(t)) - \frac{\eta}{n} \mathbf{Z}_{\mathbf{S}}(t)(\hat{\mathbf{y}}(t) - \mathbf{y}), \quad (11)$$

$$\mathbf{a}(t+1) = \mathbf{a}(t) - \frac{\eta}{n\sqrt{m}} \mathbf{A}\boldsymbol{\sigma}(\mathbf{W}(t), \mathbf{S})(\hat{\mathbf{y}}(t) - \mathbf{y}), \quad (12)$$

where $\mathbf{y}_i = y_i$, $\hat{\mathbf{y}}(t) \in \mathbb{R}^n$ with $[\hat{\mathbf{y}}(t)]_i = f(\mathbf{W}(t), \mathbf{a}(t), \vec{\mathbf{x}}_i)$. The notations with the subscript $\mathbf{S}$ indicate the dependence on the training features $\mathbf{S}$. We also denote $f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ as $f_t(\cdot)$ which is the neural network function with weights $\mathbf{W}(t)$ and $\mathbf{a}(t)$ obtained right after the t -th step of GD. We define $\mathbf{Z}_{\mathbf{S}}(t) \in \mathbb{R}^{md \times n}$ which is computed by $[\mathbf{Z}_{\mathbf{S}}(t)]_{[(r-1)d+1:rd]i} = \frac{1}{\sqrt{m}} \mathbb{I}_{\{\vec{\mathbf{w}}_r(t)^\top \vec{\mathbf{x}}_i \geq 0\}} \vec{\mathbf{x}}_i [\mathbf{A}\mathbf{a}(t)]_r$ for all $i \in [n], r \in [m]$, where $[\mathbf{Z}_{\mathbf{S}}(t)]_{[(r-1)d+1:rd]i} \in \mathbb{R}^d$ is a vector with elements in the i -th column of $\mathbf{Z}_{\mathbf{S}}(t)$ with indices in $[(r-1)d+1 : rd]$. We have $\mathbf{a} = \mathbf{0}$ at the initialization, so that $\hat{\mathbf{y}}(0) = \mathbf{0}$. We run Algorithm 1 to train the two-layer NN by GD, where T is the total number of steps for GD. Early stopping is enforced in Algorithm 1 through a bounded T via $T \leq \hat{T}$.

Appendix B. Roadmap of Proofs

We present the roadmap of our theoretical results which lead to the main results, Theorem 1 and Theorem 2. We first introduce kernel complexity in Section B.1, a key concept in our results and their proofs. Section B.2 details the roadmap, key technical results in the proofs, our novel proof strategies and insights from our theoretical results.

B.1. Kernel Complexity

The local kernel complexity has been studied by [5, 25, 32]. For the PD kernel K , we define the empirical kernel complexity $\hat{R}_K$ and the population kernel complexity R_K as

$$\begin{aligned}\hat{R}_K(\varepsilon) &:= \sqrt{\frac{1}{n} \sum_{i=1}^n \min \{\hat{\lambda}_i, \varepsilon^2\}}, \\ R_K(\varepsilon) &:= \sqrt{\frac{1}{n} \sum_{i=1}^{\infty} \min \{\lambda_i, \varepsilon^2\}}.\end{aligned}\tag{13}$$

It can be verified that both $\sigma_0 R_K(\varepsilon)$ and $\sigma_0 \hat{R}_K(\varepsilon)$ are sub-root functions [5] in terms of ε^2 . Sub-root functions are defined in Definition 9. For a given noise ratio σ_0 , the critical empirical radius $\hat{\varepsilon}_{K,n} > 0$ is the smallest positive solution to the inequality $\hat{R}_K(\varepsilon) \leq \varepsilon^2/\sigma_0$, where $\hat{\varepsilon}_{K,n}^2$ is also the fixed point of $\sigma_0 \hat{R}_K(\varepsilon)$ as a function of ε^2 : $\sigma_0 \hat{R}_K(\hat{\varepsilon}_{K,n}) = \hat{\varepsilon}_{K,n}^2$. Similarly, the critical population rate $\varepsilon_{K,n}$ is defined to be the smallest positive solution to the inequality $R_K(\varepsilon) \leq \varepsilon^2/\sigma_0$, where $\varepsilon_{K,n}^2$ is the fixed point of $\sigma_0 R_K(\varepsilon)$ as a function of ε^2 : $\sigma_0 R_K(\varepsilon_{K,n}) = \varepsilon_{K,n}^2$. Kernel complexity can also be defined for the attention kernel $K^{(\text{attn})}$, leading to the empirical kernel complexity $\hat{R}_{K^{(\text{attn})}}$ and the population kernel complexity $R_{K^{(\text{attn})}}$ for $K^{(\text{attn})}$, with the critical empirical radius $\hat{\varepsilon}_{K^{(\text{attn})},n}$ and the critical population rate $\varepsilon_{K^{(\text{attn})},n}$, respectively. For simplicity of the notations, we use ε_n and $\hat{\varepsilon}_n$ to denote $\varepsilon_{K^{(\text{attn})},n}$ and $\hat{\varepsilon}_{K^{(\text{attn})},n}$, respectively. In this paper we consider the kernel K such that $\min \{\varepsilon_{K,n}, \varepsilon_n\} \cdot n \rightarrow \infty$ as $n \rightarrow \infty$, which covers most popular positive semi-definite kernels including the kernel (2) and a broad range of data distributions [61]. Let $\eta_t := \eta t$ for all $t \geq 0$, we then define the stopping time $\hat{T}$ as

$$\hat{T} := \min \left\{ T : \hat{R}_{K^{(\text{attn})}}(\sqrt{1/\eta_t}) > (\sigma_0 \eta_t)^{-1} \right\} - 1.\tag{14}$$

The stopping time in fact limits the number of steps T for Algorithm 1, which enforces the early stopping mechanism. In fact, as will be shown later in this section, we need to have $T \leq \hat{T}$ when training the two-layer NN (1) by GD with Algorithm 1.

B.2. Detailed Roadmap, Key Results, and Novel Proof Strategies and Insights

The detailed roadmap, summary of the key technical results, and our novel proof strategies which lead to Theorem 1 are presented as follows. Theorem 2 follows from Theorem 1 by applying Theorem 1 to the case of the polynomial EDR of $\lambda_j \asymp j^{-2\alpha}$ for all $j \geq 1$.

Roadmap and Key Technical Results. First, uniform convergence of the empirical attention kernel, $\hat{K}^{(\text{attn})}$, to the attention kernel, $K^{(\text{attn})}$, is established during training the two-layer NN with channel attention (1) by GD, which is presented in the following theorem.

Theorem 3 *For every fixed $\mathbf{x}' \in \mathcal{X}$ and every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the random sample $\mathbf{Q}$, we have $\left\| \hat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right\|_{\infty} \lesssim \sqrt{\frac{\log 1/\delta}{N}}$.*

The proof of Theorem 3 adapts the martingale based concentration inequality for Banach space-valued process [37, Theorem 2] to the RKHS associated with the attention kernel $K^{(\text{attn})}$.

Based on Theorem 3, we establish a novel decomposition of the neural network function at any GD step into a function within the RKHS associated with the attention kernel $K^{(\text{attn})}$ and an error function small L^∞ -norm with high probability, as stated in Theorem 4.

Theorem 4 *Suppose $\delta \in (0, 1)$, $w \in (0, 1)$, m, N are sufficiently large and finite, and the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ is trained by GD using Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$. Then for every $t \in [T]$ with $T \leq \hat{T}$, with high probability over the random initialization $\mathbf{W}(0)$, the random noise $\mathbf{w}$, and the random sample $\mathbf{Q}$, f_t has the following decomposition on $\mathcal{X}$:*

$$f_t = h_t + e_t, \quad (15)$$

where $h_t \in \mathcal{H}_K(B_h)$ with B_h defined in (29), $e_t \in L^\infty$ with $\|e_t\|_\infty \leq w$. The lower bounds for m, N depends on w , and smaller w leads to larger lower bounds for m, N .

Second, leveraging the decomposition in Theorem 4, we introduce a new technique based on local Rademacher complexity to obtain a tight bound on the Rademacher complexity of the function class formed by all neural network functions generated through GD iterations. This development leads directly to the sharp regression risk bound presented in Theorem 5 below.

Theorem 5 *Suppose $\delta \in (0, 1)$, $w \in (0, 1)$, m, N are sufficiently large and finite, and the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ is trained by GD using Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$, and $T \leq \hat{T}$. Then for every $t \in [T]$ and every $\delta \in (0, 1)$, with high probability over the random initialization $\mathbf{W}(0)$, the random noise $\mathbf{w}$, the random training features $\mathbf{S}$, and the random sample $\mathbf{Q}$, $\mathbb{E}_P [(f_t - f^*)^2] - 2\mathbb{E}_{P_n} [(f_t - f^*)^2] \lesssim \varepsilon_n^2 + w$.*

We then obtain Theorem 1 using Theorem 5 where w is set to ε_n^2 , with the empirical loss $\mathbb{E}_{P_n} [(f_t - f^*)^2]$ bounded by $\Theta(1/(\eta t)) \asymp \varepsilon_n^2$ with high probability by Theorem 18 deferred to Section E.4 of the appendix.

Novel Proof Strategies and Insights for the Benefit of Channel Attention. As emphasized in Section 1, the main novelty of this paper is to study the generalization capability of neural networks with channel attention trained by GD through the lens of the new attention kernel $K^{(\text{attn})}$ induced by channel attention. Compared to [57] which uses local complexity-based method on the RKHS associated with the NTK of the vanilla network $f^{(\text{vanilla})}$, our results rely on two substantially novel proof strategies built upon local complexity-based analysis in the new RKHS $\mathcal{H}_{K^{(\text{attn})}}$ associated with the attention kernel. First, the uniform convergence of $\hat{K}^{(\text{attn})}$ to $K^{(\text{attn})}$ during the training process of the network with channel attention (1), established in Theorem 3, enables a new decomposition of the neural network function at any step of GD into a function in $\mathcal{H}_{K^{(\text{attn})}} = [\mathcal{H}_K]^3$ and an error function with small L^∞ -norm with high probability in Theorem 4. The uniform convergence of $\hat{K}^{(\text{attn})}$ to $K^{(\text{attn})}$ in Theorem 3 is proved by employing the martingale-based concentration inequality for Banach space-valued processes [37, Theorem 2]. Second, leveraging the decomposition in Theorem 4, we introduce a new technique based on local Rademacher complexity, which tightly bounds the Rademacher complexity of the function class consisting of all neural network functions generated by GD iterations. This leads to the sharp regression risk bound in Theorem 5.

To the best of our knowledge, our results reveal the theoretical benefit of the XCA-style channel attention [1] mechanism used in our two-layer NN (1) for nonparametric regression in an interpolation space. In particular, with sufficiently large network width m in the over-parameterized regime,

the network (1) trained by GD approximately performs kernel regression with the new attention kernel $K^{(\text{attn})}$. In a strong contrast, over-parameterized neural networks trained by GD, widely studied in existing works on such as [30, 46, 57, 58], induces the standard NTK of the form such as (2). As elaborated in ‘‘Sharper Regression Risk Bound by Theorem 1’’ in Section 3, channel attention yields a sharper regression risk bound $\mathcal{O}(\varepsilon_n^2)$ for learning the target function in the interpolation space $[\mathcal{H}_K]^3$, compared to the risk bound $\mathcal{O}(\varepsilon_{K,n}^2)$ rendered by the vanilla network $f^{(\text{vanilla})}$ (3) without such channel attention. The fundamental reason for the sharper bound is that, the network with channel attention (1) induces the attention kernel $K^{(\text{attn})}$, whose kernel complexity $R_{K^{(\text{attn})}}$ is lower than the kernel complexity of the standard NTK (2), which is induced by the vanilla network $f^{(\text{vanilla})}$.

Beyond the Regular NTK Limit of the Vanilla Network $f^{(\text{vanilla})}$ (3). We remark that our result is beyond the NTK limit or the linear region of the regular NTK (2) of the vanilla network $f^{(\text{vanilla})}$ defined in (3), since the function represented by the two-layer NN with channel attention (1) trained with our novel GD is arbitrarily close to some $h_t \in \mathcal{H}_{K^{(\text{attn})}}(B_h)$, where $\mathcal{H}_{K^{(\text{attn})}}$ is an RKHS distinct from $\mathcal{H}_K$ associated with the regular NTK (2). In particular, training the counterpart network without channel attention, $f^{(\text{vanilla})}$ (3), cannot achieve our sharp risk bound. Furthermore, it is technically nontrivial to induce the new kernel $K^{(\text{attn})}$ when training with the proposed GD algorithm, as detailed through our proof strategies described above. Our results are significantly from the existing kernel learning literature and they lead to a better lower bound on the network width m compared to the existing literature, which are detailed in Section D.3 and Section D.2 of the appendix.

Appendix C. Mathematical Tools

The basic mathematical results employed in our proofs are provided in this section.

C.1. Concentration Inequalities for Supremum of Empirical Processes

The Rademacher complexity of a function class and its empirical version are defined below.

Definition 6 Let $\sigma = \{\sigma_i\}_{i=1}^n$ be n i.i.d. random variables such that $\Pr[\sigma_i = 1] = \Pr[\sigma_i = -1] = \frac{1}{2}$. The Rademacher complexity of a function class $\mathcal{F}$ is defined as

$$\mathfrak{R}(\mathcal{F}) = \mathbb{E}_{\{\vec{\mathbf{x}}_i\}_{i=1}^n, \{\sigma_i\}_{i=1}^n} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i) \right]. \quad (16)$$

The empirical Rademacher complexity is defined as

$$\hat{\mathfrak{R}}(\mathcal{F}) = \mathbb{E}_{\{\sigma_i\}_{i=1}^n} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i) \right], \quad (17)$$

For simplicity of notations, Rademacher complexity and empirical Rademacher complexity are also denoted by $\mathbb{E} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i) \right]$ and $\mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i) \right]$ respectively.

For data $\{\vec{\mathbf{x}}\}_{i=1}^n$ and a function class $\mathcal{F}$, we define the notation $R_n \mathcal{F}$ by $R_n \mathcal{F} := \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i)$.

Theorem 7 ([5, Theorem 2.1]) *Let $\mathcal{X}, P$ be a probability space, $\{\vec{\mathbf{x}}_i\}_{i=1}^n$ be independent random variables distributed according to P . Let $\mathcal{F}$ be a class of functions that map $\mathcal{X}$ into $[a, b]$. Assume that there is some $r > 0$ such that for every $f \in \mathcal{F}$, $\text{Var}[f(\vec{\mathbf{x}}_i)] \leq r$. Then, for every $x > 0$, with probability at least $1 - e^{-x}$,*

$$\sup_{f \in \mathcal{F}} (\mathbb{E}_P[f(\mathbf{x})] - \mathbb{E}_{P_n}[f(\mathbf{x})]) \leq \inf_{\alpha > 0} \left(2(1 + \alpha) \mathbb{E}_{\{\vec{\mathbf{x}}_i\}_{i=1}^n, \{\sigma_i\}_{i=1}^n} [R_n \mathcal{F}] + \sqrt{\frac{2rx}{n}} + (b - a) \left(\frac{1}{3} + \frac{1}{\alpha} \right) \frac{x}{n} \right), \quad (18)$$

and with probability at least $1 - 2e^{-x}$,

$$\sup_{f \in \mathcal{F}} (\mathbb{E}_P[f(\mathbf{x})] - \mathbb{E}_{P_n}[f(\mathbf{x})]) \leq \inf_{\alpha \in (0,1)} \left(\frac{2(1+\alpha)}{1-\alpha} \mathbb{E}_{\{\sigma_i\}_{i=1}^n} [R_n \mathcal{F}] + \sqrt{\frac{2rx}{n}} + (b - a) \left(\frac{1}{3} + \frac{1}{\alpha} + \frac{1+\alpha}{2\alpha(1-\alpha)} \right) \frac{x}{n} \right). \quad (19)$$

P_n is the empirical distribution over $\{\vec{\mathbf{x}}_i\}_{i=1}^n$ with $\mathbb{E}_{P_n}[f(\mathbf{x})] = \frac{1}{n} \sum_{i=1}^n f(\vec{\mathbf{x}}_i)$. Moreover, the same results hold for $\sup_{f \in \mathcal{F}} (\mathbb{E}_{P_n}[f(\mathbf{x})] - \mathbb{E}_P[f(\mathbf{x})])$.

In addition, we have the contraction property for Rademacher complexity, which is due to Ledoux and Talagrand [29].

Theorem 8 *Let ϕ be a contraction, that is, $|\phi(x) - \phi(y)| \leq \mu |x - y|$ for $\mu > 0$. Then, for every function class $\mathcal{F}$,*

$$\mathbb{E}_{\{\sigma_i\}_{i=1}^n} [R_n \phi \circ \mathcal{F}] \leq \mu \mathbb{E}_{\{\sigma_i\}_{i=1}^n} [R_n \mathcal{F}], \quad (20)$$

where $\phi \circ \mathcal{F}$ is the function class defined by $\phi \circ \mathcal{F} = \{\phi \circ f : f \in \mathcal{F}\}$.

Definition 9 (Sub-root function, [5, Definition 3.1]) *A function $\psi : [0, \infty) \rightarrow [0, \infty)$ is sub-root if it is nonnegative, nondecreasing and if $\frac{\psi(r)}{\sqrt{r}}$ is nonincreasing for $r > 0$.*

Theorem 10 ([5, Theorem 3.3]) *Let $\mathcal{F}$ be a class of functions with ranges in $[a, b]$ and assume that there are some functional $T : \mathcal{F} \rightarrow \mathbb{R}^+$ and some constant $\bar{B}$ such that for every $f \in \mathcal{F}$, $\text{Var}[f] \leq T(f) \leq \bar{B}P(f)$. Let ψ be a sub-root function and let r^* be the fixed point of ψ . Assume that ψ satisfies that, for any $r \geq r^*$, $\psi(r) \geq \bar{B}\mathfrak{R}(\{f \in \mathcal{F} : T(f) \leq r\})$. Fix $x > 0$, then for any $K_0 > 1$, with probability at least $1 - e^{-x}$,*

$$\forall f \in \mathcal{F}, \quad \mathbb{E}_P[f] \leq \frac{K_0}{K_0 - 1} \mathbb{E}_{P_n}[f] + \frac{704K_0}{\bar{B}} r^* + \frac{x(11(b - a) + 26\bar{B}K_0)}{n}.$$

Also, with probability at least $1 - e^{-x}$,

$$\forall f \in \mathcal{F}, \quad \mathbb{E}_{P_n}[f] \leq \frac{K_0 + 1}{K_0} \mathbb{E}_P[f] + \frac{704K_0}{\bar{B}} r^* + \frac{x(11(b - a) + 26\bar{B}K_0)}{n}.$$

Appendix D. More Details about the Theoretical Results and Experiments

D.1. Generation of the Random Sample $\mathbf{Q}$

We note that the sample $\mathbf{Q}$ contains i.i.d. random variables $\{\vec{\mathbf{q}}_i\}_{i=1}^N$ distributed according to P , the same distribution as the training features $\mathbf{S}$. When $N \leq n$, we can directly use a subset of size N of $\mathbf{S}$ as $\mathbf{Q}$. Otherwise, $\mathbf{S}$ is used as a subset of $\mathbf{Q}$. A remaining set $\mathbf{Q}'$ of $N - n$ i.i.d. random variables distributed according to P is sampled, and $\overline{\mathbf{Q}} = \mathbf{Q}' \cup \mathbf{S}$. To be shown in the next paragraph, if P is known, $\mathbf{Q}'$ can be sampled exactly according to P so that $\overline{\mathbf{Q}}$ serves as $\mathbf{Q}$. If P is unknown, $\mathbf{Q}'$ can be sampled approximately according to P , and $\overline{\mathbf{Q}}$ can be used in practice as an approximation to $\mathbf{Q}$.

In practice, $\mathbf{Q}'$ can be sampled depending on if P is known or not. $\mathbf{Q}'$ can be sampled exactly according to P if P is known, or sampled approximately according to P if P is unknown. In particular, if P is a known distribution, $\mathbf{Q}'$ can be sampled by inverse transform sampling with invertible cumulative distribution function (CDF) of the distribution P or rejection sampling using the probability density function of P without invertible CDF. If P is unknown, we can train generative models on $\mathbf{S}$ and then generate synthetic data points distributed approximately according to P as the sample $\mathbf{Q}'$. These generative models learn an approximation $\hat{P}$ to the underlying data distribution P and enable efficient sampling from $\hat{P}$. Popular classes include (1) diffusion models, which generate samples by iteratively denoising noise using a learned reverse-time stochastic process [18, 42], (2) flow matching methods, which directly learn continuous-time vector fields that transport noise to data distributions [31], and (3) normalizing flows, which construct an invertible transformation mapping a simple base distribution (e.g., Gaussian) into the data distribution P , trained via maximum likelihood [34, 40]. These approaches provide a principled mechanism for generating synthetic data points that approximate the unknown P .

D.2. Difference from Existing Kernel Learning Theory

In this subsection, we demonstrate that our results in Section 3 are fundamentally different from the existing kernel learning theory, such as the minimax lower rate in [9] for kernel regression using the regular NTK defined in (2). In particular, our regression risk bound obtained by the network with channel attention (1) is sharper and fundamentally different from that in [9]. Under the same source condition on the target function that $f^* \in \mathcal{H}_{K^{(\text{attn})}}(\mu_0) = [\mathcal{H}_K]^3(\mu_0)$, the existing minimax lower rate in [9, Theorem 2] for kernel regression using the regular NTK (2) is $\mathcal{O}\left(\tau(\delta) n^{-\frac{6\alpha}{6\alpha+1}}\right)$ with probability $1 - \delta$, where $\tau(\delta) \rightarrow \infty$ as $\delta \rightarrow 0$, under the polynomial EDR of $\lambda_j \asymp j^{-2\alpha}$ for $\alpha > 1/2$. That is, to ensure the rate holds with probability approaching 1 (or $1 - \delta$ with $\delta \rightarrow 0$), there is an additional cost $\tau(\delta) \rightarrow \infty$ as $\delta \rightarrow 0$ [9]. This is the fundamental reason that the rate obtained by [30, Proposition 13] is $\mathcal{O}\left(n^{-\frac{6\alpha}{6\alpha+1}}\right) \log^2(1/\delta)$, which contains the additional logarithmic factor $\log^2(1/\delta)$ compared to our rate in Theorem 2.

In strong contrast, the two-layer NN with channel attention (1) trained by GD achieves the sharper and minimax optimal rate of $\mathcal{O}\left(n^{-\frac{6\alpha}{6\alpha+1}}\right)$ in Theorem 2. The fundamental reason for such a sharper rate is that canonical kernel regression methods [9, 62] only apply to kernels with the original capacity condition, such as the regular NTK (2) with the polynomial EDR $\lambda_j \asymp j^{-2\alpha}$. On the other hand, two-layer NN with channel attention (1) trained by GD approximately performs kernel regression with a completely different new kernel, namely the attention kernel $K^{(\text{attn})}$, which satisfies the smoother capacity condition $\lambda_j^{(\text{attn})} = \lambda_j^3 \asymp j^{-6\alpha}$. Our key insight is that the interpolation

space $[\mathcal{H}_K]^3$ is in fact the RKHS associated with the integral kernel, that is, $[\mathcal{H}_K]^3 = \mathcal{H}_{K^{(\text{attn})}}$. Kernel regression with the integral kernel and the target function $f^* \in \mathcal{H}_{K^{(\text{attn})}}(\mu_0)$ renders the minimax optimal rate of $\mathcal{O}\left(n^{-\frac{6\alpha}{6\alpha+1}}\right)$ according to the analytical results in [44, 60, 65], which coincides with the rate in Theorem 2.

D.3. Better Lower Bound for Network Width m

The lower bound on the network width m required for our result in Theorem 2, $m \gtrsim n^{48\alpha/(6\alpha+1)} d^3 \log^3 m$ with $\alpha = d/(2(d-1))$, is smaller than that required by the current state-of-the-art. In particular, [46, Theorem 3.11] show that $m/\log^3 m \gtrsim L^{20} n^{24}$, where L is the number of layers of the DNN in their work, which further implies $m/\log^3 m \gtrsim 2^{20} n^{24}$ even for the two-layer NN considered here with $L = 2$. Similarly, [30] require $m/(\log m)^{12} \gtrsim n^{24}$ for regression with target function $f^* \in [\mathcal{H}_K]^3$, which is the same source condition studied in this paper. Both lower bounds for m in [30, 46] are therefore much larger than ours in the regime $n \rightarrow \infty$ with fixed d , which is precisely the setting considered in prior works on training over-parameterized neural networks for nonparametric regression with sharp rates and algorithmic guarantees [20, 30, 46, 57, 58].

D.4. Existing Empirical and Theoretical Works about Channel Attention and General Attention Mechanism

Popular channel attention methods [1, 16, 51] enhances DNN representations by adaptively reweighting channels. In particular, XCI^T [1] views channel attention as cross-covariance across features, showing strong performance for classification. Covariance pooling [10, 43, 52] has been applied to channels with various theoretical results such as stability of DNNs and preconditioner effect. Kernelizable attention is studied in [12, 36, 67] with fast attention matrix approximation, and [19] studies the behavior of multi-head attention architectures as Gaussian Processes with infinite number of heads. While few works, such as [24], study the optimality of attention-based neural networks on in-context learning (ICL) tasks, the theoretical benefits of the attention mechanism, especially channel attention, on standard nonparametric regression tasks remain largely unknown.

Appendix E. Detailed Proofs

We present the detailed proofs for the theoretical results of this paper, and the basic notations are introduced are first introduced in Section E.1.

E.1. Basic Definitions

We introduce the following definitions for our analysis. We introduce the following definitions for the proof of Theorem 1 and Theorem 2. Let the gram matrix of $K^{(\text{attn})}$ over the training features $\mathbf{S}$ be $\mathbf{K}^{(\text{attn})} \in \mathbb{R}^{n \times n}$, $\mathbf{K}_{ij}^{(\text{attn})} = K^{(\text{attn})}(\vec{\mathbf{x}}_i, \vec{\mathbf{x}}_j)$ for $i, j \in [n]$, and $\mathbf{K}_n^{(\text{attn})} := \mathbf{K}^{(\text{attn})}/n$. Similarly, $\widehat{\mathbf{K}}^{(\text{attn})} \in \mathbb{R}^{n \times n}$ is the gram matrix of $\widehat{K}^{(\text{attn})}$ over $\mathbf{S}$, and $\widehat{\mathbf{K}}_n^{(\text{attn})} = \widehat{\mathbf{K}}^{(\text{attn})}/n$. Let the singular value decomposition of $\mathbf{K}_n^{(\text{attn})}$ be $\mathbf{K}_n^{(\text{attn})} = \mathbf{U}^{(\text{attn})} \mathbf{\Sigma}^{(\text{attn})} \mathbf{U}^{(\text{attn})\top}$, where $\mathbf{\Sigma}^{(\text{attn})}$ is a diagonal matrix with its diagonal elements $\{\widehat{\lambda}_i^{(\text{attn})}\}_{i=1}^n$ being the eigenvalues of $\mathbf{K}_n^{(\text{attn})}$ and sorted in a non-increasing order. We have $\widehat{\lambda}_1^{(\text{attn})} \in (0, 1)$, and we show in Proposition 36 deferred to Section F.1

that $\mathbf{K}_n^{(\text{attn})}$ is always non-singular. We define

$$\mathbf{u}(t) := \widehat{\mathbf{y}}(t) - \mathbf{y} \quad (21)$$

as the difference between the network output $\widehat{\mathbf{y}}(t)$ and the training response vector $\mathbf{y}$ right after the t -th step of GD. Let $0 < \tau \leq 1$, for $\tau, t \geq 0$, and $T \geq 1$ we define the following quantities: $c_{\mathbf{u}} := \mu_0 / \min \{ \sqrt{2e\eta}, 1 \} + \sigma_0 + \tau + 1$,

$$R := \frac{\eta c_{\mathbf{u}} (4d + 6 \log(2mn)) T}{\sqrt{m}}, \quad (22)$$

$$\mathcal{V}_t := \left\{ \mathbf{v} \in \mathbb{R}^n : \mathbf{v} = - \left(\mathbf{I}_n - \eta \mathbf{K}_n^{(\text{attn})} \right)^t f^*(\mathbf{S}) \right\}, \quad (23)$$

$$\mathcal{E}_{t,\tau} := \left\{ \mathbf{e} : \mathbf{e} = \vec{\mathbf{e}}_1 + \vec{\mathbf{e}}_2 \in \mathbb{R}^n, \vec{\mathbf{e}}_1 = - \left(\mathbf{I}_n - \eta \mathbf{K}_n^{(\text{attn})} \right)^t \mathbf{w}, \left\| \vec{\mathbf{e}}_2 \right\|_2 \leq \sqrt{n\tau} \right\}. \quad (24)$$

We define

$$\mathcal{W}_0 := \{ \mathbf{W}(0) : (31) \text{ holds} \} \quad (25)$$

as the set of all the good random initializations which satisfy (31) in Theorem 11.

Lemma 14 in Section E.4 shows that with high probability over the random initialization $\mathbf{W}(0)$ and the random noise $\mathbf{w}$, the distance of every weighting vector $\mathbf{w}_r(t)$ to its initialization $\mathbf{w}_r(0)$ is bounded by R , and the distance of every weighting vector $\mathbf{a}_r(t)$ to its initialization 0 is bounded by $2R$. In addition, $\mathbf{u}(t)$ can be composed into two vectors, $\mathbf{u}(t) = \mathbf{v}(t) + \mathbf{e}(t)$ such that $\mathbf{v}(t) \in \mathcal{V}_t$ and $\mathbf{e}(t) \in \mathcal{E}_{t,\tau}$. We then define the set of the neural network weights during the training by GD using Algorithm 1 as follows:

$$\begin{aligned} \mathcal{W}(\mathbf{S}, \mathbf{W}(0), T) &:= \left\{ (\mathbf{W}, \mathbf{a}) : \exists t \in [T] \text{ s.t. } \text{vec}(\mathbf{W}) = \text{vec}(\mathbf{W}(0)) - \sum_{t'=0}^{t-1} \frac{\eta}{n} \mathbf{Z}_{\mathbf{S}}(t') \mathbf{u}(t'), \right. \\ \mathbf{a}(t) &= \sum_{t'=0}^{t-1} -\frac{\eta}{n\sqrt{m}} \mathbf{A} \boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) \mathbf{u}(t'), \\ \left. \mathbf{u}(t') \in \mathbb{R}^n, \mathbf{u}(t') = \mathbf{v}(t') + \mathbf{e}(t'), \mathbf{v}(t') \in \mathcal{V}_{t'}, \mathbf{e}(t') \in \mathcal{E}_{t',\tau}, \text{ for all } t' \in [0, t-1] \right\}. \quad (26) \end{aligned}$$

We will also show by Lemma 14 that with high probability over $\mathbf{w}$, $\mathcal{W}(\mathbf{S}, \mathbf{W}(0), T)$ is the set of the weights of the two-layer NN (1) trained by GD on the training features $\mathbf{S}$ with the random initialization $\mathbf{W}(0)$ and the number of steps of GD not greater than T .

The set of the functions represented by the neural network with weights in $\mathcal{W}(\mathbf{S}, \mathbf{W}(0), T)$ is then defined as

$$\mathcal{F}_{\text{NN}}(\mathbf{S}, \mathbf{W}(0), T) := \{ f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot) : \exists t \in [T], (\mathbf{W}(t), \mathbf{a}(t)) \in \mathcal{W}(\mathbf{S}, \mathbf{W}(0), T) \}. \quad (27)$$

We also define the function class $\mathcal{F}(B, w)$ for any $B, w > 0$ as

$$\mathcal{F}(B, w) := \{f: f = h + e, h \in \mathcal{H}_K(B), \|e\|_\infty \leq w\}. \quad (28)$$

We will show by Theorem 15 in Section E.4 that with high probability over $\mathbf{w}$, $\mathcal{F}_{\text{NN}}(\mathbf{S}, \mathbf{W}(0), T)$ is a subset of $\mathcal{F}(B_h, w)$, where a smaller w requires a larger network width m , and $B_h > \mu_0$ is an absolute positive constant defined by

$$B_h := \mu_0 + 1 + \sqrt{2}. \quad (29)$$

E.2. Uniform Convergence to the NTK (2)

We define the following functions with $\mathbf{W} = \{\mathbf{w}_r\}_{r=1}^m$:

$$h(\mathbf{w}, \mathbf{u}, \mathbf{v}) := \sigma(\mathbf{w}^\top \mathbf{u}) \sigma(\mathbf{w}^\top \mathbf{v}), \quad \hat{h}(\mathbf{W}, \mathbf{u}, \mathbf{v}) := \frac{1}{m} \sum_{r=1}^m h(\vec{\mathbf{w}}_r, \mathbf{u}, \mathbf{v}), \quad (30)$$

where $\mathbf{u}, \mathbf{v} \in \mathcal{X}$. Then we have the following theorem stating the uniform convergence of $\hat{h}(\mathbf{W}(0), \cdot, \cdot)$ to $K(\cdot, \cdot)$.

Theorem 11 *Suppose $m \gtrsim n$ and $m/\log m \geq d$. Then with probability at least $1 - 1/n$ over the random initialization $\mathbf{W}(0) = \{\vec{\mathbf{w}}_r(0)\}_{r=1}^m$,*

$$\sup_{\mathbf{u} \in \mathcal{X}, \mathbf{v} \in \mathcal{X}} \left| K(\mathbf{u}, \mathbf{v}) - \hat{h}(\mathbf{W}(0), \mathbf{u}, \mathbf{v}) \right| \leq C_1(m, d, 1/n) \lesssim d \log m \sqrt{\frac{d \log m}{m}}, \quad (31)$$

where $C_1(m, d, 1/n)$ is a positive number depending on (m, d, n) , and its formal definition is deferred to (49) in Section E.5.

Proof This theorem follows from Theorem 19 in Section E.5. ■

Repeating the proof of Theorem 11, we have the following proposition, which states the boundedness of $\hat{h}(\mathbf{Q}, \mathbf{u}, \mathbf{v}) = 1/N \cdot \sum_{i=1}^N h(\vec{\mathbf{q}}_i, \mathbf{u}, \mathbf{v})$, with $\mathbf{Q} = \{\vec{\mathbf{q}}_i\}_{i=1}^N$ as sample of N i.i.d. random variables distributed according to arbitrary continuous distribution P supported on $\mathcal{X}$.

Proposition 12 *Suppose $m \gtrsim n$ and $N/\log N \geq d$. Then with probability at least $1 - 1/n$ over the random sample $\mathbf{Q}$,*

$$\sup_{\mathbf{u} \in \mathcal{X}, \mathbf{v} \in \mathcal{X}} \left| h(\mathbf{Q}, \mathbf{u}, \mathbf{v}) - \mathbb{E}_{\mathbf{q} \sim P} \left[\sigma(\mathbf{q}^\top \mathbf{u}) \sigma(\mathbf{q}^\top \mathbf{v}) \right] \right| \lesssim \sqrt{\frac{d \log m}{m}}, \quad (32)$$

We define

$$\mathcal{W}_0 := \{\mathbf{W}(0): (31) \text{ holds}\} \quad (33)$$

as the set of all the good random initializations which satisfy (31) in Theorem 11. Theorem 11 shows that we have good random initialization with high probability, that is, $\Pr[\mathbf{W}(0) \in \mathcal{W}_0] \geq 1 - 1/n$. When $\mathbf{W}(0) \in \mathcal{W}_0$, the uniform convergence (31) holds with high probability, which is important for the analysis of the training dynamics of the two-layer NN with channel attention (1) by GD.

E.3. Proofs for the Main Result, Theorem 1 and Theorem 2

We note that Theorem 15, Theorem 17, and Theorem 33 are the formal versions of Theorem 4, Theorem 5, and Theorem 3 in Section B.2 of this paper.

Proof [Proof of Theorem 1] We apply Theorem 17 and Theorem 18 to prove this theorem.

First, with the condition on m in this theorem, Theorem 11 hold, and $\Pr[\mathbf{W}(0) \in \mathcal{W}_0] \geq 1 - 1/n$. With $\eta = \Theta(1)$, it follows by Theorem 18 that with probability at least $1 - \exp(-\Theta(n\hat{\varepsilon}_n^2))$ over the random noise $\mathbf{w}$,

$$\mathbb{E}_{P_n} [(f_t - f^*)^2] \lesssim \frac{1}{\eta t}.$$

Plugging such bound for $\mathbb{E}_{P_n} [(f_t - f^*)^2]$ in (45) of Theorem 17 leads to

$$\mathbb{E}_P [(f_t - f^*)^2] \lesssim \frac{1}{\eta t} + \varepsilon_n^2 + w. \quad (34)$$

Due to the definition of $\hat{T}$ and $\hat{\varepsilon}_n^2$, we have

$$\hat{\varepsilon}_n^2 \leq \frac{1}{\eta \hat{T}} \leq \frac{2}{\eta(\hat{T} + 1)} \leq 2\hat{\varepsilon}_n^2. \quad (35)$$

It follows from Lemma 30 that that $\hat{\varepsilon}_n^2 \asymp \varepsilon_n^2$ with probability at least $1 - 4 \exp(-\Theta(n\varepsilon_n^2))$ over $\mathbf{S}$. In addition, combined with the fact that $T \asymp \hat{T}$, for any $t \in [c_t T, T]$, we have

$$\frac{1}{\eta t} \asymp \frac{1}{\eta \hat{T}} \asymp \frac{1}{\eta T} \asymp \hat{\varepsilon}_n^2 \asymp \varepsilon_n^2.$$

We set $w = \varepsilon_n^2$ in (34), then with $\eta = \Theta(1)$,

$$\mathbb{E}_P [(f_t - f^*)^2] \lesssim \varepsilon_n^2. \quad (36)$$

With $N \gtrsim \log(n/\delta)/\varepsilon_n^8$, the requirement on N , (41) in Theorem 15 that $N \gtrsim \max\{T^2 \log(n/\delta)/w^2, T^4 \log(n/\delta)\}$ is satisfied. In addition, with

$$m \gtrsim \max\{(d^4 + \log^4 m)/\varepsilon_n^{16}, (d \log m)^3/\varepsilon_n^8, n\},$$

and $w = \varepsilon_n^2$, the condition (40) on m in Theorem 15 is satisfied. $\blacksquare$

Proof [Proof of Theorem 2] We apply Theorem 1 to prove this theorem. First, it then follows from Theorem 35 in Section F.1 that $\lambda_j^{(\text{attn})} = \lambda_j^3 \asymp j^{-6\alpha}$ for $j \geq 1$. For such EDR of $\{\lambda_j^{(\text{attn})}\}_{j \geq 1}$, it is well known, such as [39, Corollary 3], that $\varepsilon_n^2 \asymp n^{-\frac{6\alpha}{6\alpha+1}}$. It also follows from Lemma 30 that that $\hat{\varepsilon}_n^2 \asymp \varepsilon_n^2$ with probability at least $1 - 4 \exp(-\Theta(n\varepsilon_n^2))$ over $\mathbf{S}$. This theorem is then proved by plugging in $\varepsilon_n^2 \asymp \varepsilon_n^2 \asymp n^{-\frac{6\alpha}{6\alpha+1}}$ and $w = \varepsilon_n^2 \asymp n^{-\frac{6\alpha}{6\alpha+1}}$ in Theorem 1. $\blacksquare$

Proposition 13 We have $\varepsilon_n^2 \leq \varepsilon_{K,n}^2$.

Proof First, it follows from Theorem 35 that $0 < \lambda_j \leq 1/2 < 1$ for all $j \geq 1$, and $\lambda_j^{(\text{attn})} = \lambda_j^3 < \lambda_j$ for all $j \geq 1$. As a result, we have

$$R_{K^{(\text{attn})}}(\varepsilon) \leq R_K(\varepsilon), \quad \forall \varepsilon \geq 0. \quad (37)$$

Setting $\varepsilon = \varepsilon_n$ in (37), we have $\varepsilon_n^2 = \sigma_0 R_{K^{(\text{attn})}}(\varepsilon_n) \leq \sigma_0 R_K(\varepsilon_n)$. Since $\sigma_0 R_K(\varepsilon)$ is a sub-root function of ε^2 with the unique fixed point of $\varepsilon_{K,n}^2$, it then follows from [5, Lemma 3.2] that $\varepsilon_n^2 \leq \varepsilon_{K,n}^2$. ■

E.4. Key Technical Results

We present our key technical results regarding optimization and generalization of the two-layer NN (1) trained by GD in this section. Lemma 14 is our main result about the optimization of the network (1), which states that with high probability over $\mathbf{W}(0)$ and $\mathbf{w}$, the weights of the network $(\mathbf{W}(t), \mathbf{a}(t))$ obtained right after the t -th step of GD using Algorithm 1 belongs to $\mathcal{W}(\mathbf{S}, \mathbf{W}(0), T)$. Furthermore, every weighing vector $\mathbf{w}_r$ and a_r have bounded distances to their corresponding initialized values, $\vec{\mathbf{w}}_r(0)$ and 0. The proof of Lemma 14 is based on Lemma 20, Lemma 21, Lemma 22, and Lemma 23 deferred to Section E.6 of this appendix.

Lemma 14 Suppose $\delta \in (0, 1)$, $N \gtrsim d \log N$, $m \gtrsim (d \log m)^3$,

$$m \gtrsim \max \{T^2(d \log m)^3/\tau^2, T^6(d^4 + \log^4 m)/\tau^2, n\}, \quad (38)$$

$$N \gtrsim T^2 \log(n/\delta)/\tau^2, \quad (39)$$

the neural network $f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ trained by GD using Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$, the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Then with probability at least $1 - 1/n - \delta - \exp(-\Theta(n))$ over the random initialization $\mathbf{W}(0)$, the random noise $\mathbf{w}$, and the random sample $\mathbf{Q}$, $(\mathbf{W}(t), \mathbf{a}(t)) \in \mathcal{W}(\mathbf{S}, \mathbf{W}(0), T)$ for every $t \in [T]$. Moreover, for every $t \in [0, T]$, $\mathbf{u}(t) = \mathbf{v}(t) + \mathbf{e}(t)$ where $\mathbf{u}(t) = \hat{\mathbf{y}}(t) - \mathbf{y}$, $\mathbf{v}(t) \in \mathcal{V}_t$, $\mathbf{e}(t) \in \mathcal{E}_{t,\tau}$, $\|\mathbf{u}(t)\|_2 \leq c_u \sqrt{n}$, and $\|\vec{\mathbf{w}}_r(t) - \vec{\mathbf{w}}_r(0)\|_2 \leq R$, $|a_r(t) - a_r(0)| \leq 2R$.

The following theorem, Theorem 15, states that with high probability over $\mathbf{w}$, $\mathcal{F}_{\text{NN}}(\mathbf{S}, \mathbf{W}(0), T) \subseteq \mathcal{F}(B_h, w)$, with the early stopping mechanism such that $T \leq \hat{T}$.

Theorem 15 Suppose $\delta \in (0, 1)$, $w \in (0, 1)$,

$$m \gtrsim \max \{T^4(d \log m)^3, T^8(d^4 + \log^4 m), T^2(d \log m)^3/w^2, T^4(d^4 + \log^4 m)/w^2, n\}, \quad (40)$$

$$N \gtrsim \max \{T^4 \log(n/\delta), T^2 \log(n/\delta)/w^2\}, \quad (41)$$

and the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ is trained by GD using Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$, the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Then for every $t \in [T]$ with $T \leq \hat{T}$, with probability at least $1 - 1/n - \delta - \exp(-\Theta(n)) - \exp(-\Theta(n\varepsilon_n^2))$ over the random initialization $\mathbf{W}(0)$, the random noise $\mathbf{w}$, and the random sample $\mathbf{Q}$, $f_t \in \mathcal{F}_{\text{NN}}(\mathbf{S}, \mathbf{W}(0), T) \subseteq \mathcal{F}(B_h, w)$, and f_t has the following decomposition on $\mathcal{X}$:

$$f_t = h_t + e_t, \quad (42)$$

where $h_t \in \mathcal{H}_K(B_h)$ with B_h defined in (29), $e_t \in L^\infty$ with $\|e_t\|_\infty \leq w$.

Lemma 16 below gives a sharp upper bound for the Rademacher complexity of a localized subset of the function class $\mathcal{F}(B, w)$. Based on Lemma 16, Theorem 15, and using the local Rademacher complexity based analysis [5], Theorem 17 presents a sharp upper bound for the nonparametric regression risk, $\mathbb{E}_P [(f_t - f^*)^2]$, where f_t is the function represented by the two-layer NN with channel attention (1) right after the t -th step of GD using Algorithm 1.

Lemma 16 *For every $B, w > 0$ every $r > 0$,*

$$\mathfrak{R}(\{f \in \mathcal{F}(B, w) : \mathbb{E}_P [f^2] \leq r\}) \leq \varphi_{B,w}(r), \quad (43)$$

where

$$\varphi_{B,w}(r) := \min_{Q: Q \geq 0} \left((\sqrt{r} + w) \sqrt{\frac{Q}{n}} + B \left(\frac{\sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})}}{n} \right)^{1/2} \right) + w. \quad (44)$$

We then have the following theorem giving the sharp bound for the regression risk of f_t right after every step t of GD.

Theorem 17 *Suppose $w \in (0, 1)$ and m, N satisfy (40) and (41), respectively. Suppose the neural network $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$ is trained by GD in Algorithm 1 with the learning rate $\eta = \Theta(1) \in (0, 1)$ on the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$, and $T \leq \hat{T}$. Then for every $t \in [T]$ and every $\delta \in (0, 1)$, with probability at least $1 - 1/n - \delta - \exp(-\Theta(n)) - \exp(-\Theta(n\hat{\varepsilon}_n^2)) - \exp(-\Theta(n\varepsilon_n^2))$ over the random initialization $\mathbf{W}(0)$, the random noise $\mathbf{w}$, the random training features $\mathbf{S}$, and the random sample $\mathbf{Q}$,*

$$\mathbb{E}_P [(f_t - f^*)^2] - 2\mathbb{E}_{P_n} [(f_t - f^*)^2] \lesssim \varepsilon_n^2 + w. \quad (45)$$

Theorem 18 below shows that the empirical loss $\mathbb{E}_{P_n} [(f_t - f^*)^2]$ is bounded by $\Theta(1/(\eta t))$ with high probability over $\mathbf{w}$. Such upper bound for the empirical loss by Theorem 18 will be plugged in the risk bound in Theorem 17 to prove Theorem 1 and Theorem 2.

Theorem 18 *Suppose the neural network trained after the t -th step of GD, $f_t = f(\mathbf{W}(t), \mathbf{a}(t), \cdot)$, satisfies $\mathbf{u}(t) = f_t(\mathbf{S}) - \mathbf{y} = \mathbf{v}(t) + \mathbf{e}(t)$ with $\mathbf{v}(t) \in \mathcal{V}_t$ and $\mathbf{e}(t) \in \mathcal{E}_{t,\tau}$, and $t \in [T]$ with $T \leq \hat{T}$. If*

$$\tau \lesssim \frac{1}{\eta T}, \quad (46)$$

Then for every $t \in [T]$, with probability at least $1 - \exp(-\Theta(n\hat{\varepsilon}_n^2))$ over the random noise $\mathbf{w}$, we have

$$\mathbb{E}_{P_n} [(f_t - f^*)^2] \leq \frac{3}{\eta t} \left(\frac{\mu_0^2}{2e} + \frac{1}{\eta T} + 2 \right). \quad (47)$$

E.5. Proofs for Results in Section E.2 and Section E.4

We have the following theorem, Theorem 19, regarding the uniform convergence to the PD kernel K defined in (2) on the unit sphere $\mathcal{X}$. The proof of Theorem 19 is deferred to Section F.2 of this appendix.

Theorem 19 *Let $\mathbf{W}(0) = \left\{ \vec{\mathbf{w}}_r(0) \right\}_{r=1}^m$, where each $\vec{\mathbf{w}}_r(0) \sim \mathcal{N}(\mathbf{0}, \kappa^2 \mathbf{I}_d)$ for $r \in [m]$. Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $\mathbf{W}(0)$,*

$$\sup_{\mathbf{u} \in \mathcal{X}, \mathbf{v} \in \mathcal{X}} \left| K(\mathbf{u}, \mathbf{v}) - \hat{h}(\mathbf{W}(0), \mathbf{u}, \mathbf{v}) \right| \leq C_1(m, d, \delta), \quad (48)$$

where

$$C_1(m, d, \delta) := \frac{12M_{\frac{\delta}{2(1+2m)^{2d}}}}{m} \left(2d + 3 \log \frac{6m(1+2m)^{2d}}{\delta} \right) + M_{\frac{\delta}{2(1+2m)^{2d}}}^2 \left(\sqrt{\frac{2 \log \frac{4(1+2m)^{2d}}{\delta}}{m}} + \frac{16 \log \frac{4(1+2m)^{2d}}{\delta}}{3m} \right), \quad (49)$$

$$M_{\delta'} := \kappa \sqrt{2 \log(2m)} + \kappa \sqrt{2 \log(3/\delta')} + \frac{\sqrt{2d + 3 \log(3m/\delta')}}{m}, \quad \forall \delta' > 0.$$

In addition, when $m \gtrsim n$, $m/\log m \geq d$, and $\delta \asymp 1/n$, we have $M_{\frac{\delta}{2(1+2m)^{2d}}} \lesssim \sqrt{d \log m}$ and $C_1(m, d, \delta) \lesssim d \log m \sqrt{\frac{d \log m}{m}} + \frac{(d \log m)^{3/2}}{m} \lesssim d \log m \sqrt{\frac{d \log m}{m}}$.

Proof [Proof of Lemma 14] First, $\mathbf{E}_{m, \eta, \delta}$ is defined by (81) of Lemma 21, and we have

$$\mathbf{E}_{m, \eta, \delta} \lesssim \frac{T^2 \sqrt{n} (d^2 + \log^2 m) + \sqrt{n} (d \log m)^{3/2}}{\sqrt{m}} + \sqrt{\frac{n \log(n/\delta)}{N}}.$$

When $m \gtrsim \max \{T^2(d \log m)^3/\tau^2, T^6(d^4 + \log^4 m)/\tau^2, n\}$, $N \gtrsim T^2 \log(n/\delta)/\tau^2$ with proper constants, it can be verified that $\mathbf{E}_{m, \eta, \delta} \leq \tau \sqrt{n}/T$. We then use mathematical induction to prove this theorem. We will first prove that $\mathbf{u}(t) = \mathbf{v}(t) + \mathbf{e}(t)$ where $\mathbf{v}(t) \in \mathcal{V}_t$, $\mathbf{e}(t) \in \mathcal{E}_{t, \tau}$, and $\|\mathbf{u}(t)\|_2 \leq c_{\mathbf{u}} \sqrt{n}$ for all $t \in [0, T]$.

When $t = 0$, we have

$$\mathbf{u}(0) = -\mathbf{y} = \mathbf{v}(0) + \mathbf{e}(0), \quad (50)$$

where $\mathbf{v}(0) := -f^*(\mathbf{S}) = -\left(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}\right)^0 f^*(\mathbf{S})$, $\mathbf{e}(0) = -\mathbf{w} = \vec{\mathbf{e}}_1(0) + \vec{\mathbf{e}}_2(0)$ with $\vec{\mathbf{e}}_1(0) = -\left(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}\right)^0 \mathbf{w}$ and $\vec{\mathbf{e}}_2(0) = \mathbf{0}$. Therefore, $\mathbf{v}(0) \in \mathcal{V}_0$ and $\mathbf{e}(0) \in \mathcal{E}_{0, \tau}$. Also, it follows from the proof of Lemma 20 that $\|\mathbf{u}(0)\|_2 \leq c_{\mathbf{u}} \sqrt{n}$ with probability at least $1 - \exp(-\Theta(n))$ over the random noise $\mathbf{w}$.

Suppose that for all $t_1 \in [0, t]$ with $t \in [0, T-1]$, $\mathbf{u}(t_1) = \mathbf{v}(t_1) + \mathbf{e}(t_1)$ where $\mathbf{v}(t_1) \in \mathcal{V}_{t_1}$, and $\mathbf{e}(t_1) = \vec{\mathbf{e}}_1(t_1) + \vec{\mathbf{e}}_2(t_1)$ with $\mathbf{v}(t_1) \in \mathcal{V}_{t_1}$ and $\mathbf{e}(t_1) \in \mathcal{E}_{t_1, \tau}$, and $\|\mathbf{u}(t_1)\|_2 \leq c_u \sqrt{n}$ for all $t_1 \in [0, t]$. Then it follows from Lemma 21 that the recursion $\mathbf{u}(t'+1) = (\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}) \mathbf{u}(t') + \mathbf{E}(t'+1)$ holds for all $t' \in [0, t]$. As a result, we have

$$\begin{aligned} \mathbf{u}(t+1) &= (\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}) \mathbf{u}(t) + \mathbf{E}(t+1) \\ &= -(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1} f^*(\mathbf{S}) - (\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1} \mathbf{w} + \sum_{t'=1}^{t+1} (\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1-t'} \mathbf{E}(t') \\ &= \mathbf{v}(t+1) + \mathbf{e}(t+1), \end{aligned} \quad (51)$$

where $\mathbf{v}(t+1)$ and $\mathbf{e}(t+1)$ are defined as

$$\mathbf{v}(t+1) := -(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1} f^*(\mathbf{S}) \in \mathcal{V}_{t+1}, \quad (52)$$

$$\mathbf{e}(t+1) := \underbrace{-(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1} \mathbf{w}}_{\vec{\mathbf{e}}_1(t+1)} + \underbrace{\sum_{t'=1}^{t+1} (\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^{t+1-t'} \mathbf{E}(t')}_{\vec{\mathbf{e}}_2(t+1)}. \quad (53)$$

We now prove the upper bound for $\vec{\mathbf{e}}_2(t+1)$. With $\eta \in (0, 1)$, we have $\|\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}\|_2 \in (0, 1)$. It follows that

$$\|\vec{\mathbf{e}}_2(t+1)\|_2 \leq \sum_{t'=1}^{t+1} \|\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})}\|_2^{t+1-t'} \|\mathbf{E}(t')\|_2 \leq \tau \sqrt{n}, \quad (54)$$

where the last inequality follows from the fact that $\|\mathbf{E}(t)\|_2 \leq \mathbf{E}_{m, \eta, \delta} \leq \tau \sqrt{n}/T$ for all $t \in [T]$. It follows that $\mathbf{e}(t+1) \in \mathcal{E}_{t+1, \tau}$. Also, it follows from Lemma 20 that

$$\begin{aligned} \|\mathbf{u}(t+1)\|_2 &\leq \|\mathbf{v}(t+1)\|_2 + \|\vec{\mathbf{e}}_1(t+1)\|_2 + \|\vec{\mathbf{e}}_2(t+1)\|_2 \leq \left(\frac{\mu_0}{\sqrt{2e\eta}} + \sigma_0 + \tau + 1 \right) \sqrt{n} \\ &\leq c_u \sqrt{n}. \end{aligned}$$

The above inequality completes the induction step, which also completes the proof. It is noted that $\|\vec{\mathbf{w}}_r(t) - \vec{\mathbf{w}}_r(0)\|_2 \leq R$ and $|a_r(t) - a_r(0)| \leq 2R$ hold for all $t \in [T]$ by Lemma 23. ■

Proof [Proof of Theorem 15] In this proof, we abbreviate f_t as f and $\mathbf{W}(t)$ as $\mathbf{W}$. It follows from Lemma 14 and its proof that conditioned on an event Ω with probability at least $1 - 1/n - \delta - \exp(-\Theta(n))$, $f \in \mathcal{F}_{\text{NN}}(\mathbf{S}, \mathbf{W}(0), T)$ with $\mathbf{W}(0) \in \mathcal{W}_0$. Moreover, $f = f(\mathbf{W}, \mathbf{a}, \cdot)$ with $(\mathbf{W}, \mathbf{a}) = \left(\left\{ \vec{\mathbf{w}}_r \right\}_{r=1}^m, \mathbf{a} \right) \in \mathcal{W}(\mathbf{S}, \mathbf{W}(0), T)$, and $\text{vec}(\mathbf{W}) = \text{vec}(\mathbf{W}_\mathbf{S}) = \text{vec}(\mathbf{W}(0)) - \sum_{t'=0}^{t-1} \eta/n \cdot \mathbf{Z}_\mathbf{S}(t') \mathbf{u}(t')$ for some $t \in [T]$, where $\mathbf{u}(t') \in \mathbb{R}^n$, $\mathbf{u}(t') = \mathbf{v}(t') + \mathbf{e}(t')$ with $\mathbf{v}(t') \in \mathcal{V}_{t'}$

and $\mathbf{e}(t') \in \mathcal{E}_{t', \tau}$ for all $t' \in [0, t-1]$. It also follows from Lemma 14 that conditioned on Ω , $\|\vec{\mathbf{w}}_r(t) - \vec{\mathbf{w}}_r(0)\|_2 \leq R$, $|a_r(t) - a_r(0)| \leq 2R$ hold for all $t \in [T]$.

$\vec{\mathbf{w}}_r, \mathbf{a}$ are expressed as

$$\vec{\mathbf{w}}_r = \vec{\mathbf{w}}_{\mathbf{S}, r}(t) = \vec{\mathbf{w}}_r(0) - \sum_{t'=0}^{t-1} \frac{\eta}{n} [\mathbf{Z}_{\mathbf{S}}(t')]_{[(r-1)d+1:rd]} \mathbf{u}(t'), \quad (55)$$

$$\mathbf{a} = \mathbf{a}(t) = \sum_{t'=0}^{t-1} -\frac{\eta}{n\sqrt{m}} \mathbf{A} \boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) \mathbf{u}(t'), \quad (56)$$

where the notation $\vec{\mathbf{w}}_{\mathbf{S}, r}$ emphasizes that $\vec{\mathbf{w}}_r$ depends on the training features $\mathbf{S}$. Let $\mathbf{a} = \mathbf{a}(t) = [a_1(t), \dots, a_m(t)]$, we approximate $f(\mathbf{W}, \mathbf{a}, \mathbf{x})$ by

$$g(\mathbf{x}) := \frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m a_r(t) \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r}.$$

We have

$$\begin{aligned} & |f(\mathbf{W}, \mathbf{a}, \mathbf{x}) - g(\mathbf{x})| \\ &= \frac{1}{\sqrt{m}} \left| \sum_{r'=1}^m \sum_{r=1}^m a_r(t) \sigma \left(\vec{\mathbf{w}}_{r'}(t)^\top \mathbf{x} \right) \mathbf{A}_{r'r} - \sum_{r'=1}^m \sum_{r=1}^m a_r(t) \mathbb{I}_{\left\{ \vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \geq 0 \right\}} \vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \mathbf{A}_{r'r} \right| \\ &\leq \frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m |a_r| \left\| \vec{\mathbf{w}}_{r'}(t) - \vec{\mathbf{w}}_{r'}(0) \right\|_2 \mathbf{A}_{r'r} \\ &\leq \frac{1}{\sqrt{m}} \cdot 2R\sqrt{m} \cdot \|\mathbf{A}\|_2 \cdot R\sqrt{m} \leq 2R^2\sqrt{m}, \forall \mathbf{x} \in \mathcal{X}, \end{aligned} \quad (57)$$

where last inequality follows from $\|\mathbf{A}\|_2 \leq 1$ due to (84) in the proof of Lemma 21 with $\mathbf{W}(0) \in \mathcal{W}_0$ and $m \gtrsim (d \log m)^3$. Using (56), $g(\mathbf{x})$ is expressed as

$$g(\mathbf{x}) = - \underbrace{\sum_{t'=0}^{t-1} \frac{\eta}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) \mathbf{u}(t')}_{:=G_{t'}(\mathbf{x})}. \quad (58)$$

For each $G_{t'}$ on the RHS of (58), we have

$$\begin{aligned} G_{t'}(\mathbf{x}) &= \underbrace{\frac{\eta}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) \mathbf{u}(t')}_{:=D(\mathbf{x}, t')} \\ &+ \underbrace{\frac{\eta}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \left(\boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) \right) \mathbf{u}(t')}_{:=E_1(\mathbf{x}, t')} \end{aligned}$$

$$= \frac{\eta}{n} \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') + E_1(\mathbf{x}, t') + E_2(\mathbf{x}, t'), \quad (59)$$

where $E_2(\mathbf{x}, t') := D(\mathbf{x}, t') - \eta/n \cdot \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t')$. We now analyze each term on the RHS of (59). It follows from Lemma 27 that

$$\|E_2(\mathbf{x}, t')\|_\infty \lesssim C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}. \quad (60)$$

Let $h(\cdot, t') : \mathcal{X} \rightarrow \mathbb{R}$ be defined by $h(\mathbf{x}, t') := \frac{\eta}{n} \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t')$, then $h(\cdot, t') \in \mathcal{H}_{K^{(\text{attn})}}$ for each $t' \in [0, t-1]$. We further define

$$h_t(\cdot) := - \sum_{t'=0}^{t-1} h(\cdot, t') \in \mathcal{H}_{K^{(\text{attn})}}. \quad (61)$$

We note that $E_1(\mathbf{x}, t') = \eta/(nm) \cdot \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{x}) \mathbf{A}^2 (\boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S})) \mathbf{u}(t')$, and it follows that

$$\begin{aligned} \|E_1(\mathbf{x}, t')\|_\infty &\leq \frac{\eta}{nm} \left\| \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{x}) \right\|_2 \left\| \mathbf{A} \right\|_2^2 \left\| \boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) \right\|_2 \left\| \mathbf{u}(t') \right\|_2 \\ &\stackrel{\textcircled{1}}{\leq} \frac{\eta}{nm} \cdot \sqrt{m} \cdot \sqrt{nm} R \cdot c_{\mathbf{u}} \sqrt{n} = \eta c_{\mathbf{u}} R. \end{aligned} \quad (62)$$

Since $\mathbf{W}(0) \in \mathcal{W}_0$ and $m \gtrsim (d \log m)^3$, $\left\| \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{x}) \right\|_2 \leq \sqrt{m}$. It follows from (84) in the proof of Lemma 21 that $\left\| \vec{\mathbf{w}}_r(t) - \vec{\mathbf{w}}_r(0) \right\|_2 \leq R$ for all $t \in [T]$, we have $\left\| \boldsymbol{\sigma}(\mathbf{W}(t'), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) \right\|_2 \leq \sqrt{nm} R$. As a result, $\textcircled{1}$ holds.

It follows from (59), (60), and (62), for any $t' \in [0, t-1]$,

$$\|G_{t'}(\mathbf{x}) - h(\mathbf{x}, t')\|_\infty \leq \|E_1\|_\infty + \|E_2\|_\infty \lesssim \eta c_{\mathbf{u}} R + C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}. \quad (63)$$

Define $e_t(\cdot) = f(\mathbf{W}, \mathbf{a}, \cdot) - h_t(\cdot)$. It then follows from (57), (58), and (63) that

$$\begin{aligned} \|e_t\|_\infty &\leq \|f(\mathbf{W}, \mathbf{a}, \mathbf{x}) - g(\mathbf{x})\|_\infty + \|g(\mathbf{x}) - h_t(\mathbf{x})\|_\infty \\ &\leq \|f(\mathbf{W}, \mathbf{a}, \mathbf{x}) - g(\mathbf{x})\|_\infty + \sum_{t'=0}^{t-1} \|G_{t'}(\mathbf{x}) - h(\mathbf{x}, t')\|_\infty \\ &\stackrel{\textcircled{4}}{\lesssim} 2R^2 \sqrt{m} + T \left(\eta c_{\mathbf{u}} R + C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}} \right) := \Delta_{m,n,\delta,T}. \end{aligned} \quad (64)$$

We now give an estimate for $\Delta_{m,n,\delta,T}$. With $\mathbf{W}(0) \in \mathcal{W}_0$, $C_1(m, d, 1/n) \lesssim d \log m \sqrt{\frac{d \log m}{m}}$. As a result, plugging $R = \eta c_{\mathbf{u}} (2d + 3 \log(2mn)) T / \sqrt{m}$ on the RHS of (64), we have

$$\Delta_{m,n,\delta,T} \lesssim \frac{T^2(d^2 + \log^2 m) + (d \log m)^{3/2} T}{\sqrt{m}} + T \sqrt{\frac{\log(n/\delta)}{N}}.$$

By direct calculations, for any $w > 0$, when

$$m \gtrsim \max \{T^2(d \log m)^3/w^2, T^4(d^4 + \log^4 m)/w^2, n\}, \quad (65)$$

$$N \gtrsim T^2 \log(n/\delta)/w^2, \quad (66)$$

we have $\Delta_{m,n,\delta,T} \lesssim w$.

It follows from Lemma 26 that with probability at least $1 - \exp(-\Theta(n\hat{\varepsilon}_n^2))$ over the random noise $\mathbf{w}$, $\|h_t\|_{\mathcal{H}_{K(\text{attn})}} \leq B_h$, where B_h is defined in (29), and τ is required to satisfy $\tau \leq 1/(\eta T)$. Lemma 14 requires that

$$\begin{aligned} m &\gtrsim \max \{T^2(d \log m)^3/\tau^2, T^6(d^4 + \log^4 m)/\tau^2, n\}, \\ N &\gtrsim T^2 \log(n/\delta)/\tau^2. \end{aligned}$$

As a result, we have $m \gtrsim \max \{T^4(d \log m)^3, T^8(d^4 + \log^4 m), n\}$ and $N \gtrsim T^4 \log(n/\delta)$, which lead to the conditions on m, N , (40) and (41), when combined with (65)-(66). $\blacksquare$

Proof [Proof of Theorem 17] We first remark that the conditions on m, N are required by Theorem 15. It also follows from Theorem 15 that conditioned on an event Ω with probability at least $1 - 1/n - \delta - \exp(-\Theta(n)) - \exp(-\Theta(n\hat{\varepsilon}_n^2))$ over $\mathbf{w}$ and $\mathbf{Q}$, we have $(\mathbf{W}(t), \mathbf{a}(t)) \in \mathcal{W}(\mathbf{S}, \mathbf{Q}, \mathbf{W}(0), T)$, and

$$f(\mathbf{W}(t), \mathbf{a}(t), \cdot) = f_t = h + e \in \mathcal{F}(B_h, w)$$

with $h \in \mathcal{H}_{K(\text{attn})}(B_h)$ and $\|e\|_\infty \leq w$. The proof then follows a similar strategy to that of [58, Theorem VI.5] and [57, Theorem C.10]. We then derive the sharp upper bound for $\mathbb{E}_P[(f_t - f^*)^2]$ by applying Theorem 10 to the function class

$$\mathcal{F} = \left\{ F = (f - f^*)^2 : f \in \mathcal{F}(B_h, w) \right\}.$$

With $B_0 = B_h + 1 + \mu_0 \geq B_h + w + \mu_0$, we have $\|F\|_\infty \leq B_0^2$ with $F \in \mathcal{F}$, so that $\mathbb{E}_P[F^2] \leq B_0^2 \mathbb{E}_P[F]$. Let $T(F) = B_0^2 \mathbb{E}_P[F]$ for $F \in \mathcal{F}$. Then $\text{Var}[F] \leq \mathbb{E}_P[F^2] \leq T(F) = B_0^2 \mathbb{E}_P[F]$. We have

$$\begin{aligned} \mathfrak{R}(\{F \in \mathcal{F} : T(F) \leq r\}) &= \mathfrak{R}\left(\left\{(f - f^*)^2 : f \in \mathcal{F}(B_h, w), \mathbb{E}_P[(f - f^*)^2] \leq \frac{r}{B_0^2}\right\}\right) \\ &\stackrel{\textcircled{1}}{\leq} 2B_0 \mathfrak{R}\left(\left\{f - f^* : f \in \mathcal{F}(B_h, w), \mathbb{E}_P[(f - f^*)^2] \leq \frac{r}{B_0^2}\right\}\right) \\ &\stackrel{\textcircled{2}}{\leq} 4B_0 \mathfrak{R}\left(\left\{f \in \mathcal{F}(B_h, w) : \mathbb{E}_P[f^2] \leq \frac{r}{4B_0^2}\right\}\right), \end{aligned} \quad (67)$$

where $\textcircled{1}$ is due to the contraction property of Rademacher complexity in Theorem 8. Since $f^* \in \mathcal{F}(B_h, w)$, $f \in \mathcal{F}(B_h, w)$, we have $\frac{f - f^*}{2} \in \mathcal{F}(B_h, w)$ due to the fact that $\mathcal{F}(B_h, w)$ is symmetric and convex, and it follows that $\textcircled{2}$ holds.

It follows from (67) and Lemma 16 that

$$B_0^2 \mathfrak{R}(\{F \in \mathcal{F} : T(F) \leq r\}) \leq 4B_0^3 \mathfrak{R}\left(\left\{f : f \in \mathcal{F}(B_h, w), \mathbb{E}_P[f^2] \leq \frac{r}{4B_0^2}\right\}\right)$$

$$\leq 4B_0^3 \varphi_{B_h, w} \left(\frac{r}{4B_0^2} \right) := \psi(r). \quad (68)$$

It follows from the definition of $\varphi_{B_h, w}$ in (44) and the Cauchy-Schwarz inequality that

$$\begin{aligned} \psi(r) &= 4B_0^3 \min_{Q: Q \geq 0} \left(\left(\frac{\sqrt{r}}{2B_0} + w \right) \sqrt{\frac{Q}{n}} + B_h \left(\frac{\sum_{q=Q+1}^{\infty} \lambda_q}{n} \right)^{1/2} \right) + 4B_0^3 w \\ &\leq 4B_0^3 B_h \min_{Q: Q \geq 0} \left(\sqrt{\frac{Qr}{n}} + \left(\frac{\sum_{q=Q+1}^{\infty} \lambda_q}{n} \right)^{1/2} \right) + 8B_0^3 w \\ &\leq \frac{4\sqrt{2}B_0^3 B_h}{\sigma_0} \cdot \sigma_0 R_{K(\text{attn})}(\sqrt{r}) + 8B_0^3 w := \psi_1(r), \end{aligned}$$

where the last inequality follows from the definition of the kernel complexity. It can be verified that $\psi_1(r)$ is a sub-root function. Let the fixed point of $\psi_1(r)$ be r_1^* . Because the fixed point of $\sigma_0 R_K(\sqrt{r})$ as a function of r is ε_n^2 , it follows from Lemma 31 that

$$r_1^* \leq \max \left\{ \frac{32B_0^6 B_h^2}{\sigma_0^2}, 1 \right\} \varepsilon_n^2 + 16B_0^3 w. \quad (69)$$

It then follows from Theorem 10 with $K_0 = 2$ that with probability at least $1 - \exp(-x)$,

$$\mathbb{E}_P [(f_t - f^*)^2] - 2\mathbb{E}_{P_n} [(f_t - f^*)^2] \lesssim r_1^* + \frac{x}{n}.$$

Letting $x = n\varepsilon_n^2$, then plugging the upper bound (69) for r_1^* in the above inequality leads to

$$\mathbb{E}_P [(f_t - f^*)^2] - 2\mathbb{E}_{P_n} [(f_t - f^*)^2] \lesssim \varepsilon_n^2 + w, \quad (70)$$

which proves (45). ■

Proof [Proof of Theorem 18] We have

$$f_t(\mathbf{S}) = f^*(\mathbf{S}) + \mathbf{w} + \mathbf{v}(t) + \mathbf{e}(t), \quad (71)$$

where $\mathbf{v}(t) \in \mathcal{V}_t$, $\mathbf{e}(t) \in \mathcal{E}_{t, \tau}$, $\vec{\mathbf{e}}(t) = \vec{\mathbf{e}}_1(t) + \vec{\mathbf{e}}_2(t)$ with $\vec{\mathbf{e}}_1(t) = -(\mathbf{I}_n - \eta \mathbf{K}_n^{(\text{attn})})^t \mathbf{w}$ and $\|\vec{\mathbf{e}}_2(t)\|_2 \lesssim \sqrt{n}\tau$. It follows from (71) that

$$\begin{aligned} \mathbb{E}_{P_n} [(f_t - f^*)^2] &= \frac{1}{n} \|f_t(\mathbf{S}) - f^*(\mathbf{S})\|_2^2 = \frac{1}{n} \|\mathbf{v}(t) + \mathbf{w} + \mathbf{e}(t)\|_2^2 \\ &= \frac{1}{n} \left\| -(\mathbf{I}_n - \eta \mathbf{K}_n^{(\text{attn})})^t f^*(\mathbf{S}) + \left(\mathbf{I}_n - (\mathbf{I}_n - \eta \mathbf{K}_n^{(\text{attn})})^t \right) \mathbf{w} + \vec{\mathbf{e}}_2(t) \right\|_2^2 \end{aligned}$$

$$\begin{aligned}
 &\stackrel{\textcircled{1}}{\leq} \frac{3}{n} \sum_{i=1}^n \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^{2t} [\mathbf{U}^\top f^*(\mathbf{S})]_i^2 + \frac{3}{n} \sum_{i=1}^n \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^2 [\mathbf{U}^\top \mathbf{w}]_i^2 + \frac{3}{n} \|\vec{\mathbf{e}}_2(t)\|_2^2 \\
 &\stackrel{\textcircled{2}}{\leq} \frac{3\mu_0^2}{2\eta t} + \frac{3}{n} \sum_{i=1}^n \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^2 [\mathbf{U}^\top \mathbf{w}]_i^2 + 3\tau^2 \\
 &\leq \frac{3}{\eta t} \left(\frac{\mu_0^2}{2e} + \frac{1}{\eta T}\right) + 3 \cdot \underbrace{\frac{1}{n} \sum_{i=1}^n \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^2 [\mathbf{U}^\top \mathbf{w}]_i^2}_{:=E_\varepsilon}
 \end{aligned} \tag{72}$$

Here ① follows from the Cauchy-Schwarz inequality, ② follows from (79) in the proof of Lemma 20, and ③ follows from the conditions on N, τ in (46).

We then derive the upper bound for E_ε on the RHS of (72). We define the diagonal matrix $\mathbf{R} \in \mathbb{R}^{n \times n}$ with $\mathbf{R}_{ii} = \left(1 - (1 - \eta \lambda_i)^t\right)^2$. Then we have $E_\varepsilon = 1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top)$. It follows from [55] that

$$\begin{aligned}
 &\Pr \left[1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top) - \mathbb{E} \left[1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top)\right] \geq u\right] \\
 &\leq \exp \left(-c \min \left\{nu/\|\mathbf{R}\|_2, n^2 u^2/\|\mathbf{R}\|_F^2\right\}\right)
 \end{aligned} \tag{73}$$

holds for all $u > 0$, and c is a positive constant. With $\eta_t = \eta t$ for all $t \geq 0$, we have

$$\begin{aligned}
 \mathbb{E} \left[1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top)\right] &\leq \frac{\sigma_0^2}{n} \sum_{i=1}^n \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^2 \stackrel{\textcircled{1}}{\leq} \frac{\sigma_0^2}{n} \sum_{i=1}^n \min \left\{1, \eta_t^2 (\hat{\lambda}_i^{(\text{attn})})^2\right\} \\
 &\leq \frac{\sigma_0^2 \eta_t}{n} \sum_{i=1}^n \min \left\{\frac{1}{\eta_t}, \eta_t (\hat{\lambda}_i^{(\text{attn})})^2\right\} \stackrel{\textcircled{2}}{\leq} \frac{\sigma_0^2 \eta_t}{n} \sum_{i=1}^n \min \left\{\frac{1}{\eta_t}, \hat{\lambda}_i^{(\text{attn})}\right\} = \sigma_0^2 \eta_t \hat{R}_{K^{(\text{attn})}}^2(\sqrt{1/\eta_t}) \leq \frac{1}{\eta_t}.
 \end{aligned} \tag{74}$$

Here ① follows from the fact that $(1 - \eta \hat{\lambda}_i^{(\text{attn})})^t \geq \max \{0, 1 - t\eta \hat{\lambda}_i^{(\text{attn})}\}$, and ② follows from $\min \{a, b\} \leq \sqrt{ab}$ for any nonnegative numbers a, b . Because $t \leq T \leq \hat{T}$, we have $R_{K^{(\text{attn})}}(\sqrt{1/\eta_t}) \leq 1/(\sigma \eta_t)$, so the last inequality holds.

Moreover, we have the upper bounds for $\|\mathbf{R}\|_2$ and $\|\mathbf{R}\|_F$ as follows. First, we have

$$\|\mathbf{R}\|_2 \leq \max_{i \in [n]} \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^2 \leq \min \left\{1, \eta_t^2 (\hat{\lambda}_i^{(\text{attn})})^2\right\} \leq 1. \tag{75}$$

We also have

$$\begin{aligned}
 \frac{1}{n} \|\mathbf{R}\|_F^2 &= \frac{1}{n} \sum_{i=1}^n \left(1 - \left(1 - \eta \hat{\lambda}_i^{(\text{attn})}\right)^t\right)^4 \leq \frac{\eta_t}{n} \sum_{i=1}^n \min \left\{\frac{1}{\eta_t}, \eta_t^3 (\hat{\lambda}_i^{(\text{attn})})^4\right\} \\
 &\leq \frac{\eta_t}{n} \sum_{i=1}^n \min \left\{\hat{\lambda}_i^{(\text{attn})}, \frac{1}{\eta_t}\right\} = \eta_t \hat{R}_{K^{(\text{attn})}}^2(\sqrt{1/\eta_t}) \leq \frac{1}{\sigma_0^2 \eta_t}.
 \end{aligned} \tag{76}$$

Combining (73)-(76), we have

$$\Pr \left[1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top) - \mathbb{E} \left[1/n \cdot \text{tr}(\mathbf{URU}^\top \mathbf{ww}^\top)\right] \geq u\right] \leq \exp \left(-cn \min \{u, u^2 \sigma_0^2 \eta_t\}\right).$$

Let $u = 1/(\eta t)$ in the above inequality, we have

$$\exp(-cn \min\{u, u^2 \sigma_0^2 \eta_t\}) = \exp(-c' n / \eta_t) \leq \exp(-c' n \hat{\varepsilon}_n^2)$$

where $c' = c \min\{1, \sigma_0^2\}$, and the last inequality is due to the fact that $1/\eta_t \geq \hat{\varepsilon}_n^2$ since $t \leq T \leq \hat{T}$. It follows that with probability at least $1 - \exp(-\Theta(n \hat{\varepsilon}_n^2))$,

$$E_\varepsilon \leq u + \frac{1}{\eta_t} = \frac{2}{\eta_t}. \quad (77)$$

It then follows from (72), (73)-(77) that

$$\mathbb{E}_{P_n} [(f_t - f^*)^2] \leq \frac{3}{\eta t} \left(\frac{\mu_0^2}{2e} + \frac{1}{\eta T} + 2 \right)$$

with probability at least $1 - \exp(-c' n \hat{\varepsilon}_n^2)$. ■

E.6. Proof of the Lemmas Required for the Proofs in Section E.5

Lemma 20 *Let $t \in [0, T]$, $\mathbf{v} = -(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^t f^*(\mathbf{S})$, $\mathbf{e} = -(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})})^t \mathbf{w}$, and $\eta = \Theta(1) \in (0, 1)$. Suppose $\delta \in (0, 1/2)$, then with probability at least $1 - \exp(-\Theta(n))$ over the random noise $\mathbf{w}$,*

$$\|\mathbf{v}\|_2 + \|\mathbf{e}\|_2 \leq \left(\frac{\mu_0}{\min\{\sqrt{2e\eta}, 1\}} + \sigma_0 + 1 \right) \cdot \sqrt{n}. \quad (78)$$

Proof When $t \geq 1$, we have

$$\|\mathbf{v}\|_2^2 = \sum_{i=1}^n (1 - \eta \hat{\lambda}_i)^{2t} [\mathbf{U}^\top f^*(\mathbf{S})]_i^2 \stackrel{\textcircled{1}}{\leq} \sum_{i=1}^n \frac{1}{2e\eta \hat{\lambda}_i t} [\mathbf{U}^\top f^*(\mathbf{S})]_i^2 \stackrel{\textcircled{2}}{\leq} \frac{n\mu_0^2}{2e\eta t} \leq \frac{\mu_0^2}{2e\eta} \cdot n. \quad (79)$$

Here ① follows from Lemma 29, ② follows by Lemma 28. Moreover, it follows from the concentration inequality about quadratic forms of sub-Gaussian random variables in [55] that $\Pr \left[\|\mathbf{w}\|_2^2 - \mathbb{E} [\|\mathbf{w}\|_2^2] > n \right] \leq \exp(-\Theta(n))$, so that $\|\mathbf{e}\|_2 \leq \|\mathbf{w}\|_2 \leq \sqrt{\mathbb{E} [\|\mathbf{w}\|_2^2]} + \sqrt{n} = \sqrt{n}(\sigma_0 + 1)$ with probability at least $1 - \exp(-\Theta(n))$. As a result, (78) follows from this inequality and (79) for $t \geq 1$. When $t = 0$, $\|\mathbf{v}\|_2 \leq \mu_0 \sqrt{n}$, so that (78) still holds. ■

Lemma 21 *Let $\eta \in (0, 1)$, $0 \leq t \leq T - 1$ for $T \geq 1$, and suppose that $\|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq c_u \sqrt{n}$ holds for all $0 \leq t' \leq t$, $N \gtrsim d \log N$, $m \gtrsim \max\{(d^2 + \log^2 m)T^2, (d \log m)^3, n\}$, and the*

random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Let $\delta \in (0, 1)$, then with probability at least $1 - 1/n - \delta$ over $\mathbf{Q}$ and $\mathbf{W}(0)$,

$$\hat{\mathbf{y}}(t+1) - \mathbf{y} = \left(\mathbf{I} - \eta \mathbf{K}_n^{(\text{attn})} \right) (\hat{\mathbf{y}}(t) - \mathbf{y}) + \mathbf{E}(t+1), \quad (80)$$

where $\|\mathbf{E}(t+1)\|_2 \lesssim \mathbf{E}_{m,\eta,\delta}$, and $\mathbf{E}_{m,\eta,\delta}$ is defined by

$$\mathbf{E}_{m,\eta,\delta} := R^2 \sqrt{mn} + \eta c_{\mathbf{u}} \sqrt{n} \left(R + C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}} \right). \quad (81)$$

Proof

Because $\|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq \sqrt{n} c_{\mathbf{u}}$ holds for all $t' \in [0, t]$, it follows from Lemma 23 that for every $t' \in [0, t+1]$ and $r \in [m]$,

$$\left\| \vec{\mathbf{w}}_{\mathbf{S},r}(t') - \vec{\mathbf{w}}_r(0) \right\|_2 \leq R, \quad |a_r(t')| \leq 2R. \quad (82)$$

We have

$$\begin{aligned} \hat{\mathbf{y}}_i(t+1) - \hat{\mathbf{y}}_i(t) &= \frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m (a_r(t+1) - a_r(t)) \sigma \left(\vec{\mathbf{w}}_{r'}(t)^\top \vec{\mathbf{x}}_i \right) \mathbf{A}_{r'r} \\ &\quad + \underbrace{\frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m a_r(t+1) \left(\sigma \left(\vec{\mathbf{w}}_{r'}(t+1)^\top \vec{\mathbf{x}}_i \right) - \sigma \left(\vec{\mathbf{w}}_{r'}(t)^\top \vec{\mathbf{x}}_i \right) \right) \mathbf{A}_{r'r}}_{:= \mathbf{E}_i^{(1)}} \\ &= -\frac{\eta}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(t)^\top \vec{\mathbf{x}}_i \right) \mathbf{A}_{r'r} [\mathbf{A}]_r \sigma(\mathbf{W}(t), \mathbf{S}) (\hat{\mathbf{y}}(t) - \mathbf{y}) + \mathbf{E}_i^{(1)} \\ &= \underbrace{-\frac{\eta}{nm} \sigma^\top(\mathbf{W}(0), \vec{\mathbf{x}}_i) \mathbf{A}^2 \sigma(\mathbf{W}(t), \mathbf{S}) (\hat{\mathbf{y}}(t) - \mathbf{y})}_{:= \mathbf{D}_i^{(1)}} + \mathbf{E}_i^{(1)}, \end{aligned} \quad (83)$$

and $\mathbf{D}^{(1)}, \mathbf{E}^{(1)} \in \mathbb{R}^n$ are vectors with their i -th element being $\mathbf{D}_i^{(1)}$ and $\mathbf{E}_i^{(1)}$ defined on the RHS of (83). Now we derive the upper bound for $\mathbf{E}_i^{(1)}$. Define $\mathbf{H}_{\mathbf{Q}}(0) \in \mathbb{R}^{N \times N} = \sigma(\mathbf{W}(0), \mathbf{Q})^\top \sigma(\mathbf{W}(0), \mathbf{Q}) / (Nm)$, then it follows from Theorem 11 that $\|\mathbf{H}_{\mathbf{Q}}(0) - \mathbf{K}_N\|_2 \leq C_1(m, d, 1/n)$ with $\mathbf{W}(0) \in \mathcal{W}_0$, where $\mathbf{K}_{\mathbf{Q},N} \in \mathbb{R}^{N \times N}$ and $[\mathbf{K}_{\mathbf{Q},N}]_{ij} = K(\vec{\mathbf{q}}_i, \vec{\mathbf{q}}_j)/N$. It follows that

$$\|\mathbf{A}\|_2 = \|\mathbf{H}_{\mathbf{Q}}(0)\|_2 \leq \|\mathbf{K}_{\mathbf{Q},N}\|_2 + C_1(m, d, 1/n) \leq \frac{1}{2} + C_1(m, d, 1/n) \leq 1, \quad (84)$$

since $m \gtrsim (d \log m)^3$. For all $i \in [n]$ we have

$$\begin{aligned} |\mathbf{E}_i^{(1)}| &= \left| \frac{1}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m a_r(t+1) \left(\sigma \left(\vec{\mathbf{w}}_{r'}(t+1)^\top \vec{\mathbf{x}}_i \right) - \sigma \left(\vec{\mathbf{w}}_{r'}(t)^\top \vec{\mathbf{x}}_i \right) \right) \mathbf{A}_{r'r} \right| \\ &\leq \frac{R}{\sqrt{m}} \sum_{r'=1}^m \sum_{r=1}^m |a_r(t+1)| \mathbf{A}_{r'r} \leq 2R^2 \sqrt{m} \cdot \|\mathbf{A}\|_2 \leq 2R^2 \sqrt{m}, \end{aligned} \quad (85)$$

where the last inequality follow from (84). $\mathbf{D}_i^{(1)}$ on the RHS of (83) is expressed by

$$\begin{aligned} \mathbf{D}_i^{(1)} &= -\frac{\eta}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \vec{\mathbf{x}}_i) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(t), \mathbf{S}) (\hat{\mathbf{y}}(t) - \mathbf{y}) \\ &= -\frac{\eta}{nm} \underbrace{\boldsymbol{\sigma}^\top(\mathbf{W}(0), \vec{\mathbf{x}}_i) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) (\hat{\mathbf{y}}(t) - \mathbf{y})}_{:=\mathbf{D}_i^{(2)}} \\ &\quad + \underbrace{\frac{\eta}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \vec{\mathbf{x}}_i) \mathbf{A}^2 (\boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(t), \mathbf{S})) (\hat{\mathbf{y}}(t) - \mathbf{y})}_{:=\mathbf{E}_i^{(2)}} = \mathbf{D}_i^{(2)} + \mathbf{E}_i^{(2)}. \end{aligned} \quad (86)$$

It follows from (82) that $\|\mathbf{E}^{(2)}\|_2$ is bounded by

$$\begin{aligned} \|\mathbf{E}^{(2)}\|_2 &\leq \frac{\eta}{nm} \left\| \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \right\|_2 \|\mathbf{A}\|_2^2 \|\boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) - \boldsymbol{\sigma}(\mathbf{W}(t), \mathbf{S})\|_2 \|\hat{\mathbf{y}}(t) - \mathbf{y}\|_2 \\ &\stackrel{\textcircled{1}}{\leq} \frac{\eta}{nm} \cdot \sqrt{nm} \cdot \sqrt{nm} R \cdot c_{\mathbf{u}} \sqrt{n} = \eta c_{\mathbf{u}} R \sqrt{n}. \end{aligned} \quad (87)$$

With $\mathbf{W}(0) \in \mathcal{W}_0$ and $m \gtrsim (d \log m)^3$, it follows from (102) in the proof of Lemma 24 that $\|\boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S})\|_2 \leq \sqrt{nm}$. It also follows from (84) that $\|\mathbf{A}\|_2 \leq 1$, so that $\textcircled{1}$ holds.

$\mathbf{D}_i^{(2)}$ on the RHS of (90) is expressed by

$$\begin{aligned} \mathbf{D}^{(2)} &= -\frac{\eta}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) (\hat{\mathbf{y}}(t) - \mathbf{y}) \\ &= \underbrace{-\eta \mathbf{K}_n^{(\text{attn})} (\hat{\mathbf{y}}(t) - \mathbf{y})}_{:=\mathbf{D}^{(3)}} + \underbrace{\eta \left(\frac{1}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) - \mathbf{K}_n^{(\text{attn})} \right) (\hat{\mathbf{y}}(t) - \mathbf{y})}_{:=\mathbf{E}^{(3)}}. \end{aligned} \quad (88)$$

It follows from Lemma 22 that $\|\mathbf{E}^{(3)}\|_2$ is bounded by

$$\|\mathbf{E}^{(3)}\|_2 \lesssim \eta c_{\mathbf{u}} \sqrt{n} \left(C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}} \right). \quad (89)$$

It follows from (86) and (88) that

$$\mathbf{D}_i^{(1)} = \mathbf{D}_i^{(3)} + \mathbf{E}_i^{(2)} + \mathbf{E}_i^{(3)}. \quad (90)$$

It then follows from (83) and (90) that

$$\hat{\mathbf{y}}(t+1) - \hat{\mathbf{y}}(t) = \mathbf{D}^{(3)} + \underbrace{\mathbf{E}^{(1)} + \mathbf{E}^{(2)} + \mathbf{E}^{(3)}}_{:=\mathbf{E}_i} = -\eta \mathbf{K}_n^{(\text{attn})} (\hat{\mathbf{y}}(t) - \mathbf{y}) + \mathbf{E}, \quad (91)$$

where $\mathbf{E} \in \mathbb{R}^n$ with its i -th element being $\mathbf{E}_i$, and $\mathbf{E} = \mathbf{E}^{(1)} + \mathbf{E}^{(2)} + \mathbf{E}^{(3)}$. It then follows from (85), (87), and (89) that

$$\|\mathbf{E}\|_2 \lesssim R^2 \sqrt{mn} + \eta c_{\mathbf{u}} \sqrt{n} \left(R + C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}} \right), \quad (92)$$

which together with (91) completes the proof. ■

Lemma 22 Suppose the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$, $m \gtrsim (d \log m)^3$. Then with probability at least $1 - \delta$ over $\mathbf{Q}$,

$$\left\| \frac{1}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) - \mathbf{K}_n^{(\text{attn})} \right\|_2 \lesssim C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}. \quad (93)$$

Proof We have

$$\frac{1}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) = \frac{1}{nN^2} \hat{K}_{\mathbf{S}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{S}, \mathbf{Q}}^\top, \quad (94)$$

where $\hat{K}_{\mathbf{S}, \mathbf{Q}} := \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{Q})/m$, $\hat{K}_{\mathbf{Q}, \mathbf{Q}} := \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{Q})^\top \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{Q})/m$. We define $\mathbf{K}_{\mathbf{S}, \mathbf{Q}} \in \mathbb{R}^{n \times N}$ with $[\mathbf{K}_{\mathbf{S}, \mathbf{Q}}]_{ij} = K(\vec{\mathbf{x}}_i, \vec{\mathbf{q}}_j)$ for all $i \in [n]$, $j \in [N]$, and $\mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \in \mathbb{R}^{N \times N}$ with $[\mathbf{K}_{\mathbf{Q}, \mathbf{Q}}]_{ij} = K(\vec{\mathbf{q}}_i, \vec{\mathbf{q}}_j)$ for all $i, j \in [N]$.

Since $\mathbf{W}(0) \in \mathcal{W}_0$, we have $\|\hat{K}_{\mathbf{S}, \mathbf{Q}} - \mathbf{K}_{\mathbf{S}, \mathbf{Q}}\|_2 \leq \sqrt{nN} C_1(m, d, 1/n)$ and $\|\hat{K}_{\mathbf{Q}, \mathbf{Q}} - \mathbf{K}_{\mathbf{Q}, \mathbf{Q}}\|_2 \leq N C_1(m, d, 1/n)$. As a result,

$$\begin{aligned} \hat{K}_{\mathbf{S}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{S}, \mathbf{Q}}^\top &= \underbrace{(\hat{K}_{\mathbf{S}, \mathbf{Q}} - \mathbf{K}_{\mathbf{S}, \mathbf{Q}}) \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{S}, \mathbf{Q}}^\top}_{E_1} + \underbrace{\mathbf{K}_{\mathbf{S}, \mathbf{Q}} (\hat{K}_{\mathbf{Q}, \mathbf{Q}} - \mathbf{K}_{\mathbf{Q}, \mathbf{Q}}) \hat{K}_{\mathbf{S}, \mathbf{Q}}^\top}_{E_2} \\ &\quad + \underbrace{\mathbf{K}_{\mathbf{S}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} (\hat{K}_{\mathbf{S}, \mathbf{Q}} - \mathbf{K}_{\mathbf{S}, \mathbf{Q}})^\top}_{E_3} + \mathbf{K}_{\mathbf{S}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \mathbf{K}_{\mathbf{S}, \mathbf{Q}}^\top, \end{aligned} \quad (95)$$

where

$$\max \{\|E_1\|_2, \|E_2\|_2, \|E_3\|_2\} \leq nN^2 C_1(m, d, 1/n), \quad (96)$$

since $\max \left\{ \|\hat{K}_{\mathbf{Q}, \mathbf{Q}}\|_\infty, \|\hat{K}_{\mathbf{S}, \mathbf{Q}}\|_\infty, \|\mathbf{K}_{\mathbf{S}, \mathbf{Q}}\|_\infty, \|\mathbf{K}_{\mathbf{Q}, \mathbf{Q}}\|_\infty \right\} \leq 1$ with $m \gtrsim (d \log m)^3$. We also have $\mathbf{K}_{\mathbf{S}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \mathbf{K}_{\mathbf{S}, \mathbf{Q}}^\top / (nN^2) = \hat{\mathbf{K}}_n^{(\text{attn})}$. It follows from Theorem 33 that with probability at least $1 - \delta$ over $\mathbf{Q}$, $\|\hat{\mathbf{K}}^{(\text{attn})} - \mathbf{K}^{(\text{attn})}\|_2 \lesssim n \sqrt{\frac{\log(n/\delta)}{N}}$. Combining such result with (94)- (96), we have

$$\begin{aligned} \left\| \frac{1}{nm} \boldsymbol{\sigma}^\top(\mathbf{W}(0), \mathbf{S}) \mathbf{A}^2 \boldsymbol{\sigma}(\mathbf{W}(0), \mathbf{S}) - \mathbf{K}_n^{(\text{attn})} \right\|_2 &\leq \left\| \frac{1}{nN^2} \hat{K}_{\mathbf{S}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{S}, \mathbf{Q}}^\top - \mathbf{K}_n^{(\text{attn})} \right\|_2 \\ &\lesssim C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}, \end{aligned}$$

which completes the proof. $\blacksquare$

Lemma 23 Suppose that $t \in [0, T - 1]$ for $T \geq 1$, $\|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq \sqrt{n} c_u$ for all $0 \leq t' \leq t$, $N \gtrsim d \log N$, $m \gtrsim \max \{(d^2 + \log^2 m) T^2, (d \log m)^3, n\}$, and the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Then with probability at least $1 - 1/n$ over $\mathbf{Q}$ and $\mathbf{W}(0)$, for all $0 \leq t' \leq t + 1$,

$$\left\| \vec{\mathbf{w}}_{\mathbf{S}, r}(t') - \vec{\mathbf{w}}_r(0) \right\|_2 \leq R, \quad (97)$$

$$|a_r(t')| \leq 2R. \quad (98)$$

Proof We prove (97) and (98) by induction. First, (97) and (98) hold trivially when $t' = 0$. Suppose that (97) and (98) hold for all $t' \in [0, t'']$ with $t'' \in [0, t]$, we now prove that they also hold for $t' = t'' + 1$.

With $m \gtrsim (d^2 + \log^2 m)T^2$, we have $|a_r(t')| \leq 1$ for all $t' \in [0, t'']$. It follows from Lemma 25 that (97) holds with $t' = t'' + 1$. Also, (98) holds with $t' = t'' + 1$ since $R \leq 1$ with such m . As a result, (97) and (98) hold for all $t' \in [0, t + 1]$. $\blacksquare$

Lemma 24 Suppose that $t \in [0, T-1]$ for $T \geq 1$, $\|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq \sqrt{n}c_{\mathbf{u}}$ and $\|\vec{\mathbf{w}}_{\mathbf{S},r}(t') - \vec{\mathbf{w}}_r(0)\|_2 \leq 1$ hold for all $0 \leq t' \leq t$, $N \gtrsim d \log N$, $m \gtrsim (d \log m)^3$, and the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Then with probability at least $1 - 1/n$ over $\mathbf{Q}$ and $\mathbf{W}(0)$,

$$|a_r(t')| \leq \frac{4\eta c_{\mathbf{u}}(2d + 3 \log(2mn))}{\sqrt{m}} = 2R, \quad \forall 0 \leq t' \leq t + 1, \forall r \in [m]. \quad (99)$$

Proof First, for every $t' \in [0, t]$ and every $r \in [m]$, we have

$$|a_r(t' + 1) - a_r(t')| \leq \frac{\eta}{n\sqrt{m}} \|\mathbf{A}_r\|_2 \|\sigma(\mathbf{W}(t'), \mathbf{S})\|_2 \|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2, \quad (100)$$

where $\mathbf{A}_r$ is the r -th row of $\mathbf{A}$. It follows from Proposition 12 that with probability at least $1 - 1/n$ over $\mathbf{Q}$ and $\mathbf{W}(0)$,

$$\|\mathbf{A}_r\|_2 \leq \frac{4d + 6 \log(2mn)}{\sqrt{m}}, \quad \forall r \in [m], \quad (101)$$

with $N \gtrsim d \log N$.

Define $\mathbf{H}(0) \in \mathbb{R}^{n \times n} = \sigma(\mathbf{W}(0), \mathbf{S})^\top \sigma(\mathbf{W}(0), \mathbf{S}) / (nm)$, then it follows from Theorem 11 that $\|\mathbf{H}(0) - \mathbf{K}_n\|_2 \leq C_1(m, d, 1/n)$ since $\mathbf{W}(0) \in \mathcal{W}_0$. It follows that

$$\begin{aligned} \|\mathbf{H}(0)\|_2 &\leq \|\mathbf{K}_n\|_2 + C_1(m, d, 1/n) \leq \frac{1}{2} + C_1(m, d, 1/n) \leq 1, \\ \|\sigma(\mathbf{W}(0), \mathbf{S})\|_2 &\leq \sqrt{mn} \sqrt{\|\mathbf{H}(0)\|_2} \leq \sqrt{mn}, \end{aligned} \quad (102)$$

since $m \gtrsim (d \log m)^3$. We then have

$$\begin{aligned} \|\sigma(\mathbf{W}(t'), \mathbf{S})\|_2 &\leq \|\sigma(\mathbf{W}(0), \mathbf{S})\|_2 + \|\sigma(\mathbf{W}(t'), \mathbf{S}) - \sigma(\mathbf{W}(0), \mathbf{S})\|_2 \\ &\leq \sqrt{mn} + \sqrt{mn} = 2\sqrt{mn}. \end{aligned} \quad (103)$$

It then follows from (100)-(103) that for every $t' \in [0, t]$ and every $r \in [m]$,

$$\|a_r(t' + 1) - a_r(t')\|_2 \leq \frac{4\eta c_{\mathbf{u}}(2d + 3 \log(2mn))}{\sqrt{m}} = 2R,$$

which proves (99). $\blacksquare$

Lemma 25 Suppose that $t \in [0, T-1]$ for $T \geq 1$, $\eta = \Theta(1) \in (0, 1)$, and $\|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq \sqrt{n}c_{\mathbf{u}}$, $|a_r(t')| \leq 1$ hold for all $0 \leq t' \leq t$ and every $r \in [m]$, $N \gtrsim d \log N$, and the random initialization $\mathbf{W}(0) \in \mathcal{W}_0$. Then with probability at least $1 - 1/n$ over $\mathbf{Q}$ and $\mathbf{W}(0)$,

$$\left\| \vec{\mathbf{w}}_{\mathbf{S},r}(t') - \vec{\mathbf{w}}_r(0) \right\|_2 \leq R, \quad \forall 0 \leq t' \leq t+1. \quad (104)$$

Proof First, it follows from (101) in the proof of Lemma 24 that with probability at least $1 - 1/n$ over $\mathbf{Q}$ and $\mathbf{W}(0)$, for every $t' \in [0, t]$ and every $r \in [m]$, we have

$$\left| \|\mathbf{A}\mathbf{a}(t')\|_r \right| \leq \|\mathbf{A}_r\|_2 \|\mathbf{a}(t')\|_2 \leq 4d + 6 \log(2mn). \quad (105)$$

Let $[\mathbf{Z}_{\mathbf{S}}(t)]_{[(r-1)d+1:rd]}$ denote the submatrix of $\mathbf{Z}_{\mathbf{S}}(t)$ formed by the the rows of $\mathbf{Z}_{\mathbf{Q}}(t)$ with row indices in $[(r-1)d+1 : rd]$. By the GD update rule we have for $t \in [0, T-1]$ that

$$\vec{\mathbf{w}}_{\mathbf{S},r}(t'+1) - \vec{\mathbf{w}}_{\mathbf{S},r}(t') = -\frac{\eta}{n} [\mathbf{Z}_{\mathbf{S}}(t)]_{[(r-1)d+1:rd]} (\hat{\mathbf{y}}(t') - \mathbf{y}). \quad (106)$$

It follows from (105) that $\left\| [\mathbf{Z}_{\mathbf{S}}(t')]_{[(r-1)d+1:rd]} \right\|_2 \leq (4d + 6 \log(2mn)) \cdot \sqrt{n/m}$ for all $t' \in [0, t]$. It then follows from (106) that

$$\left\| \vec{\mathbf{w}}_{\mathbf{S},r}(t'+1) - \vec{\mathbf{w}}_{\mathbf{S},r}(t') \right\|_2 \leq \frac{\eta}{n} \left\| [\mathbf{Z}_{\mathbf{S}}(t')]_{[(r-1)d+1:rd]} \right\|_2 \|\hat{\mathbf{y}}(t') - \mathbf{y}\|_2 \leq \frac{\eta c_{\mathbf{u}} (4d + 6 \log(2mn))}{\sqrt{m}}. \quad (107)$$

Note that (104) trivially holds for $t' = 0$. For $t' \in [1, t+1]$, it follows from (107) that

$$\left\| \vec{\mathbf{w}}_{\mathbf{S},r}(t') - \vec{\mathbf{w}}_r(0) \right\|_2 \leq \sum_{t''=0}^{t'-1} \left\| \vec{\mathbf{w}}_{\mathbf{S},r}(t''+1) - \vec{\mathbf{w}}_{\mathbf{S},r}(t'') \right\|_2 \leq \frac{\eta c_{\mathbf{u}} (4d + 6 \log(2mn)) T}{\sqrt{m}} = R,$$

which completes the proof. $\blacksquare$

Lemma 26 Let $h(\cdot) = \sum_{t'=0}^{t-1} h(\cdot, t')$ for $t \in [T]$, $T \leq \hat{T}$ where

$$\begin{aligned} h(\cdot, t') &= v(\cdot, t') + \hat{e}(\cdot, t'), \\ v(\cdot, t') &= \frac{\eta}{n} \sum_{j=1}^n K^{(\text{attn})}(\vec{\mathbf{x}}_j, \mathbf{x}) \mathbf{v}_j(t'), \\ \hat{e}(\cdot, t') &= \frac{\eta}{n} \sum_{j=1}^n K^{(\text{attn})}(\vec{\mathbf{x}}_j, \mathbf{x}) \vec{\mathbf{e}}_j(t'), \end{aligned}$$

and $\mathbf{v}(t') \in \mathcal{V}_{t', \tau}$, $\mathbf{e}(t') \in \mathcal{E}_{t', \tau}$ for all $0 \leq t' \leq t-1$. Suppose that

$$\tau \lesssim 1/(\eta T). \quad (108)$$

Then with probability at least $1 - \exp(-\Theta(n\hat{\varepsilon}_n^2))$ over $\mathbf{w}$,

$$\|h\|_{\mathcal{H}_{K^{(\text{attn})}}} \leq B_h = \mu_0 + 1 + \sqrt{2}, \quad (109)$$

where B_h is also defined in (29).

Proof The proof is similar to the proof of [59, Lemma B.5]. ■

Lemma 27 Suppose $\mathbf{W}(0) \in \mathcal{W}_0$, $m \gtrsim (d \log m)^3$, and $\|\mathbf{u}(t')\|_2 \leq c_u \sqrt{n}$ with $c_u = \Theta(1)$. Then with probability at least $1 - \delta$ over $\mathbf{Q}$,

$$\left\| \frac{1}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \sigma(\mathbf{W}(0), \mathbf{S}) \mathbf{u}(t') - \frac{1}{n} \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') \right\|_\infty \lesssim C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}. \quad (110)$$

Proof For every $j \in [n]$, we have

$$\frac{1}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \sigma(\mathbf{W}(0), \vec{\mathbf{x}}_j) = \frac{1}{nN^2} \hat{K}_{\mathbf{x}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j}, \quad (111)$$

where $\hat{K}_{\mathbf{x}, \mathbf{Q}} := \sigma^\top(\mathbf{W}(0), \mathbf{x}) \sigma(\mathbf{W}(0), \mathbf{Q})/m$ and $\hat{K}_{\mathbf{Q}, \mathbf{x}} = \hat{K}_{\mathbf{x}, \mathbf{Q}}^\top$. The following notations defined in the proof of Lemma 22 are also used in this proof. In particular, $\hat{K}_{\mathbf{S}, \mathbf{Q}} = \sigma^\top(\mathbf{W}(0), \mathbf{S}) \sigma(\mathbf{W}(0), \mathbf{Q})/m$, $\hat{K}_{\mathbf{Q}, \mathbf{Q}} = \sigma(\mathbf{W}(0), \mathbf{Q})^\top \sigma(\mathbf{W}(0), \mathbf{Q})/m$, $\mathbf{K}_{\mathbf{S}, \mathbf{Q}} \in \mathbb{R}^{n \times N}$ with $[\mathbf{K}_{\mathbf{S}, \mathbf{Q}}]_{ij} = K(\vec{\mathbf{x}}_i, \vec{\mathbf{q}}_j)$ for all $i \in [n]$, $j \in [N]$, and $\mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \in \mathbb{R}^{N \times N}$ with $[\mathbf{K}_{\mathbf{Q}, \mathbf{Q}}]_{ij} = K(\vec{\mathbf{q}}_i, \vec{\mathbf{q}}_j)$ for all $i, j \in [N]$. We further define $K(\mathbf{Q}, \mathbf{x}) \in \mathbb{R}^N$ with $[K(\mathbf{Q}, \mathbf{x})]_i = K(\mathbf{x}, \vec{\mathbf{q}}_i)$ for all $i \in [N]$, and $K(\mathbf{x}, \mathbf{Q}) = K^\top(\mathbf{Q}, \mathbf{x})$. $K^{(\text{attn})}(\mathbf{Q}, \mathbf{x}) \in \mathbb{R}^N$ is defined similarly for the kernel $\hat{K}^{(\text{attn})}$.

Since $\mathbf{W}(0) \in \mathcal{W}_0$, we have $\|\hat{K}_{\mathbf{x}, \mathbf{Q}} - K_{\mathbf{x}, \mathbf{Q}}\|_2 \leq \sqrt{N} C_1(m, d, 1/n)$ and $\|\hat{K}_{\mathbf{Q}, \mathbf{Q}} - \mathbf{K}_{\mathbf{Q}, \mathbf{Q}}\|_2 \leq NC_1(m, d, 1/n)$. As a result,

$$\begin{aligned} \hat{K}_{\mathbf{x}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j} &= \underbrace{(\hat{K}_{\mathbf{x}, \mathbf{Q}} - K_{\mathbf{x}, \mathbf{Q}}) \hat{K}_{\mathbf{Q}, \mathbf{Q}} \hat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j}}_{E_1} + \underbrace{K_{\mathbf{x}, \mathbf{Q}} (\hat{K}_{\mathbf{Q}, \mathbf{Q}} - \mathbf{K}_{\mathbf{Q}, \mathbf{Q}}) \hat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j}}_{E_2} \\ &\quad + \underbrace{K_{\mathbf{x}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} (\hat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j} - \mathbf{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j})}_{E_3} + \mathbf{K}_{\mathbf{x}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j}, \end{aligned} \quad (112)$$

where

$$\max \{\|E_1\|_2, \|E_2\|_2, \|E_3\|_2\} \leq N^2 C_1(m, d, 1/n), \quad (113)$$

due to the fact that $\max \{\|\hat{K}_{\mathbf{Q}, \mathbf{Q}}\|_\infty, \|\hat{K}_{\mathbf{S}, \mathbf{Q}}\|_\infty, \|\mathbf{K}_{\mathbf{S}, \mathbf{Q}}\|_\infty, \|\mathbf{K}_{\mathbf{Q}, \mathbf{Q}}\|_\infty\} \leq 1$ with $m \gtrsim (d \log m)^3$.

We also have $\mathbf{K}_{\mathbf{x}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \mathbf{Q}} \mathbf{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j} / (nN^2) = 1/n \cdot \hat{K}^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j)$. It follows from Theorem 33 that with probability at least $1 - \delta$ over $\mathbf{Q}$, $\sup_{\mathbf{x} \in \mathcal{X}, j \in [n]} |K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) - \hat{K}^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j)| \lesssim \sqrt{\frac{\log(n/\delta)}{N}}$. Combining such result with (111)-(113), we have

$$\left\| \frac{1}{nm} \sum_{r'=1}^m \sum_{r=1}^m \sigma \left(\vec{\mathbf{w}}_{r'}(0)^\top \mathbf{x} \right) \mathbf{A}_{r'r} \mathbf{A}_r \sigma(\mathbf{W}(0), \mathbf{S}) \mathbf{u}(t') - \frac{1}{n} \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') \right\|_\infty$$

$$\begin{aligned} &\leq \left\| \frac{1}{nN^2} \sum_{j=1}^n \widehat{K}_{\mathbf{x}, \mathbf{Q}} \widehat{K}_{\mathbf{Q}, \mathbf{Q}} \widehat{K}_{\mathbf{Q}, \vec{\mathbf{x}}_j} \mathbf{u}_j(t') - \frac{1}{n} \sum_{j=1}^n \widehat{K}^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') \right\|_{\infty} \\ &+ \left\| \frac{1}{n} \sum_{j=1}^n \widehat{K}^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') - \frac{1}{n} \sum_{j=1}^n K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_j) \mathbf{u}_j(t') \right\|_{\infty} \lesssim C_1(m, d, 1/n) + \sqrt{\frac{\log(n/\delta)}{N}}, \end{aligned}$$

which completes the proof. $\blacksquare$

Lemma 28 (In the proof of [39, Lemma 8]) For any $f \in \mathcal{H}_K(\mu_0)$, we have

$$\frac{1}{n} \sum_{i=1}^n \frac{[\mathbf{U}^\top f(\mathbf{S}')]_i^2}{\widehat{\lambda}_i} \leq \mu_0^2. \quad (114)$$

Lemma 29 For any positive real number $a \in (0, 1)$ and natural number t , we have

$$(1 - a)^t \leq e^{-ta} \leq \frac{1}{eta}. \quad (115)$$

Proof The result follows from the facts that $\log(1 - a) \leq -a$ for $a \in (0, 1)$ and $\sup_{u \in \mathbb{R}} ue^{-u} \leq 1/e$. $\blacksquare$

Lemma 30 ([58, Lemma B.7]) With probability at least $1 - 4 \exp(-\Theta(n\varepsilon_n^2))$,

$$\varepsilon_n^2 \lesssim \widehat{\varepsilon}_n^2, \quad \widehat{\varepsilon}_n^2 \lesssim \varepsilon_n^2. \quad (116)$$

Similarly, with probability at least $1 - 4 \exp(-\Theta(n\varepsilon_{K,n}^2))$,

$$\varepsilon_{K,n}^2 \lesssim \widehat{\varepsilon}_{K,n}^2, \quad \widehat{\varepsilon}_{K,n}^2 \lesssim \varepsilon_{K,n}^2. \quad (117)$$

Proof [Proof of Lemma 16] We first decompose the Rademacher complexity of the function class $\{f \in \mathcal{F}(B, w) : \mathbb{E}_P[f^2] \leq r\}$ into two terms as follows:

$$\begin{aligned} &\mathfrak{R}(\{f : f \in \mathcal{F}(B, w), \mathbb{E}_P[f^2] \leq r\}) \\ &\leq \underbrace{\frac{1}{n} \mathbb{E} \left[\sup_{f \in \mathcal{F}(B, w) : \mathbb{E}_P[f^2] \leq r} \sum_{i=1}^n \sigma_i h(\vec{\mathbf{x}}_i) \right]}_{:= \mathcal{R}_1} + \underbrace{\frac{1}{n} \mathbb{E} \left[\sup_{f \in \mathcal{F}(B, w) : \mathbb{E}_P[f^2] \leq r} \sum_{i=1}^n \sigma_i e(\vec{\mathbf{x}}_i) \right]}_{:= \mathcal{R}_2}. \quad (118) \end{aligned}$$

We now analyze the upper bounds for $\mathcal{R}_1, \mathcal{R}_2$ on the RHS of (118).

Derivation for the upper bound for $\mathcal{R}_1$.

According to definition of $\mathcal{F}(B, w)$ in (27), for any $f \in \mathcal{F}(B, w)$, we have $f = h + e$ with $h \in \mathcal{H}_{K^{(\text{attn})}}(B)$, $e \in L^\infty$, $\|e\|_\infty \leq w$.

When $\mathbb{E}_P[f^2] \leq r$, it follows from the triangle inequality that $\|h\|_{L^2} \leq \|f\|_{L^2} + \|e\|_{L^2} \leq \sqrt{r} + w := r_h$. We now consider $h \in \mathcal{H}_{K^{(\text{attn})}}(B)$ with $\|h\|_{L^2} \leq r_h$ in the remaining of this proof. We have

$$\begin{aligned} \sum_{i=1}^n \sigma_i f(\vec{\mathbf{x}}_i) &= \sum_{i=1}^n \sigma_i \left(h(\vec{\mathbf{x}}_i) + e(\vec{\mathbf{x}}_i) \right) \\ &= \left\langle h, \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i) \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} + \sum_{i=1}^n \sigma_i e(\vec{\mathbf{x}}_i). \end{aligned} \quad (119)$$

Because $\left\{ v_q^{(\text{attn})} = \sqrt{\lambda_q^{(\text{attn})}} e_q \right\}_{q \geq 1}$ is an orthonormal basis of $\mathcal{H}_{K^{(\text{attn})}}$, for any $0 \leq Q \leq n$, we further express the first term on the RHS of (119) as

$$\begin{aligned} \left\langle h, \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i) \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} &= \\ &= \left\langle \sum_{q=1}^Q \sqrt{\lambda_q^{(\text{attn})}} \left\langle h, v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})}, \sum_{q=1}^Q \left\langle \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} \frac{v_q^{(\text{attn})}}{\sqrt{\lambda_q^{(\text{attn})}}} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} \\ &+ \left\langle h, \sum_{q>Q} \left\langle \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}}. \end{aligned} \quad (120)$$

Due to the fact that $h \in \mathcal{H}_{K^{(\text{attn})}}$, $h = \sum_{q=1}^{\infty} \beta_q^{(h)} v_q^{(\text{attn})} = \sum_{q=1}^{\infty} \sqrt{\lambda_q^{(\text{attn})}} \beta_q^{(h)} e_q$ with $v_q^{(\text{attn})} = \sqrt{\lambda_q^{(\text{attn})}} e_q$. Therefore, $\|h\|_{L^2}^2 = \sum_{q=1}^{\infty} \lambda_q^{(\text{attn})} \beta_q^{(h)2}$, and

$$\begin{aligned} \left\| \sum_{q=1}^Q \sqrt{\lambda_q^{(\text{attn})}} \left\langle h, v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} &= \left\| \sum_{q=1}^Q \sqrt{\lambda_q^{(\text{attn})}} \beta_q^{(h)} v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} \\ &= \sqrt{\sum_{q=1}^Q \lambda_q^{(\text{attn})} \beta_q^{(h)2}} \leq \|h\|_{L^2} \leq r_h. \end{aligned} \quad (121)$$

According to Mercer's Theorem, because the kernel K is continuous symmetric positive definite, it has the decomposition

$$K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i) = \sum_{j=1}^{\infty} \lambda_j^{(\text{attn})} e_j(\cdot) e_j(\vec{\mathbf{x}}_i),$$

so that we have

$$\left\langle \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} = \left\langle \sum_{i=1}^n \sigma_i \sum_{j=1}^{\infty} \lambda_j^{(\text{attn})} e_j e_j(\vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}}$$

$$\begin{aligned}
 &= \left\langle \sum_{i=1}^n \sigma_i \sum_{j=1}^{\infty} \sqrt{\lambda_j^{(\text{attn})}} e_j(\vec{\mathbf{x}}_i) \cdot v_j^{(\text{attn})}, v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} \\
 &= \sum_{i=1}^n \sigma_i \sqrt{\lambda_q^{(\text{attn})}} e_q(\vec{\mathbf{x}}_i). \tag{122}
 \end{aligned}$$

Combining (120), (121), and (122), we have

$$\begin{aligned}
 \left\langle h, \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i) \right\rangle &\stackrel{\textcircled{1}}{\leq} \left\| \sum_{q=1}^Q \sqrt{\lambda_q^{(\text{attn})}} \left\langle h, v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} \\
 &\quad \left\| \sum_{q=1}^Q \frac{1}{\sqrt{\lambda_q^{(\text{attn})}}} \left\langle \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} \\
 &\quad + \|h\|_{\mathcal{H}_{K^{(\text{attn})}}} \cdot \left\| \sum_{q=Q+1}^{\infty} \left\langle \sum_{i=1}^n \sigma_i K^{(\text{attn})}(\cdot, \vec{\mathbf{x}}_i), v_q^{(\text{attn})} \right\rangle_{\mathcal{H}_{K^{(\text{attn})}}} v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} \\
 &\leq \|h\|_{L^2} \left\| \sum_{q=1}^Q \sum_{i=1}^n \sigma_i e_q(\vec{\mathbf{x}}_i) v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} + B \left\| \sum_{q=Q+1}^{\infty} \sum_{i=1}^n \sigma_i \sqrt{\lambda_q^{(\text{attn})}} e_q(\vec{\mathbf{x}}_i) v_q^{(\text{attn})} \right\|_{\mathcal{H}_{K^{(\text{attn})}}} \\
 &\leq r_h \sqrt{\sum_{q=1}^Q \left(\sum_{i=1}^n \sigma_i e_q(\vec{\mathbf{x}}_i) \right)^2} + B \sqrt{\sum_{q=Q+1}^{\infty} \left(\sum_{i=1}^n \sigma_i \sqrt{\lambda_q^{(\text{attn})}} e_q(\vec{\mathbf{x}}_i) \right)^2}, \tag{123}
 \end{aligned}$$

where $\textcircled{1}$ is due to Cauchy-Schwarz inequality. Moreover, by Jensen's inequality we have

$$\begin{aligned}
 \mathbb{E} \left[\sqrt{\sum_{q=1}^Q \left(\sum_{i=1}^n \sigma_i e_q(\vec{\mathbf{x}}_i) \right)^2} \right] &\leq \sqrt{\mathbb{E} \left[\sum_{q=1}^Q \left(\sum_{i=1}^n \sigma_i e_q(\vec{\mathbf{x}}_i) \right)^2 \right]} \\
 &\leq \sqrt{\mathbb{E} \left[\sum_{q=1}^Q \sum_{i=1}^n e_q^2(\vec{\mathbf{x}}_i) \right]} = \sqrt{nQ}. \tag{124}
 \end{aligned}$$

and similarly,

$$\mathbb{E} \left[\sqrt{\sum_{q=Q+1}^{\infty} \left(\sum_{i=1}^n \sigma_i \sqrt{\lambda_q^{(\text{attn})}} e_q(\vec{\mathbf{x}}_i) \right)^2} \right] \leq \sqrt{\mathbb{E} \left[\sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})} \sum_{i=1}^n e_q^2(\vec{\mathbf{x}}_i) \right]} = \sqrt{n \sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})}}. \tag{125}$$

Since (123)-(125) hold for all $Q \geq 0$, it follows that

$$\mathbb{E} \left[\sup_{h \in \mathcal{H}_{K^{(\text{attn})}}(B), \|h\|_{L^2} \leq r_h} \frac{1}{n} \sum_{i=1}^n \sigma_i h(\vec{\mathbf{x}}_i) \right] \leq \min_{Q: Q \geq 0} \left(r_h \sqrt{nQ} + B \sqrt{n \sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})}} \right). \tag{126}$$

It follows from (118), (119), and (126) that

$$\begin{aligned} \mathcal{R}_1 &\leq \frac{1}{n} \mathbb{E} \left[\sup_{h \in \mathcal{H}_{K^{(\text{attn})}}(B), \|h\|_{L^2} \leq r_h} \sum_{i=1}^n \sigma_i h(\vec{\mathbf{x}}_i) \right] \\ &\leq \min_{Q: Q \geq 0} \left(r_h \sqrt{\frac{Q}{n}} + B \left(\frac{\sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})}}{n} \right)^{1/2} \right). \end{aligned} \quad (127)$$

Derivation for the upper bound for $\mathcal{R}_2$.

Because $\left| 1/n \sum_{i=1}^n \sigma_i e(\vec{\mathbf{x}}_i) \right| \leq w$ when $\|e\|_{\infty} \leq w$, we have

$$\mathcal{R}_2 \leq \frac{1}{n} \mathbb{E} \left[\sup_{e \in L^{\infty}: \|e\|_{\infty} \leq w} \sum_{i=1}^n \sigma_i e(\vec{\mathbf{x}}_i) \right] \leq w. \quad (128)$$

It follows from (127) and (128) that

$$\mathfrak{R}(\{f: f \in \mathcal{F}(B, w), \mathbb{E}_P[f^2] \leq r\}) \leq \min_{Q: Q \geq 0} \left(r_h \sqrt{\frac{Q}{n}} + B \left(\frac{\sum_{q=Q+1}^{\infty} \lambda_q^{(\text{attn})}}{n} \right)^{1/2} \right) + w.$$

Plugging r_h in the RHS of the above inequality completes the proof. $\blacksquare$

Lemma 31 ([58, Lemma B.9]) Suppose $\psi: [0, \infty) \rightarrow [0, \infty)$ is a sub-root function with the unique fixed point r^* . Then the following properties hold.

- (1) Let $a \geq 0$, then $\psi(r) + a$ as a function of r is also a sub-root function with fixed point r_a^* , and $r^* \leq r_a^* \leq r^* + 2a$.
- (2) Let $b \geq 1$, $c \geq 0$ then $\psi(br + c)$ as a function of r is also a sub-root function with fixed point r_b^* , and $r_b^* \leq br^* + 2c/b$.
- (3) Let $b \geq 1$, then $\psi_b(r) = b\psi(r)$ is also a sub-root function with fixed point r_b^* , and $r_b^* \leq b^2 r^*$.

E.7. Proofs for the Approximate Uniform Convergence for the Kernel $K^{(\text{attn})}$

In this subsection, we present the main theorem, Theorem 33, regarding the approximate uniform convergence of $\widehat{K}^{(\text{attn})}(\cdot, \mathbf{x}')$ to $K^{(\text{attn})}(\cdot, \mathbf{x}')$ for every fixed $\mathbf{x}' \in \mathcal{X}$. Theorem 33 is the formal version of Theorem 3 in Section B.2. We first present below the concentration inequality for independent random variables taking values in a Hilbert space $\mathcal{B}$ of functions defined on a measurable space (S, Σ_S, μ_S) . Let $\{f_k\}_{k=0}^{\infty}$ be a martingale a separable Banach space $(\mathcal{B}, \|\cdot\|)$ with respect to an increasing sequence of σ -algebras $\{\mathcal{F}_k\}_{k=0}^{\infty}$ and $f_0 = 0$. Define $d_k := f_k - f_{k-1}$ for $k \geq 1$, $d_0 = 0$, and $f^* := \sup_{k \geq 0} \|f_k\|$.

For a function $g: \mathcal{B} \rightarrow \mathbb{R}$, The first Gâteaux derivative of g at a point $x \in \mathcal{B}$ along a direction $h \in \mathcal{B}$ is defined as

$$g'(x)(h) := \lim_{t \rightarrow 0} \frac{\|g(x + th)\| - \|g(x)\|}{t}.$$

The second Gâteaux derivative of g at a point $x \in \mathcal{B}$ along two directions $h_1, h_2 \in \mathcal{B}$ is defined as

$$g''(x)(h_1, h_2) := \lim_{t \rightarrow 0} \frac{g'(x + th_2)(h_1) - g'(x)(h_1)}{t}.$$

The class $D(A_1, A_2)$ consists of Banach spaces $\mathcal{B}$ such that $\|x\|'(\Delta) \leq A_1 \|\Delta\|$ and $\|x\|''(\Delta, \Delta) \leq A_2 \|\Delta\|^2 / \|x\|$ hold for all $x, \Delta \in \mathcal{B}$ and $x \neq 0$.

Lemma 32 (Martingale based concentration inequality for Banach space-valued process [37, Theorem 2])

Suppose that $\sum_{k=1}^{\infty} \text{esssup} \|d_k\|^2 \leq 1$ where $\text{esssup}(f) = \inf_{a \in \mathbb{R}} \{\mu(f^{-1}(a, +\infty)) = 0\}$ for a function denotes the essential supremum of a function, and $\mathcal{B} \in D(A_1, A_2)$ or $\mathcal{B} \subseteq L^p(S, \Sigma, \mu)$ with $p \geq 2$. Then for every $r > 0$,

$$\Pr[f^* > r] \leq 2 \exp\left(-\frac{r^2}{2B}\right) \quad (129)$$

with $B = A_1^2 + A_2$ for $\mathcal{B} \in D(A_1, A_2)$, and $B = p - 1$ for $\mathcal{B} \subseteq L^p(S, \Sigma_S, \mu_S)$.

Remark It is pointed out in [37] that when $\mathcal{B} \subseteq L^p(S, \Sigma, \mu)$, $\mathcal{B} \in D(1, p - 1)$, so that $B = A_1^2 + A_2 = p$ with $A_1 = 1, A_2 = p - 1$. However, for such specific case that $\mathcal{B} \subseteq L^p(S, \Sigma, \mu)$, a sharp bound with $B = p - 1$ can be achieved [37].

Theorem 33 For every fixed $\mathbf{x}' \in \mathcal{X}$ and every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $\mathbf{Q} = \{\vec{\mathbf{q}}_i\}_{i=1}^N$, we have

$$\left\| \widehat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right\|_{\infty} \lesssim \sqrt{\frac{\log 1/\delta}{N}}. \quad (130)$$

As a result, with probability at least $1 - \delta$ over $\mathbf{Q}$,

$$\sup_{\mathbf{x} \in \mathcal{X}, i \in [n]} \left| \widehat{K}^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_i) - K^{(\text{attn})}(\mathbf{x}, \vec{\mathbf{x}}_i) \right| \lesssim \sqrt{\frac{\log(n/\delta)}{N}}, \quad (131)$$

$$\left\| \widehat{\mathbf{K}}^{(\text{attn})} - \mathbf{K}^{(\text{attn})} \right\|_2 \lesssim n \sqrt{\frac{\log(n/\delta)}{N}}. \quad (132)$$

Proof We define

$$p(\mathbf{q}, \mathbf{x}') := \frac{1}{N} \sum_{j=1}^N K(\mathbf{q}, \vec{\mathbf{q}}_j) K(\vec{\mathbf{q}}_j, \mathbf{x}'), \quad \forall \mathbf{q}, \mathbf{x}' \in \mathcal{X}. \quad (133)$$

Since $\sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{X}} |K(\mathbf{x}, \mathbf{x}')| \leq \frac{1}{2} < 1$, we have $\sup_{\mathbf{q}, \mathbf{x}' \in \mathcal{X}} p(\mathbf{q}, \mathbf{x}') = \Theta(1)$. We now fix $\mathbf{x}' \in \mathcal{X}$ in the following arguments. It follows from (138) of Lemma 34 that for every $t > 0$ and every $i \in [N]$,

$$\Pr \left[\left\| \frac{1}{N} \sum_{j=1}^N K(\cdot, \vec{\mathbf{q}}_j) K(\vec{\mathbf{q}}_j, \mathbf{x}') - \bar{K}(\cdot, \mathbf{x}') \right\|_{\mathcal{H}_K} < t \right] \geq 1 - 2 \exp(-\Theta(Nt^2)), \quad (134)$$

where $\bar{K}(\cdot, \mathbf{x}') := \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) K(\mathbf{q}, \mathbf{x}')]$. The following arguments are conditioned on the event that (134) holds.

For each $i \in [N]$, we have $K(\cdot, \vec{\mathbf{q}}_i) \in \mathcal{H}_K$, and $\bar{K}(\cdot, \mathbf{x}') \in \mathcal{H}_K$. It follows from (134) that, for all $\mathbf{q} \in \mathcal{X}$,

$$\begin{aligned} |p(\mathbf{q}, \mathbf{x}') - \bar{K}(\mathbf{q}, \mathbf{x}')| &= \langle p(\cdot, \mathbf{x}') - \bar{K}(\cdot, \mathbf{x}'), K(\cdot, \mathbf{q}) \rangle \\ &\leq \left\| \frac{1}{N} \sum_{j=1}^N K(\cdot, \vec{\mathbf{q}}_j) K(\vec{\mathbf{q}}_j, \mathbf{x}') - \bar{K}(\cdot, \mathbf{x}') \right\|_{\mathcal{H}_K} \cdot \|K(\cdot, \mathbf{q})\|_{\mathcal{H}_K} \leq t. \end{aligned} \quad (135)$$

Define

$$\bar{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') := \frac{1}{N} \sum_{j=1}^N K(\mathbf{x}, \vec{\mathbf{q}}_j) \bar{K}(\vec{\mathbf{q}}_j, \mathbf{x}'), \quad \forall \mathbf{x}, \mathbf{x}' \in \mathcal{X}.$$

Then it follows from the definition of $\hat{K}^{(\text{attn})}$ in (5) that for all $\mathbf{x} \in \mathcal{X}$,

$$\begin{aligned} \left| \hat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - \bar{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right| &= \left| \frac{1}{N} \sum_{j=1}^N K(\mathbf{x}, \vec{\mathbf{q}}_j) p(\vec{\mathbf{q}}_j, \mathbf{x}') - \frac{1}{N} \sum_{j=1}^N K(\mathbf{x}, \vec{\mathbf{q}}_j) \bar{K}(\vec{\mathbf{q}}_j, \mathbf{x}') \right| \\ &\leq \frac{1}{N} \sum_{j=1}^N \left| K(\mathbf{x}, \vec{\mathbf{q}}_j) \right| \left| p(\vec{\mathbf{q}}_j, \mathbf{x}') - \bar{K}(\vec{\mathbf{q}}_j, \mathbf{x}') \right| \leq t, \end{aligned} \quad (136)$$

where the last inequality follows from (135).

Given the fixed $\mathbf{x}' \in \mathcal{X}$, we now approximate $K^{(\text{attn})}(\cdot, \mathbf{x}')$ by $\bar{K}^{(\text{attn})}(\cdot, \mathbf{x}')$. First, it can be verified from the definition of $\bar{K}$ that $\sup_{\mathbf{q}, \mathbf{x}' \in \mathcal{X}} |\bar{K}(\mathbf{q}, \mathbf{x}')| = \Theta(1)$. It then follows from (138) of Lemma 34 that

$$\Pr \left[\left\| \bar{K}^{(\text{attn})}(\cdot, \mathbf{x}') - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) \bar{K}(\mathbf{q}, \mathbf{x}')] \right\|_{\mathcal{H}_K} > t \right] \leq 2 \exp(-\Theta(Nt^2)), \quad (137)$$

and we have $\mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) \bar{K}(\mathbf{q}, \mathbf{x}')] = K^{(\text{attn})}(\cdot, \mathbf{x}')$. It follows from (136) and (137) that for with probability at least $1 - 4 \exp(-\Theta(Nt^2))$, for all $\mathbf{x} \in \mathcal{X}$,

$$\begin{aligned} &\left| \hat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right| \\ &\leq \left| \hat{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - \bar{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right| + \left| \bar{K}^{(\text{attn})}(\mathbf{x}, \mathbf{x}') - K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') \right| \\ &\leq t + \left\| \bar{K}^{(\text{attn})}(\cdot, \mathbf{x}') - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) \bar{K}(\mathbf{q}, \mathbf{x}')] \right\|_{\mathcal{H}_K} \cdot \|K(\cdot, \mathbf{x})\|_{\mathcal{H}_K} \leq 2t, \end{aligned}$$

which proves (130). (131) and (132) follow from (130) by the union bound. $\blacksquare$

Lemma 34 Suppose that p is a function defined on $\mathcal{X}$ and $\|p(\mathbf{x})\|_\infty = \Theta(1)$. Then for every $r > 0$,

$$\Pr \left[\left\| \frac{1}{N} \sum_{i=1}^N K(\cdot, \vec{\mathbf{q}}_i) p(\vec{\mathbf{q}}_i) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right\|_{\mathcal{H}_K} > r \right] \leq 2 \exp(-\Theta(Nr^2)). \quad (138)$$

Proof Let $\mathcal{B} = \mathcal{H}_K \subseteq L^2(\mathbb{S}^{d-1}, \mu)$, then $\mathcal{B} \in D(1, 1)$ [37]. Let $p_0 = \|p(\mathbf{x})\|_\infty = \Theta(1)$. We then construct the martingale $\{f_k\}_{k \in [N]}$. For each $k \in [N]$, we define

$$f_k := \mathbb{E} \left[\frac{1}{2p_0\sqrt{N}} \sum_{i=1}^N \left(K(\cdot, \vec{\mathbf{q}}_i) p(\vec{\mathbf{q}}_i) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right) \middle| \mathcal{F}_k \right], \forall k \in [N],$$

where $\{\mathcal{F}_k\}_{k=0}^N$ is an increasing sequence of σ -algebras, $\mathcal{F}_k$ is the σ -algebra generated by $\{\vec{\mathbf{q}}_t\}_{t=1}^k$. $\mathcal{F}_0$ is the trivial σ -algebra so that $f_0 = 0$. We note that

$$\begin{aligned} f_N &= \frac{1}{2p_0\sqrt{N}} \sum_{i=1}^N \left(K(\cdot, \vec{\mathbf{q}}_i) p(\vec{\mathbf{q}}_i) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right), \\ d_k &= f_k - f_{k-1} = \frac{1}{2p_0\sqrt{N}} \left(K(\cdot, \vec{\mathbf{q}}_k) p(\vec{\mathbf{q}}_k) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right), \forall k \in [N], \end{aligned}$$

and $f^* = \max_{k \in [N]} \|f_k\|$. For every $k \in [N]$, we have

$$\begin{aligned} \|d_k\|_{\mathcal{H}_K} &= \left\| \frac{1}{2p_0\sqrt{N}} \left(K(\cdot, \vec{\mathbf{q}}_k) p(\vec{\mathbf{q}}_k) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right) \right\|_{\mathcal{H}_K} \\ &\stackrel{\textcircled{1}}{\leq} \frac{1}{2p_0\sqrt{N}} \left(p_0 \|K(\cdot, \vec{\mathbf{q}}_k)\|_{\mathcal{H}_K} + p_0 \mathbb{E}_{\mathbf{q}} [\|K(\cdot, \mathbf{q})\|_{\mathcal{H}_K}] \right) \stackrel{\textcircled{2}}{\leq} \frac{1}{\sqrt{N}}, \end{aligned} \quad (139)$$

where $\textcircled{1}$ follows from the triangle inequality and the Jensen's inequality, and $\textcircled{2}$ follows from the fact that $\|K(\cdot, \vec{\mathbf{q}}_k)\|_{\mathcal{H}_K} \leq \frac{1}{2} < 1$.

It follows from (139) that $\sum_{k=1}^N \|d_k\|^2 \leq 1$. Applying Lemma 32 with the martingale $\{f_k\}_{k=0}^N$ and $\mathcal{B} = \mathcal{H}_K \subseteq L^2(\mathbb{S}^{d-1}, \mu)$, $B = 1$, we have $\Pr[f^* = \max_{k \in [N]} \|f_k\| > r] \leq 2 \exp\left(-\frac{r^2}{2}\right)$, and it follows that for every $r > 0$,

$$\Pr \left[\left\| \frac{1}{2p_0\sqrt{N}} \sum_{i=1}^N \left(K(\cdot, \vec{\mathbf{q}}_i) p(\vec{\mathbf{q}}_i) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right) \right\|_{\mathcal{H}_K} > r \right] \leq 2 \exp\left(-\frac{r^2}{2}\right),$$

and it follows that

$$\Pr \left[\left\| \frac{1}{N} \sum_{i=1}^N K(\cdot, \vec{\mathbf{q}}_i) p(\vec{\mathbf{q}}_i) - \mathbb{E}_{\mathbf{q}} [K(\cdot, \mathbf{q}) p(\mathbf{q})] \right\|_{\mathcal{H}_K} > r \right] \leq 2 \exp(-\Theta(Nr^2)),$$

which completes the proof of (138) and the constant in $\Theta(Nr^2)$ depends on $p_0 = \Theta(1)$. ■

Appendix F. More Results about University Convergence and Integral Operators

F.1. Results about Eigenvalues of the Integral Operators

Theorem 35 Let $\{e_j\}_{j \geq 1} \subseteq L^2(\mathcal{X}, \mu)$ be a countable orthonormal basis of $L^2(\mathcal{X}, \mu)$ which comprise the eigenfunctions of the integral operator $T_K: L^2(\mathcal{X}, \mu) \rightarrow L^2(\mathcal{X}, \mu)$, $(T_K f)(\mathbf{x}) := \int_{\mathcal{X}} K(\mathbf{x}, \mathbf{x}') f(\mathbf{x}') d\mu(\mathbf{x}')$, a positive, self-adjoint, and compact operator on $L^2(\mathcal{X}, \mu)$. Let $\{\lambda_j\}_{j \geq 1}$ with $\frac{1}{2} \geq \lambda_1 \geq \lambda_2 \geq \dots > 0$ such that e_j is the eigenfunction of T_K with λ_j being the corresponding eigenvalue. Then e_j is the eigenfunction of $T_{K(\text{attn})}$ with λ_j^2 being the corresponding eigenvalue. That is, $T_{K(\text{attn})} e_j = \lambda_j^3 e_j$, so that $\lambda_j^{(\text{attn})} = \lambda_j^3$ for all $j \geq 1$.

Proof First, since $T_K e_1 = \lambda_1 e_1$ it follows from the Cauchy-Schwarz inequality that

$$\lambda_1^2 = \|\lambda_1 e_1\|_{L^2(\mathcal{X}, \mu)}^2 \leq \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{x}')^2 e_1^2(\mathbf{x}') d\mu(\mathbf{x}') d\mu(\mathbf{x}) \leq 1/4,$$

which proves that $\lambda_j \in (0, 1/2]$ for all $j \geq 1$. It follows from Mercer's theorem that

$$K(\mathbf{v}, \mathbf{v}') = \sum_{j \geq 1} \lambda_j e_j(\mathbf{v}) e_j(\mathbf{v}'), \quad \forall \mathbf{v}, \mathbf{v}' \in \mathcal{X},$$

and the convergence on the RHS of the above equality is uniform and absolute. Then it follows from the definition of $K^{(\text{attn})}$ in (4) that

$$\begin{aligned} K^{(\text{attn})}(\mathbf{x}, \mathbf{x}') &= \int_{\mathcal{X} \times \mathcal{X}} K(\mathbf{x}, \mathbf{v}) K(\mathbf{v}, \mathbf{v}') K(\mathbf{v}', \mathbf{x}') d\mu(\mathbf{v}) \otimes \mu(\mathbf{v}') \\ &\stackrel{\textcircled{1}}{=} \int_{\mathcal{X}} \left(\int_{\mathcal{X}} \sum_{j \geq 1} \lambda_j e_j(\mathbf{x}) e_j(\mathbf{v}) \cdot \sum_{j \geq 1} \lambda_j e_j(\mathbf{v}) e_j(\mathbf{v}') d\mu(\mathbf{v}) \right) \cdot \sum_{j \geq 1} \lambda_j e_j(\mathbf{v}') e_j(\mathbf{x}') \mu(\mathbf{v}') \\ &\stackrel{\textcircled{2}}{=} \int_{\mathcal{X}} \left(\sum_{j \geq 1} \lambda_j^2 e_j(\mathbf{x}) e_j(\mathbf{v}') \right) \cdot \sum_{j \geq 1} \lambda_j e_j(\mathbf{v}') e_j(\mathbf{x}') \mu(\mathbf{v}') \stackrel{\textcircled{3}}{=} \sum_{j \geq 1} \lambda_j^3 e_j(\mathbf{x}) e_j(\mathbf{x}'), \end{aligned} \quad (140)$$

where ① follows from the Fubini's Theorem, and ②, ③ follow by the orthogonality of the orthogonal basis $\{e_j\}_{j \geq 1}$.

It follows from (140) that for all $j \geq 1$,

$$(T_{K(\text{attn})} e_j)(\mathbf{x}) = \int_{\mathcal{X}} \left(\sum_{j' \geq 1} \lambda_{j'}^3 e_{j'}(\mathbf{x}) e_{j'}(\mathbf{x}') \right) e_j(\mathbf{x}') d\mu(\mathbf{x}') = \lambda_j^3 e_j(\mathbf{x}),$$

which proves that $\lambda_j^{(\text{attn})} = \lambda_j^3$ for all $j \geq 1$. ■

It is known, such as [15, Theorem 3.1], that $\mathbf{K}_n$ is non-singular. Based on this fact, we have the following propositions showing that $\mathbf{K}_n^{(\text{attn})}$ is also non-singular.

Proposition 36 If $\vec{\mathbf{x}}_i \neq \vec{\mathbf{x}}_j$ for all $i, j \in [n]$ and $i \neq j$, then $\mathbf{K}_n^{(\text{attn})}$ is also non-singular.

Proof [15, Theorem 3.1] shows that $\mathbf{K}_n$ is non-singular. Define the feature mapping $\Phi(\mathbf{x}) := [\sqrt{\lambda_1}e_1(\mathbf{x}), \sqrt{\lambda_2}e_2(\mathbf{x}), \dots]$. Since $[\mathbf{K}_n]_{ij} = 1/n \cdot \Phi(\vec{\mathbf{x}}_i)^\top \Phi(\vec{\mathbf{x}}_j)$, the non-singularity of $\mathbf{K}$ indicates that the feature maps on the data $\mathbf{S}$, $\{\Phi(\vec{\mathbf{x}}_i)\}_{i=1}^n$, are linearly independent.

On the other hand, Theorem 35 shows that the $\{\lambda_j^3, e_j\}_{j \geq 1}$ are the eigenvalues and the corresponding eigenfunctions of the integral operator $T_{K(\text{attn})}$. Let $\tilde{\Phi} := [\lambda_1^{\frac{3}{2}}e_1(\mathbf{x}), \lambda_2^{\frac{3}{2}}e_2(\mathbf{x}), \dots]$. Then $[\mathbf{K}_n^{(\text{attn})}]_{ij} = 1/n \cdot \tilde{\Phi}(\vec{\mathbf{x}}_i)^\top \tilde{\Phi}(\vec{\mathbf{x}}_j)$. Because $\{\Phi(\vec{\mathbf{x}}_i)\}_{i=1}^n$ are linearly independent, it can be verified by definition that $\{\tilde{\Phi}(\vec{\mathbf{x}}_i)\}_{i=1}^n$ are also linearly independent, so that $\mathbf{K}_n^{(\text{attn})}$ is not singular. ■

F.2. Proofs of Theorem 19

We need the definition of ε -net in Definition 37 for the proof of Theorem 19.

Definition 37 (ε -net) Let (X, d) be a metric space and let $\varepsilon > 0$. A subset $N_\varepsilon(X, d)$ is called an ε -net of X if for every point $x \in X$, there exists some point $y \in N_\varepsilon(X, d)$ such that $d(x, y) \leq \varepsilon$. The minimal cardinality of an ε -net of X , if finite, is denoted by $N(X, d, \varepsilon)$ and is called the covering number of X at scale ε .

Proof [Proof of Theorem 19] First, we have $\mathbb{E}_{\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \kappa^2 \mathbf{I}_d)} [h(\mathbf{w}, \mathbf{u}, \mathbf{v})] = K(\mathbf{u}, \mathbf{v})$. For any $\mathbf{u} \in \mathcal{X}$, $\mathbf{v} \in \mathcal{X}$, and $s > 0$, define function class

$$\mathcal{H}_{\mathbf{u}, \mathbf{v}, s} := \left\{ h_\delta(\cdot, \mathbf{u}', \mathbf{v}') : \mathbb{R}^d \rightarrow \mathbb{R} : \mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X} \right\}, \quad (141)$$

where $h_\delta(\mathbf{w}, \mathbf{u}', \mathbf{v}') = \sigma\left([\mathbf{w}^\top \mathbf{u}']_{M_\delta}\right) \sigma\left([\mathbf{w}^\top \mathbf{v}']_{M_\delta}\right)$ for all $\mathbf{w} \in \mathbb{R}^d$ and $\mathbf{u}', \mathbf{v}' \in \mathcal{X}$ with

$$M_\delta := \tilde{M}_\delta + \frac{\sqrt{2d + 3 \log(3m/\delta)}}{m}, \quad \tilde{M}_\delta := \kappa \sqrt{2 \log(2m)} + \kappa \sqrt{2 \log(3/\delta)},$$

for $\delta \in (0, 1)$. Also, for all $a > 0$ $[\cdot]_a$ is defined as $[\cdot]_a := \text{sgn}(\cdot) \min\{|\cdot|, a\}$. We first build an s -net for the unit sphere $\mathcal{X}$. By [48, Lemma 5.2], there exists an s -net $N_s(\mathcal{X}, \|\cdot\|_2)$ of $\mathcal{X}$ such that $N(\mathcal{X}, \|\cdot\|_2, s) \leq (1 + \frac{2}{s})^d$.

In the sequel, a function in the class $\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$ is also denoted as $h_\delta(\mathbf{w})$, omitting the presence of variables $\mathbf{u}'$ and $\mathbf{v}'$ when no confusion arises. Let P_m be the empirical distribution over $\{\vec{\mathbf{w}}_r(0)\}$ so that $\mathbb{E}_{\mathbf{w} \sim P_m} [h_\delta(\mathbf{w})] = 1/m \cdot \sum_{r=1}^m h_\delta(\vec{\mathbf{w}}_r(0))$. Given $\mathbf{u} \in N(\mathcal{X}, s)$, we aim to estimate the upper bound for the supremum of empirical process $\mathbb{E}_{\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \kappa^2 \mathbf{I}_d)} [h(\mathbf{w})] - \mathbb{E}_{\mathbf{w} \sim P_m} [h(\mathbf{w})]$ when function h ranges over the function class $\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$. To this end, we apply Theorem 7 to the function class $\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$ with $\mathbf{W}(0) = \{\vec{\mathbf{w}}_r(0)\}_{r=1}^m$. Since $h_\delta(\cdot, \mathbf{u}', \mathbf{v}') \in [0, M_\delta^2]$ for any $h_\delta \in \mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$, we set $a = 0, b = M_\delta^2, \alpha = 1/2$ in Theorem 7, and $\text{Var}[h_\delta] \leq M_\delta^4$. As a result, with probability at least

$1 - \delta$ over the random initialization $\mathbf{W}(0)$,

$$\sup_{\substack{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \\ \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}}} |K_\delta(\mathbf{u}', \mathbf{v}') - \mathbb{E}_{\mathbf{w} \sim P_m} [h_\delta(\mathbf{w}, \mathbf{u}', \mathbf{v}')] | \leq 6\hat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) + M_\delta^2 \sqrt{\frac{2\log \frac{2}{\delta}}{m}} + \frac{16M_\delta^2 \log \frac{2}{\delta}}{3m}, \quad (142)$$

where $K_\delta(\mathbf{u}', \mathbf{v}') := \mathbb{E}_{\mathbf{w}} [h_\delta(\mathbf{w}, \mathbf{u}', \mathbf{v}')] , \hat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) = \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{h_\delta \in \mathcal{H}_{\mathbf{u}, \mathbf{v}, s}} \frac{1}{m} \sum_{r=1}^m \sigma_r h_\delta(\vec{\mathbf{w}}_r(0)) \right]$ is the empirical Rademacher complexity of the function class $\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$, $\{\sigma_r\}_{r=1}^m$ are i.i.d. Rademacher random variables taking values of ± 1 with equal probability. By Lemma 38, $\hat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) \leq 2s\tilde{M}_\delta \max_{r \in [m]} \|\vec{\mathbf{w}}_r(0)\|_2$. Plugging such upper bound for $\hat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s})$ in (142) and setting $s = \frac{1}{m}$, we have

$$\sup_{\substack{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \\ \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}}} |K_\delta(\mathbf{u}', \mathbf{v}') - \mathbb{E}_{\mathbf{w} \sim P_m} [h_\delta(\mathbf{w}, \mathbf{u}', \mathbf{v}')] | \leq \frac{12M_\delta \max_{r \in [m]} \|\vec{\mathbf{w}}_r(0)\|_2}{m} + M_\delta^2 \sqrt{\frac{2\log \frac{2}{\delta}}{m}} + \frac{16M_\delta^2 \log \frac{2}{\delta}}{3m}. \quad (143)$$

It follows from Lemma 40 that with probability at least $1 - \delta$ over $\mathbf{W}(0)$,

$$\begin{aligned} \max_{r \in [m]} \max \left\{ \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{v} \right|, \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{u} \right| \right\} &\leq \tilde{M}_\delta, \\ \max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2^2 &\leq d + 2\sqrt{d \log(3m/\delta)} + 2\log(3m/\delta) \leq 2d + 3\log(3m/\delta). \end{aligned}$$

As a result, with probability at least $1 - \delta$ over $\mathbf{W}(0)$,

$$\max_{r \in [m]} \sup_{\substack{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \\ \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}}} \left\{ \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{v}' \right|, \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{u}' \right| \right\} \leq \tilde{M}_\delta + \frac{\sqrt{2d + 3\log(3m/\delta)}}{m} = M_\delta. \quad (144)$$

When (144) holds, $h_\delta = h$ and $\mathbb{E}_{\mathbf{w} \sim P_m} [h_\delta(\mathbf{w}, \mathbf{u}', \mathbf{v}')] = \hat{h}(\mathbf{W}(0), \mathbf{u}', \mathbf{v}')$, $K_\delta(\mathbf{u}', \mathbf{v}') = K(\mathbf{u}', \mathbf{v}')$. It then follows from (143), (144), and the union bound that with probability at least $1 - 2\delta$ over $\mathbf{W}(0)$,

$$\begin{aligned} \sup_{\substack{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \\ \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}}} \left| K(\mathbf{u}', \mathbf{v}') - \hat{h}(\mathbf{W}(0), \mathbf{u}', \mathbf{v}') \right| &\leq \frac{12M_\delta(2d + 3\log(3m/\delta))}{m} + M_\delta^2 \sqrt{\frac{2\log \frac{2}{\delta}}{m}} \\ &\quad + \frac{16M_\delta^2 \log \frac{2}{\delta}}{3m}. \end{aligned} \quad (145)$$

By the union bound, with probability at least $1 - 2(1 + 2m)^{2d} \delta$ over $\mathbf{W}(0)$, (145) holds for arbitrary $\mathbf{u}, \mathbf{v} \in N(\mathcal{X}, s)$. In this case, for any $\mathbf{u}' \in \mathcal{X}, \mathbf{v}' \in \mathcal{X}$, there exists $\mathbf{u}, \mathbf{v} \in N_s(\mathcal{X}, \|\cdot\|_2)$ such that $\|\mathbf{u}' - \mathbf{u}\|_2 \leq s, \|\mathbf{v}' - \mathbf{v}\|_2 \leq s$, so that $\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}$, and (145) holds.

Changing the notations $\mathbf{u}', \mathbf{v}'$ to $\mathbf{u}, \mathbf{v}$, (48) is proved by the union bound. ■

Lemma 38 Let $\widehat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) := \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{h \in \mathcal{H}_{\mathbf{u}, \mathbf{v}, s}} \frac{1}{m} \sum_{r=1}^m \sigma_r h(\vec{\mathbf{w}}_r(0)) \right]$ be the Rademacher complexity of the function class $\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}$, and B is a positive constant. Then

$$\widehat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) \leq 2sM_\delta \max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2. \quad (146)$$

Proof

We have

$$\widehat{\mathcal{R}}(\mathcal{H}_{\mathbf{u}, \mathbf{v}, s}) = \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}} \frac{1}{m} \sum_{r=1}^m \sigma_r h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}', \mathbf{v}') \right] \leq \mathcal{R}_1 + \mathcal{R}_2 + \mathcal{R}_3, \quad (147)$$

where

$$\begin{aligned} \mathcal{R}_1 &= \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}} \frac{1}{m} \sum_{r=1}^m \sigma_r \left(h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}', \mathbf{v}') - h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}') \right) \right], \\ \mathcal{R}_2 &= \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}} \frac{1}{m} \sum_{r=1}^m \sigma_r \left(h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}') - h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}) \right) \right], \\ \mathcal{R}_3 &= \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}} \frac{1}{m} \sum_{r=1}^m \sigma_r h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}) \right]. \end{aligned} \quad (148)$$

Here (147) follows from the subadditivity of supremum. Now we bound $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$ separately. First, $\mathcal{R}_3 = 0$ by the definition of the Rademacher variables. For $\mathcal{R}_1$, we have

$$\begin{aligned} & \left| h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}', \mathbf{v}') - h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}') \right| \\ & \leq \left| \sigma \left(\left[\vec{\mathbf{w}}_r(0)^\top \mathbf{u}' \right]_{M_\delta} \right) \sigma \left(\left[\vec{\mathbf{w}}_r(0)^\top \mathbf{v}' \right]_{M_\delta} \right) - \sigma \left(\left[\vec{\mathbf{w}}_r(0)^\top \mathbf{u} \right]_{M_\delta} \right) \sigma \left(\left[\vec{\mathbf{w}}_r(0)^\top \mathbf{v}' \right]_{M_\delta} \right) \right| \\ & \leq s \left\| \vec{\mathbf{w}}_r(0) \right\|_2 \left| \sigma \left(\left[\vec{\mathbf{w}}_r(0)^\top \mathbf{v}' \right]_{M_\delta} \right) \right| \leq sM_\delta \max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2. \end{aligned} \quad (149)$$

It follows from (149) that

$$\begin{aligned} \mathcal{R}_1 &= \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\sup_{\mathbf{u}' \in \mathbf{B}(\mathbf{u}; s) \cap \mathcal{X}, \mathbf{v}' \in \mathbf{B}(\mathbf{v}; s) \cap \mathcal{X}} \frac{1}{m} \sum_{r=1}^m \sigma_r \left(h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}', \mathbf{v}') - h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}') \right) \right] \\ &\leq \mathbb{E}_{\{\sigma_r\}_{r=1}^m} \left[\frac{1}{m} \sum_{r=1}^m \left| h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}', \mathbf{v}') - h_\delta(\vec{\mathbf{w}}_r(0), \mathbf{u}, \mathbf{v}') \right| \right] \leq sM_\delta \max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2. \end{aligned} \quad (150)$$

Applying the argument for $\mathcal{R}_1$ to $\mathcal{R}_2$, we have $\mathcal{R}_2 \leq sM_\delta \max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2$. Plugging such upper bound for $\mathcal{R}_2$, (150), and $\mathcal{R}_3 = 0$ in (147), (146) is proved. ■

Lemma 39 *Let $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \kappa^2 \mathbf{I}_d)$ with $\kappa > 0$. Then for any $\varepsilon \in (0, 1)$ and fixed $\mathbf{u} \in \mathcal{X}$, $\Pr \left[\frac{|\mathbf{u}^\top \mathbf{w}|}{\|\mathbf{w}\|_2} \leq \varepsilon \right] \leq B\sqrt{d}\varepsilon$ where B is an absolute positive constant, and B can be set to $\pi^{-1/2}$.*

Proof Let $z = \frac{\mathbf{u}^\top \mathbf{w}}{\|\mathbf{w}\|_2}$. It can be verified that $z^2 \sim z_1$ where z_1 is a random variable following the Beta distribution $\text{Beta}(\frac{1}{2}, \frac{d-1}{2})$. Therefore, the distribution of z has the following continuous probability density function p_z with respect to the Lebesgue measure,

$$p_z(x) = (1 - x^2)^{\frac{d-3}{2}} \mathbb{1}_{\{|x| \leq 1\}} / B', \quad (151)$$

where $B' = \int_{-1}^1 (1 - x^2)^{\frac{d-3}{2}} dx$ is the normalization factor. It can be verified by standard calculation that $1/B' \leq B\sqrt{d}/2$ for an absolute positive constant B . Since $1 - x^2 \leq 1$ over $x \in [-1, 1]$, we have

$$\Pr \left[\frac{|\mathbf{u}^\top \mathbf{w}|}{\|\mathbf{w}\|_2} \leq \varepsilon \right] = \Pr[-\varepsilon \leq z \leq \varepsilon] = \frac{1}{B'} \int_{-\varepsilon}^{\varepsilon} (1 - x^2)^{\frac{d-3}{2}} dx \leq B\sqrt{d}\varepsilon, \quad (152)$$

where the last inequality is due to the fact that $1 - x^2 \leq 1$ for $x \in [-\varepsilon, \varepsilon]$ with $\varepsilon \in (0, 1)$. $\blacksquare$

For every $\mathbf{v} \in \mathbb{S}^{d-1}$, noting that $\vec{\mathbf{w}}_r(0)^\top \mathbf{v} \sim \mathcal{N}(0, \kappa^2)$, we have the following standard upper bound for $\max_{r \in [m]} \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{v} \right|$.

Lemma 40 *For every fixed $\mathbf{v} \in \mathbb{R}^d$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $\mathbf{W}(0)$,*

$$\max_{r \in [m]} \left| \vec{\mathbf{w}}_r(0)^\top \mathbf{v} \right| \leq \kappa \sqrt{2 \log(2m)} + \kappa \sqrt{2 \log \frac{1}{\delta}}. \quad (153)$$

Moreover, it follows from Lemma 41 that with probability at least $1 - \delta$ over $\mathbf{W}(0)$, $\max_{r \in [m]} \left\| \vec{\mathbf{w}}_r(0) \right\|_2^2 \leq d + 2\sqrt{d \log(m/\delta)} + 2 \log(m/\delta)$.

Lemma 41 ([27, Lemma 1]) *Let $\{X_i\}_{i=1}^k$ be i.i.d. standard Gaussian random variables and $X = \sum_{i=1}^k X_i^2$, then*

$$\begin{aligned} \Pr \left[X - k \geq 2\sqrt{kx} + 2x \right] &\leq \exp(-x) \\ \Pr \left[k - X \geq 2\sqrt{kx} \right] &\leq \exp(-x) \end{aligned} \quad (154)$$

Appendix G. Simulation Results

We provide simulation results on both synthetic data and real data in this section.

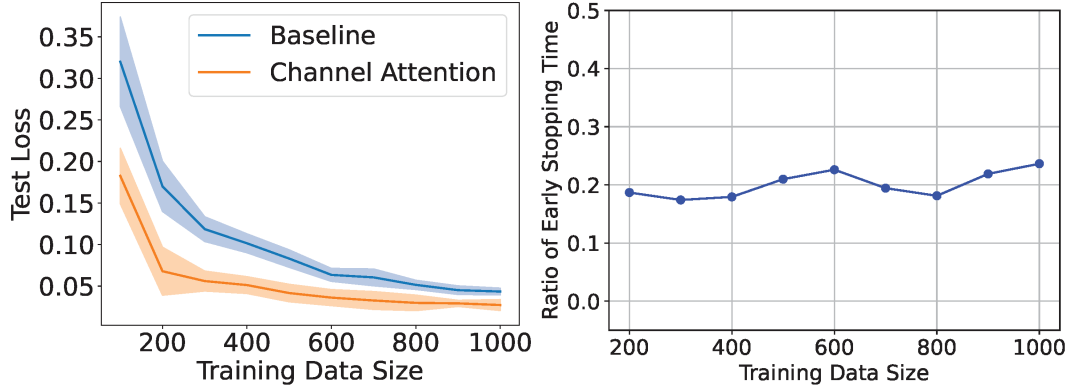


Figure 1: Left: illustration of the test loss by the vanilla network and the network with the proposed channel attention for varying n in $[100, 1000]$ with a step size of 100. The shaded area in each plot indicates the standard deviation across 10 random initializations of the neural network. Right: illustration of the ratio of early stopping time.

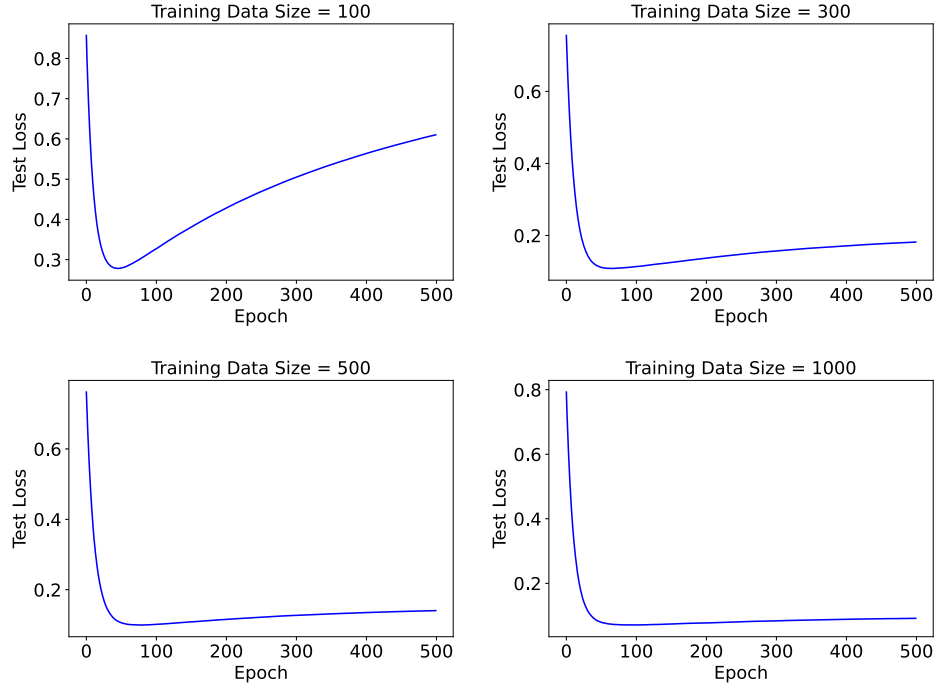


Figure 2: Illustration of the test loss by GD, averaged over 10 random initializations of the neural network.

G.1. Results on Synthetic Data

We present simulation results on synthetic data in this section. We randomly sample n points $\{\vec{\mathbf{x}}_i\}_{i=1}^n$ distributed uniformly on the unit sphere $\mathbb{S}^{49}$ in $\mathbb{R}^{50}$. The sample size n ranges from 100 to 1000 with a step size of 100. We set the target function as $f^*(\mathbf{x}) = \mathbf{s}^\top \mathbf{x}$, where $\mathbf{x} \in \mathbb{S}^{49}$ and $\mathbf{s} \sim \text{Unif}(\mathcal{X})$ is randomly sampled. The noise variance is set to $\sigma_0^2 = 1$. We also uniformly and independently sample 1000 points on the unit sphere in $\mathbb{R}^{50}$ to form the test set. The two-layer NN with channel attention (1) is trained by Algorithm 1 with network width $m = 10000$, sample size $N = 10000$ for $\mathbf{Q}$, and learning rate $\eta = 1$. The training is executed on an NVIDIA A100 GPU, and the test loss is reported in Figure 1 and Figure 2. From Figure 1, it is evident that the network with channel attention (1) consistently exhibits better generalization than the vanilla two-layer neural network without such an attention mechanism, that is, $f^{(\text{vanilla})}$ (3), by achieving lower test losses across different training data sizes. Figure 2 presents the test loss as a function of GD steps for $n = 100, 300, 500, 1000$. As shown in Figure 2, early stopping reliably improves generalization in neural network training, since the test loss initially decreases and later increases due to overfitting.

For each $n \in \{100, 200, \dots, 1000\}$, we denote the GD step that attains the minimum test loss as $\hat{t}_n$, which acts as the empirical early stopping time. We note that the early stopping time theoretically predicted by Theorem 2 scales as $\hat{T} \asymp n^{\frac{6\alpha}{6\alpha+1}} \asymp n^{\frac{3d}{4d-1}}$ with $2\alpha = d/(d-1)$. We compute the ratio of early stopping time, defined as $\hat{t}_n/n^{\frac{3d}{4d-1}}$ and averaged over 10 random neural network initializations for each n , and display it in the right plot of Figure 1. It can be observed that the ratio of early stopping time is relatively stable and lies within $[0.17, 0.23]$ with respect to different training data sizes, suggesting that the theoretically predicted early stopping time is indeed empirically proportional to the empirical early stopping time.

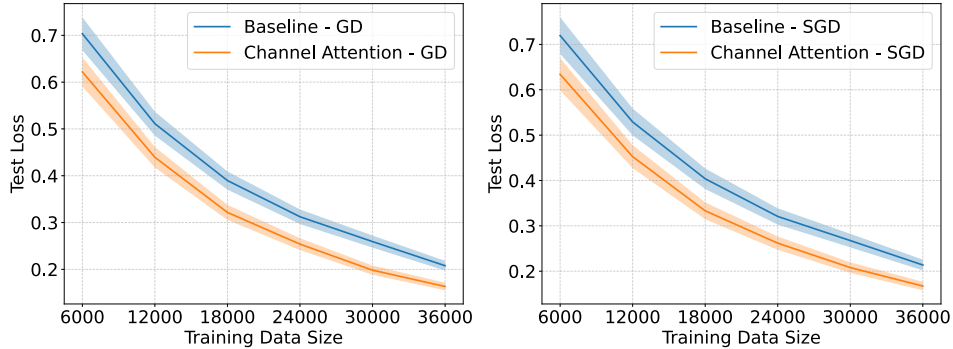


Figure 3: Illustration of the test loss by the vanilla network and the network with the proposed channel attention for varying n in $[100, 1000]$ with a step size of 100. The shaded area in each plot indicates the standard deviation across 10 random initializations of the neural network.

G.2. Results on Real Data (Mini-ImageNet)

We herein provide additional empirical results for the two-layer NN with channel attention (1) trained on a real dataset, mini-ImageNet [49] with 60000 images from 100 classes, where the training features follow a complex distribution rather than the spherical uniform distribution and the target function may not lie in a RKHS ball of bounded radius associated with the neural network. The literature such as [64] shows that the class labels of such dataset cannot be explained by the NTK of the neural network, so that the target function is not in the RKHS associated with the NTK, and it is not in the interpolation space studied in this paper either. We use 60 classes comprising 36000 images as the training features and the remaining classes as the test data, and the size of the sample $\mathbf{Q}$ is set to be three times the size of the training data. We sample $\mathbf{Q}$ from a DiT [35] trained on the training data. Figure 3 illustrates the test loss of the vanilla network, $f^{(\text{vanilla})}$, and our two-layer NN with attention channel (1) with respect to different training data sizes where the one-hot class labels serve as the response vectors for regression. It can be observed that network with channel attention always outperforms the vanilla network without channel attention with lower test losses, when both networks are trained by GD or standard SGD.